ArtComputer

DESIGNER User Manual

Automated Test with the Robot

Designer User Manual

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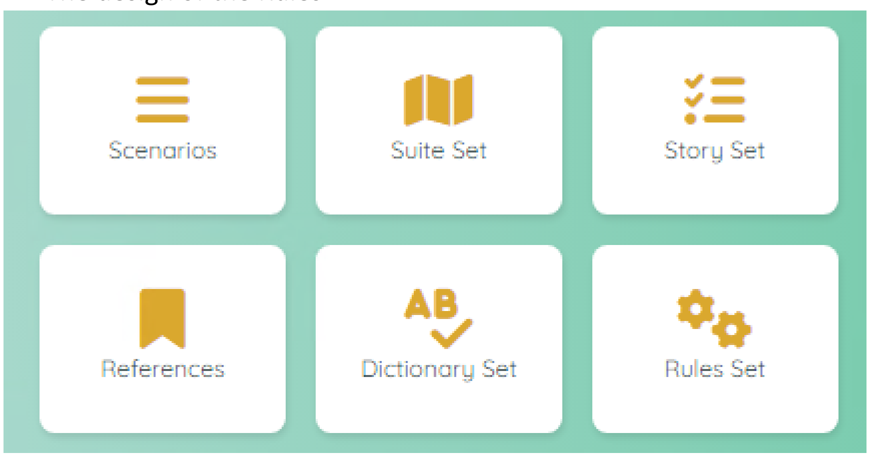
Foreword

This user manual will give you information on how to use the robot as a Designer.

Designer:

The Designer is responsible for:

- The design of the Scenarios
- The design of the Suites
- The design of the Stories
- The management of the References
- The management of the Dictionary
- The design of the Rules

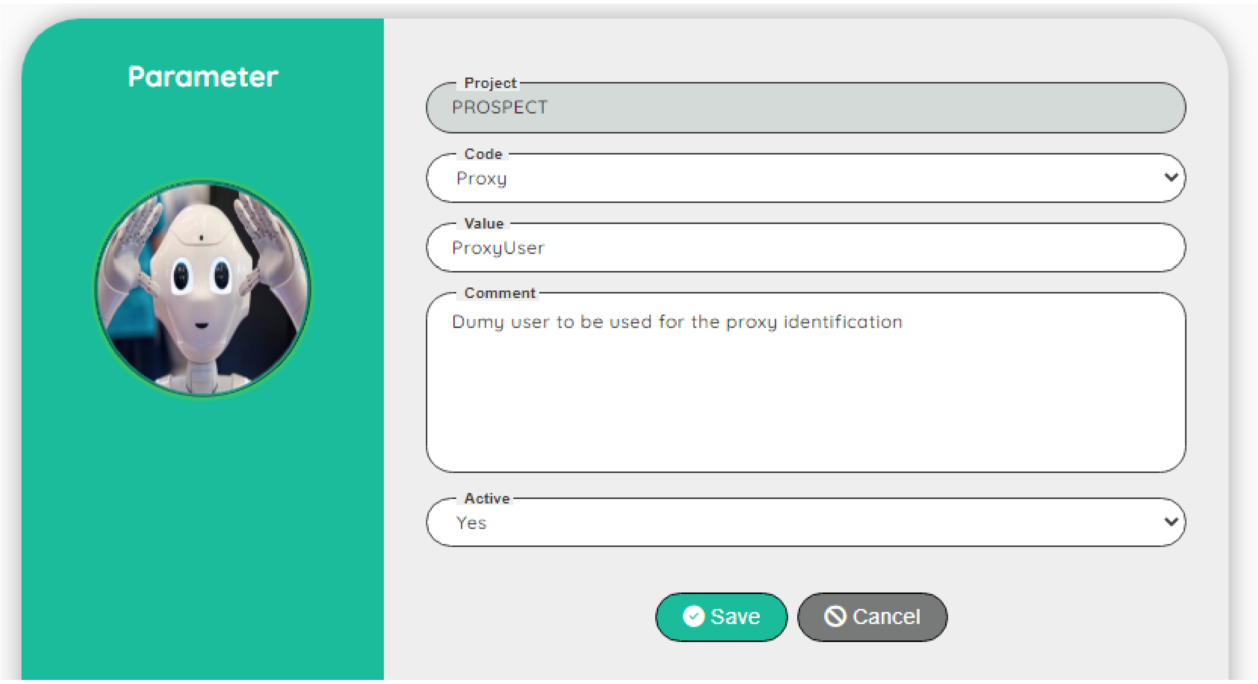


Working with Proxy

If you need to test a project that requires a proxy, please use the following method.

<u>Job to be performed by the Administrator:</u>

In the parameters of the project, create a new entry for the parameter Proxy

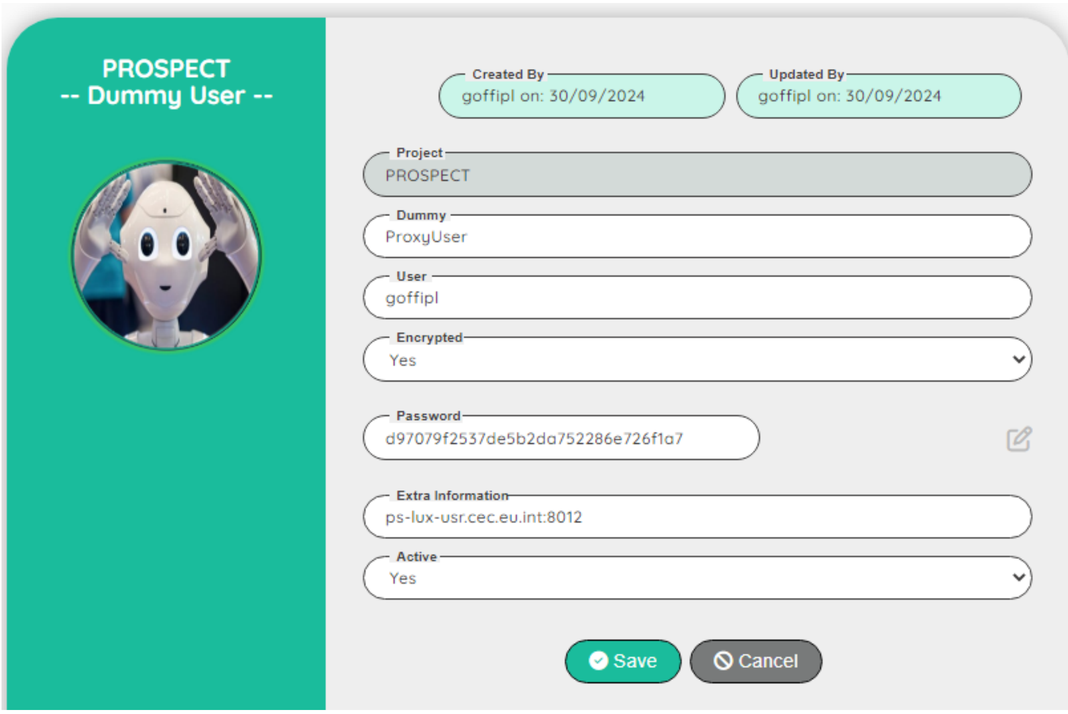


The value must be a valid dummy user

Job to be performed by the Tester (or Designer, Administrator):

Go to the dummy users and add a new entry for the value defined in the parameter (here: ProxyUser)

In the field Extra information, fill the proxy link including the port number



Note: You can encrypt the password

If you don't need a proxy login, you don't need to define a specific parameter and dummy user.

If you try to connect to a proxy website without defining the parameter and the dummy user, the application will display a window to ask you to enter a proxy login/password.
This popup window cannot be manipulated by normal functions!

If you define a proxy, the Robot will be able to login to a normal website or a website with proxy.
For information: the time to launch the script will be shorter with a proxy defined!

Default certificate selection

With Chrome (and only with Chrome), you can define the certificate to select by default.
Note: even if you select a certificate by default, the secret code will be required!
However, if you execute different scenarios in a suite, the secret code needs to be keyed only once!

The selection of the certificate by default can be configured with the Environment values

Configuration for a default certificate

Action	Role	Comment
Create an Environment: **CertificateRole**	Designer	ISSUER or SUBJECT (*)
Create an Environment: **CertificateValue**	Designer	E.g.: Citizen CA

(*) Only these two values are allowed !

I recommend to use **ISSUER** because the value can be a generic name like "Citizen CA".
If you need to select a very specific certificate, you can use **SUBJECT** with for instance the name of the person: "Mergo D'Heer Alaune (Authentication))"

It's very important that the filter you create with the role and the value returns only one certificate. Otherwise, a popup windows will be displayed to ask you to choose the right certificate!

To show your certificates, you can open **Chrome**, go to Settings, Privacy & Security, Security, Manage Certificates and finally "Your Certificates"
Check the certificates with …. (Authentication), click on the eye icon.
In Details, the first info can be used for the **SUBJECT** value (Certificate Hierarchy)
In the middle box (Certificate Fields), click on Issuer. The Field Value starting with "CN = ….." can be used for the **ISSUER** value.

I suggest you define a dataset containing the Role and Value.
Then, you create a simple scenario to retrieve the Role and Value from this dataset (getData) and store them in a Reference (setReference).
This method offers you great flexibility for certificate selection.
You can even create a story within this scenario to allow the tester to choose the certificate for testing.

Working with a certificate file (security)

You can use the following certificates: .p12, .pfx and .crt

If you need to test a project that requires a certificate to perform an authentication, you have to do the following task:

- Upload the right certificates (certificate and key)
- Define reference to define the full certificate behaviour

Configuration for a .p12 or .pfx certificate

Action	Role	Comment
Upload the **.p12** or **.pfx** certificate	Admin	Use the Upload menu
Create an Environment: **CertificateUrl**	Designer	E.g.: https://myDossier.rrn
Create an Environment: **Certificate**	Designer	E.g.: myCertificate.p12
Create an Environment: **CertificateCode**	Designer	E.g.: 1234

Configuration for a .crt certificate

Action	Role	Comment
Upload the **.crt** certificate	Admin	Use the Upload menu
Upload the **.key** certificate	Admin	Use the Upload menu
Create an Environment: **CertificateUrl**	Designer	E.g.: https://myDossier.rrn
Create an Environment: **Certificate**	Designer	E.g.: myCertificate.crt
Create an Environment: **CertificateCode**	Designer	**Optional**, only if the key file is encrypted, otherwise use **<N/A>**

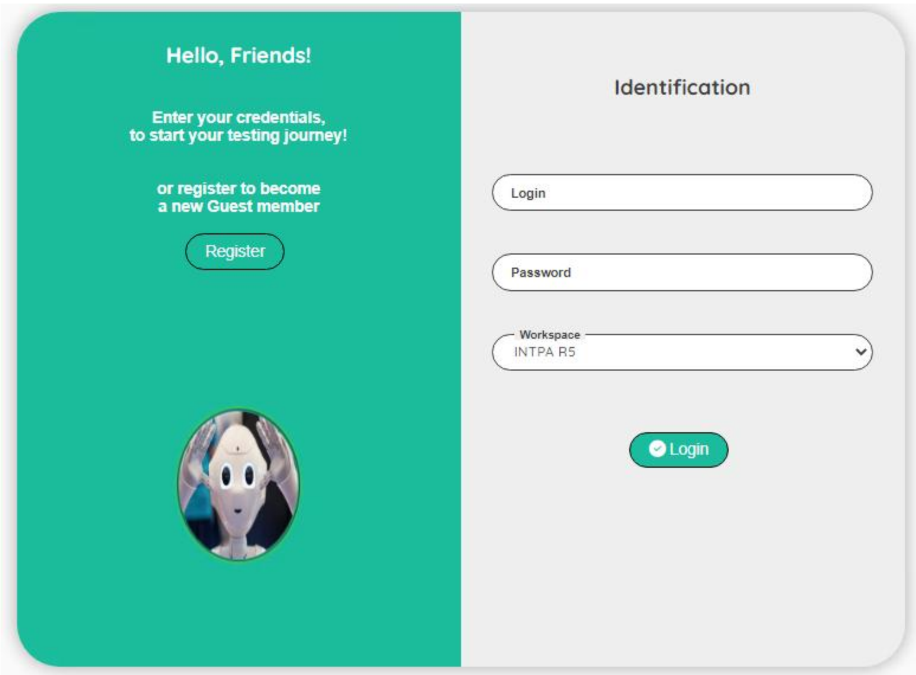
As the certificate is used at the very beginning of the test, if there is an error, you will not be notified, you have to check on the console (RobotVx) to view the error message.
If something goes wrong, please check the configuration steps below!

I suggest you define a dataset containing all the required values.
Then, you create a simple scenario to retrieve the data from this dataset (getData) and store them in a Reference (setReference).

Login

User Interface

To access to the Robot, you need to have a login.
After you successfully register to the application as a new user (Guess), you have to wait that the Administrator will assign you a Role to a project (or multiple projects).



Topic	Icon	Comment
Register	Register	Click to Register as a new user
Login	Login	Enter your login
Password	Password	Enter your password
Workspace	Workspace INTPA R5	Enter your workspace
Login	Login	Click to login to the Robot

User interface for the editing

User Interface

We use the same user interface when editing a record.
To avoid repeating the same explanation all the time, you will find here a generic explanation on how to use the edit interface.
When you have the permission to edit a record, you will see the following icons:

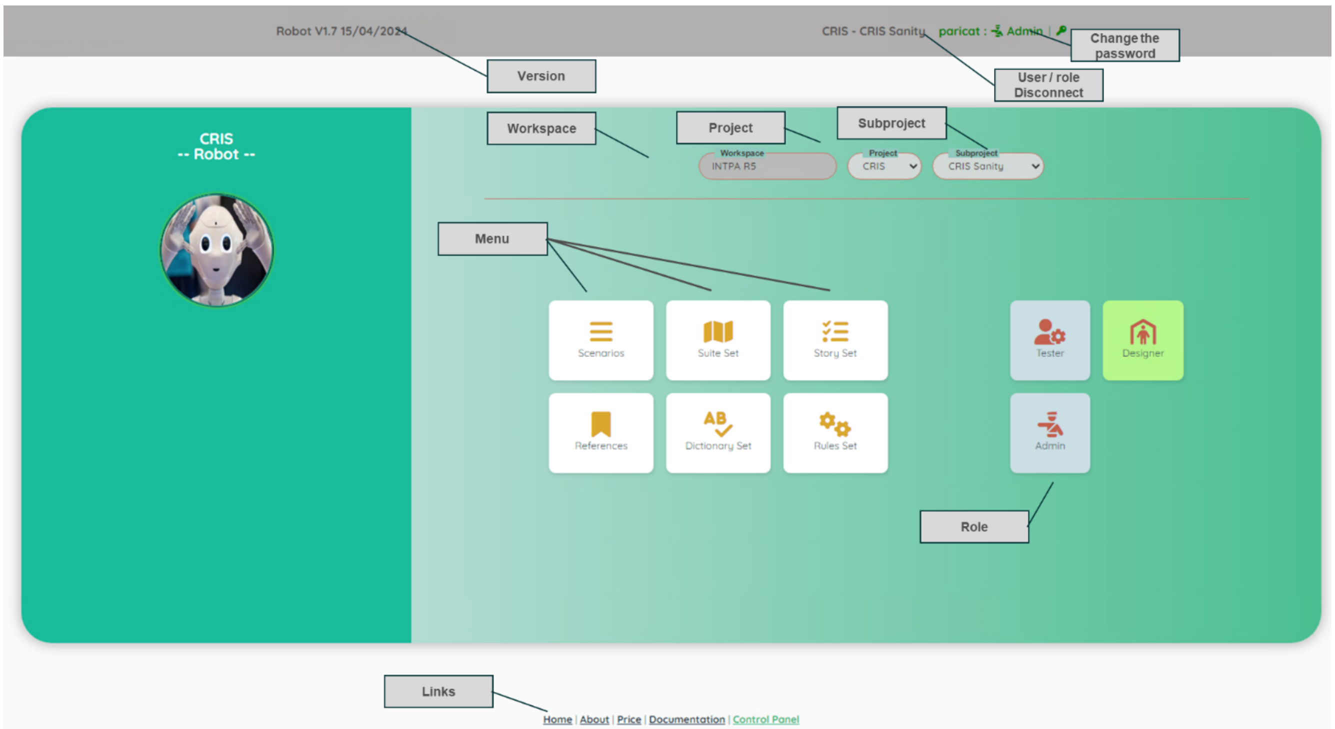
Topic	Icon	Comment
Delete	or	Delete a record (a confirmation is required!) To delete multiple records, select the records first.
Edit		Go to a screen to update the record.
Copy		First select the record to copy and then click on the copy icon. Record(s) will be set after the copy icon.
Move		First select the record to move and then click on the move icon. Record(s) will be set after the move icon.
Select / Unselect	/	Select or unselect a record

Depending of the context, some extra icon(s) can be available.
The extra icon(s) will be explain in the section dedicated to the context.

Control Panel

User Interface

The dashboard will allow you to execute pre-defined stories.
If you are assign to a **DESIGNER** or **ADMIN** role, your user interface will look like this:

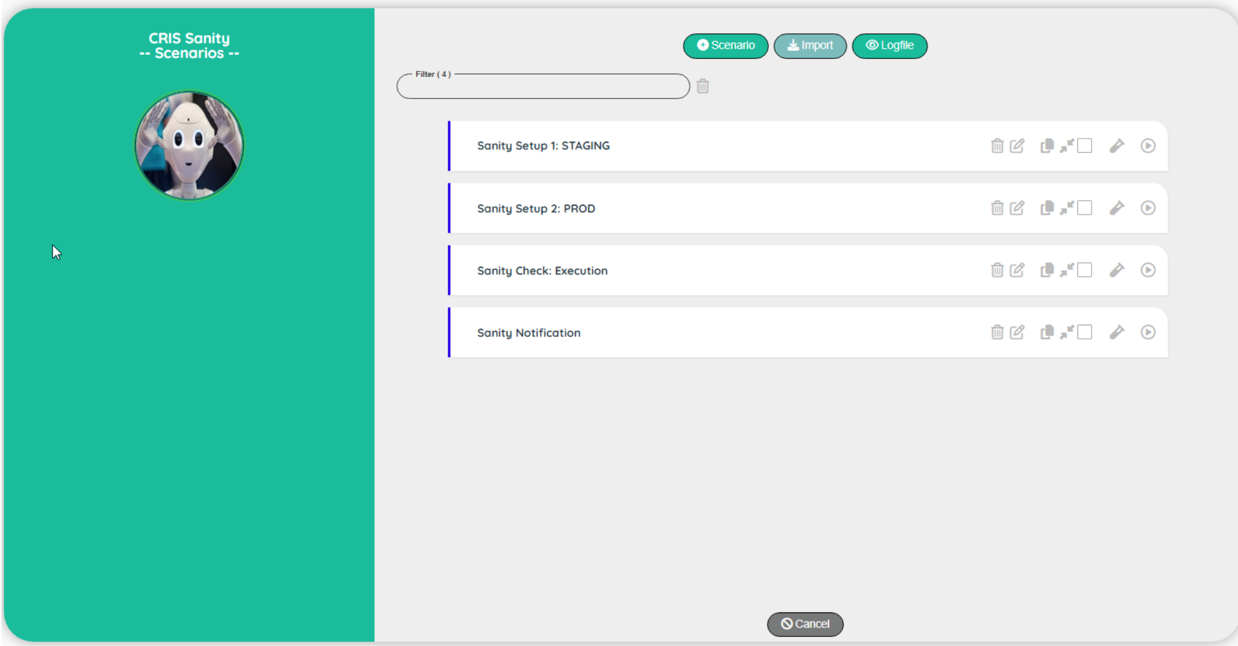


Topic	Comment
Version	Version of the Robot.
Workspace	For info: the name of the workspace of the customer.
Project	List of the project(s) that you have access (assignment to a project is done by an Administrator).
Subproject	Access to all the subproject(s) of the selected project.
	When executing a story, you can stop the execution.
User/Role Disconnect	For info: your login and your role. If you click on the 'User/Role', you will be disconnected from the application.
Change the password	Click on the icon to change your current password
Menu	Click on the items to open a new section of the application
Links	Useful links

Scenario

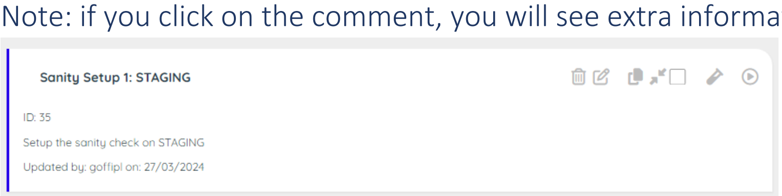
User Interface

The Scenario will allow you to manage the scenarios and the tests

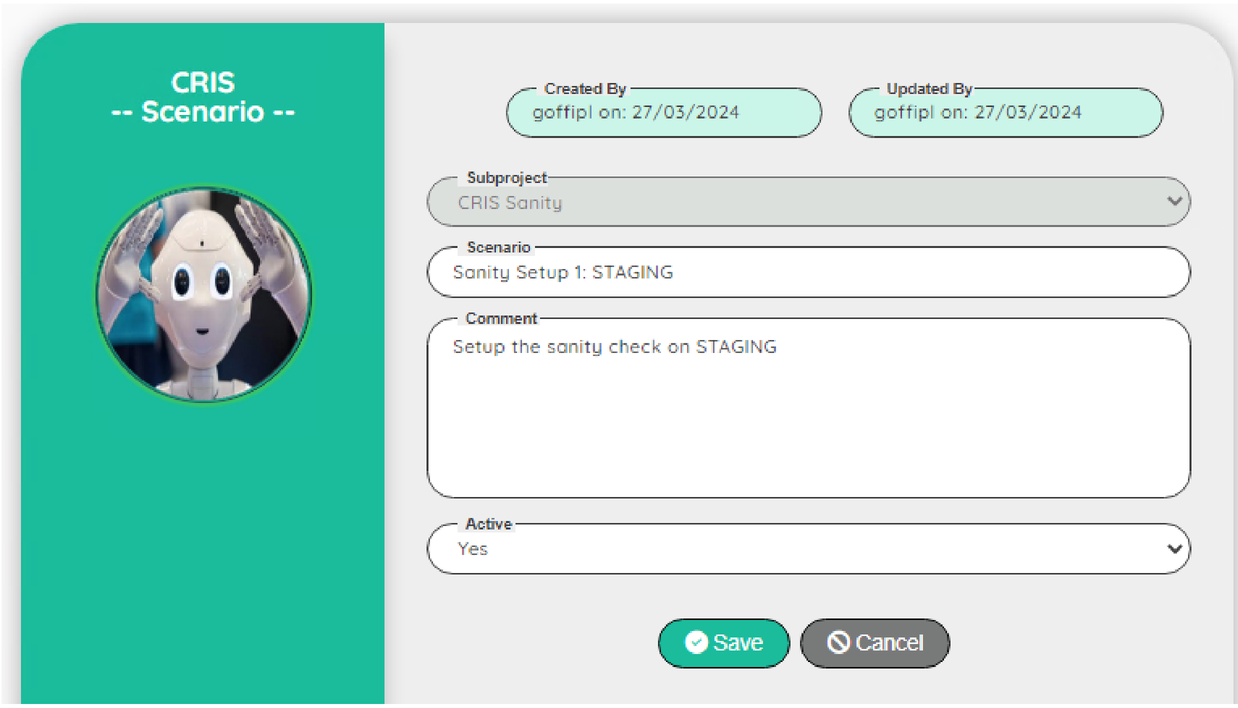


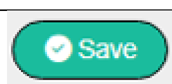
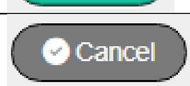
Topic	Icon	Comment
Test		Go to the screen Test.
Execute		Execute a scenario.
Add Scenario		Add a new scenario at the end of the list.
Cancel		Back to the Dashboard screen.
Logfile		Go to the screen to view the Log file.
Import		Import a scenario from any project (same workspace).

Note: if you click on the comment, you will see extra information

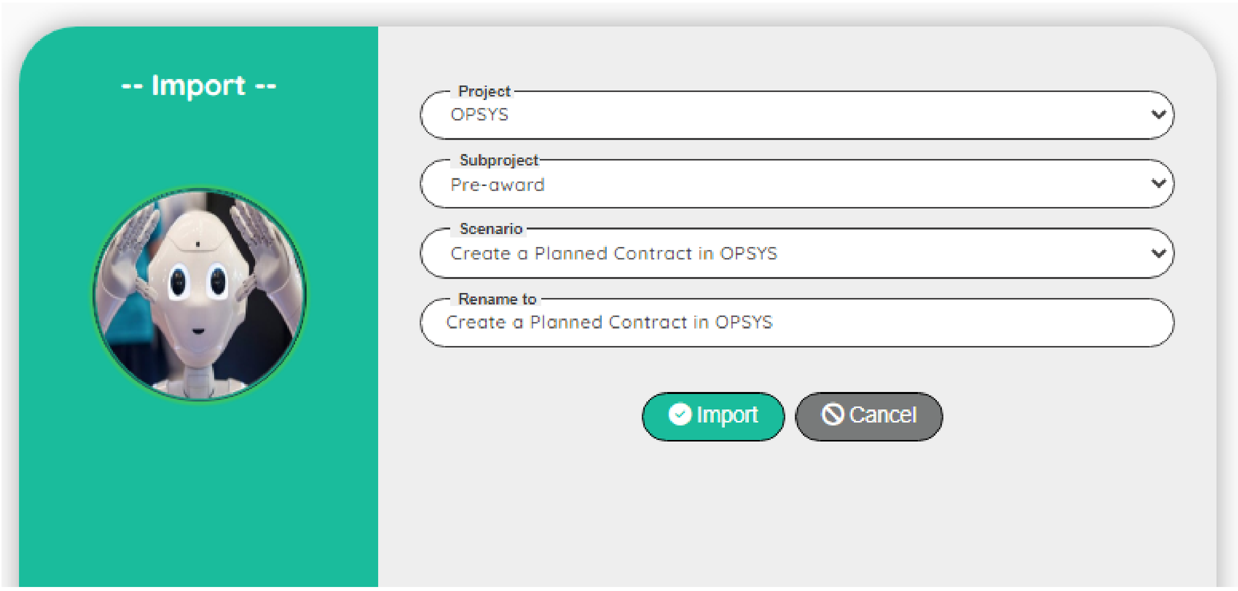


Scenario Edit



Topic	Icon	Comment
Subproject		For info: Name of the subproject
Scenario		Name of the scenario
Comment		Comment on the scenario
Active		Yes, No
Save		Save the edit
Cancel		Discard the edit

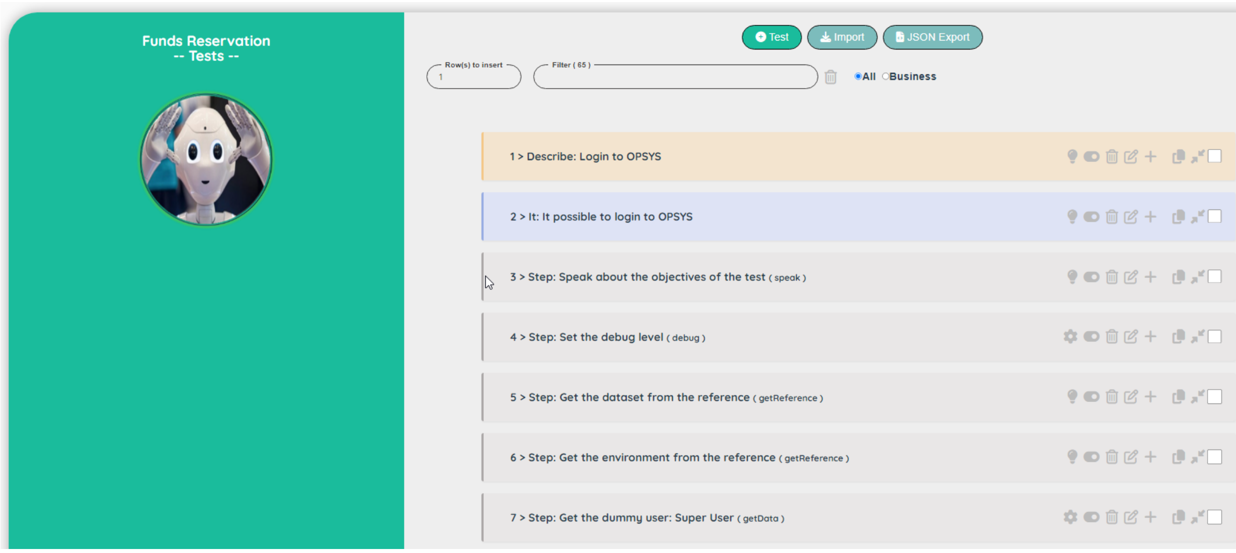
Scenario Import

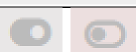
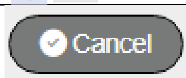


Topic	Icon	Comment
Project		Select a project in the list
Subproject		Select a subproject in the list
Scenario		Select a scenario to import
Rename		You can update the name if necessary
Import		Import the scenario
Cancel		Discard the import

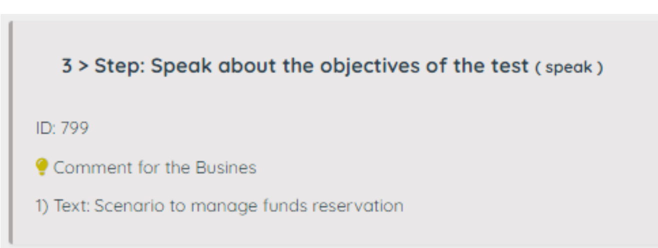
Test

The Test will allow you to manage the tests (steps)



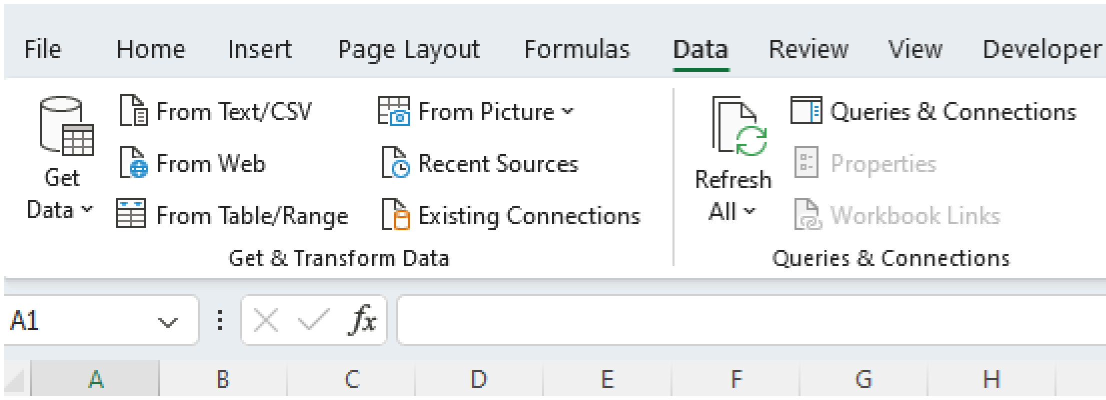
Topic	**Icon**	**Comment**
All / Business		Display all comments or only the ones for the Business
Add Test		Add a new test at the beginning of the list.
Active/Inactive		Set the test Inactive/Active
Business/Designer		Set the comment type to Business / Designer (Technical)
Cancel		Back to the Dashboard screen.
Import		Import a test(s) from another scenario (but in the same project)
JSON Export		Export the current test(s) into a .json file
Download	goffipl_test.json	Click on the link to download the .json file

Note: if you click on the comment, you will see extra information

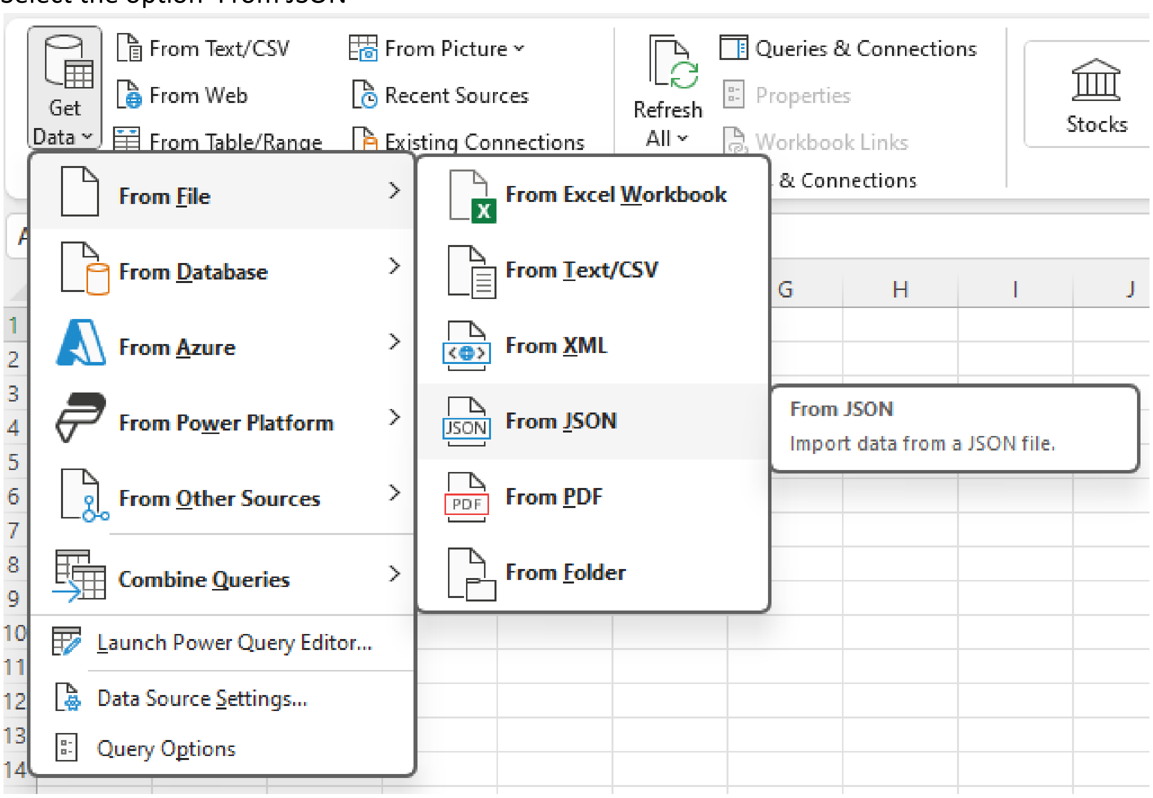


How to process the .json file into Excel

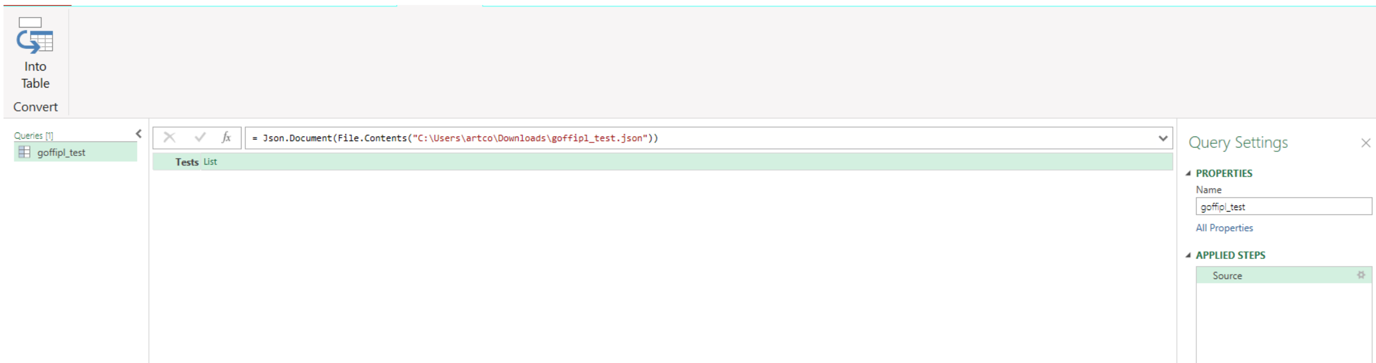
- Open a blank Excel workbook
- Go to the ‘Data’ section and click on ‘Get Data’

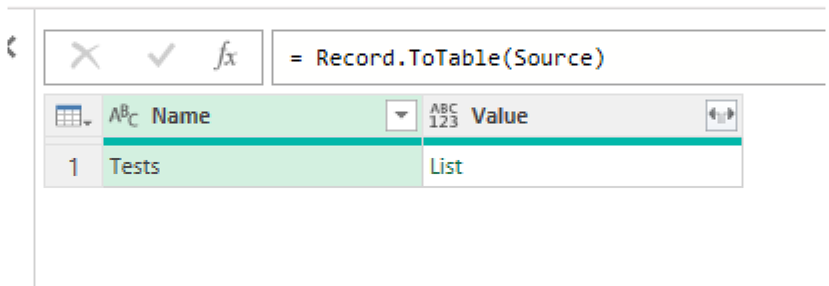


- Select the option ‘From JSON’

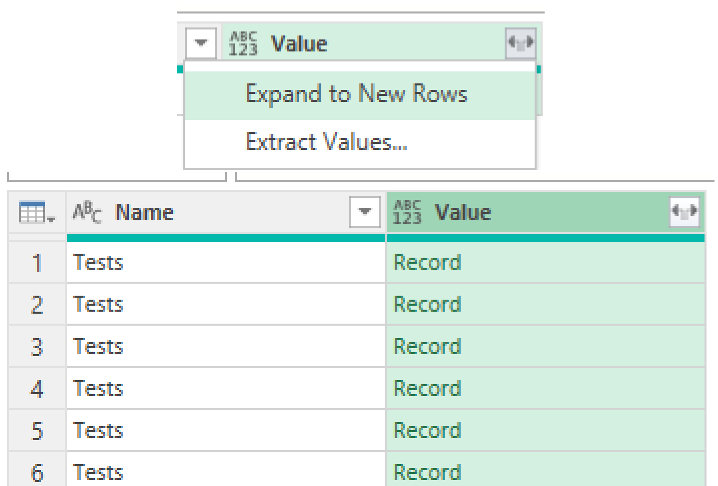


- Select the .json file from the folder 'Download' on your local driver.
  a new window is now opened.
  Click in the item of the menu 'Into Table Convert'

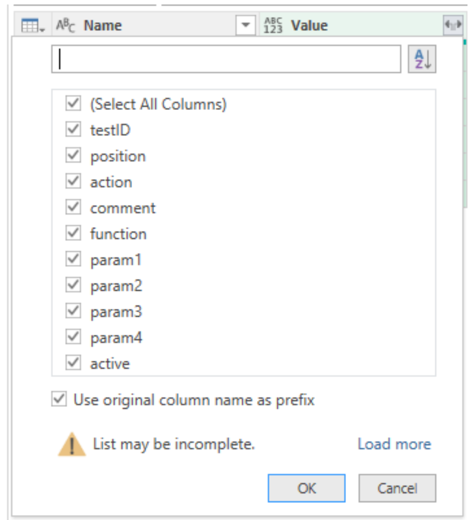




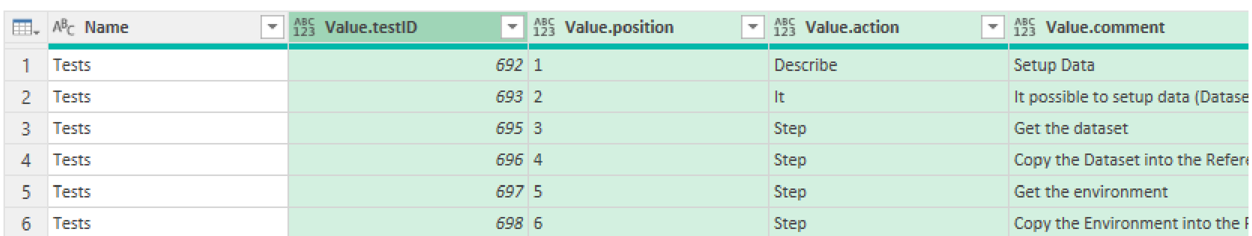
- Click on the right icon of the cell Value and select Expand to New rows



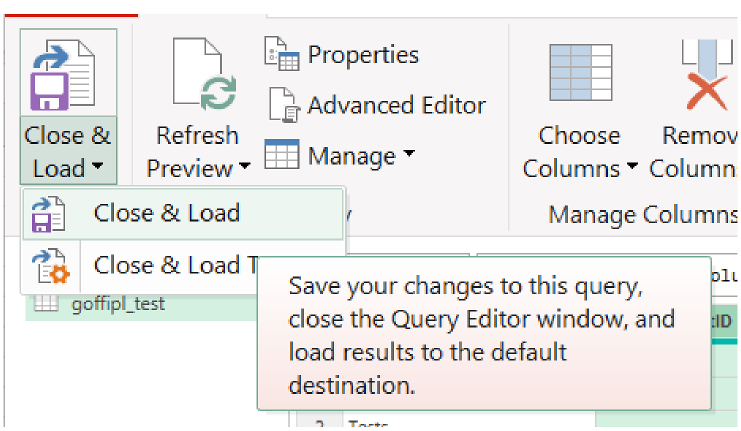
Click again on the right icon of the cell Value and click OK



The data are now displayed into a table.
Click the first item in the menu 'Close & Load'



	Name	Value.testID	Value.position	Value.action	Value.comment
1	Tests	692	1	Describe	Setup Data
2	Tests	693	2	It	It possible to setup data (Datase
3	Tests	695	3	Step	Get the dataset
4	Tests	696	4	Step	Copy the Dataset into the Refere
5	Tests	697	5	Step	Get the environment
6	Tests	698	6	Step	Copy the Environment into the F

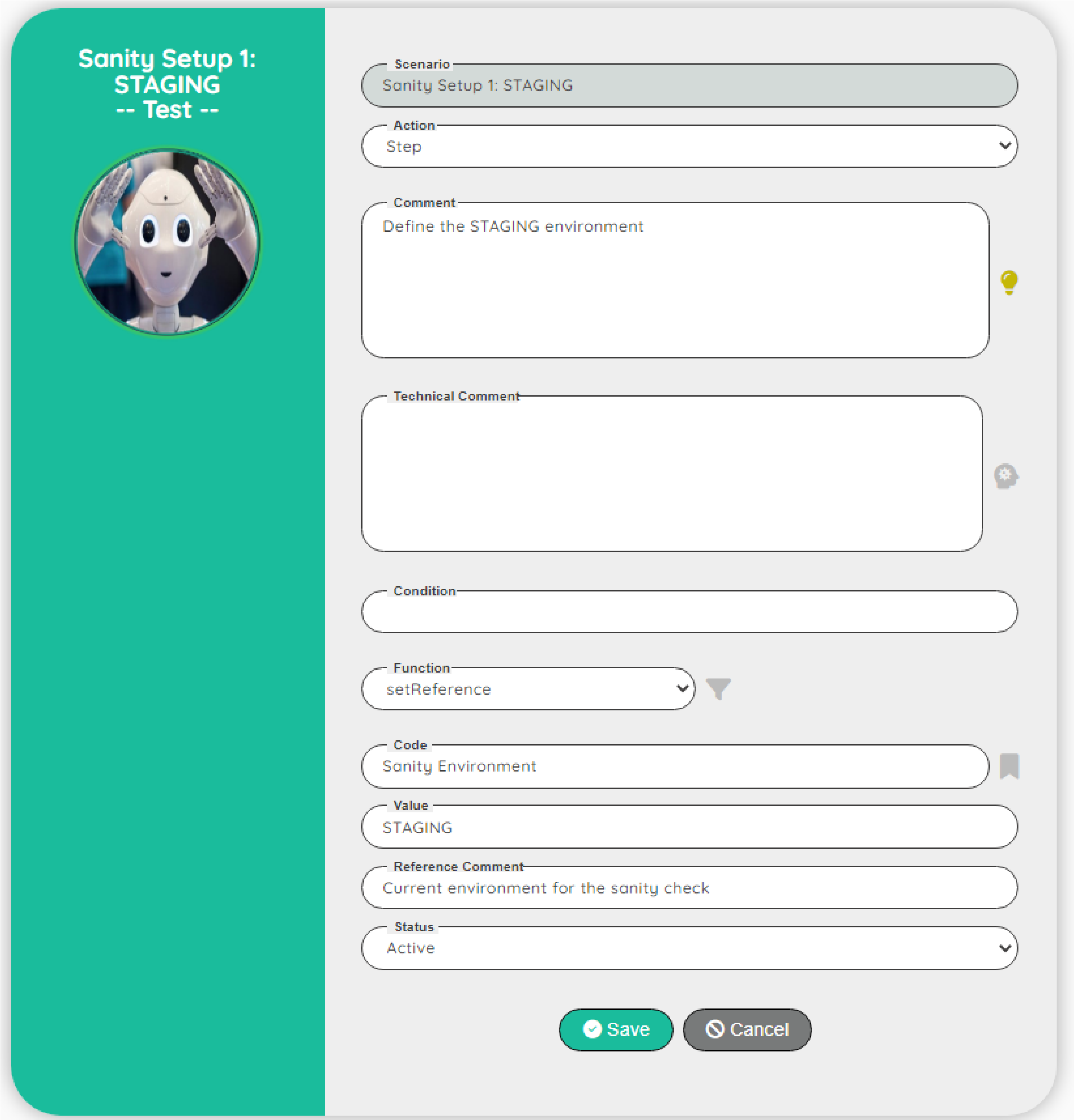


And now, you have a nice table with the tests of a scenario.

You can now manage this table to share with the Business Team.
For instance, you can remove the name of the function and the parameters and eventually, remove the technical test (like wait a little bit or wait for the refresh of the screen…)

Name	Value.testID	Value.position	Value.action	Value.comment	Value.function	Value.param1
Tests	692	1	Describe	Setup Data	Not selected	
Tests	693	2	It	It possible to setup data (Dataset, Environment…)	Not selected	
Tests	695	3	Step	Get the dataset	getData	#Data_Dataset
Tests	696	4	Step	Copy the Dataset into the Reference	setReference	Dataset
Tests	697	5	Step	Get the environment	getData	#Data_Environme
Tests	698	6	Step	Copy the Environment into the Reference	setReference	Environment

Test Edit


Sanity Setup 1:
STAGING
-- Test --
Scenario
Sanity Setup 1: STAGING
Action
Step
Comment
Define the STAGING environment
Technical Comment
Condition
Function
setReference
Code
Sanity Environment
Value
STAGING
Reference Comment
Current environment for the sanity check
Status
Active
Save
Cancel

Topic	Icon	Comment
Scenario		For info: Name of the scenario
Comment	Comment for the Business Comment for the Designer	Comment on the test
Technical		Text used for the natural language process
Condition		Any valid JavaScript expression. If the condition is false the test will be skipped!
Function	/	Select the name of the function (you can also filter by name or comment)
Parameters		Parameters depending of the selected function
Active		Yes, No
Save	Save	Save the edit
Cancel	Cancel	Discard the edit

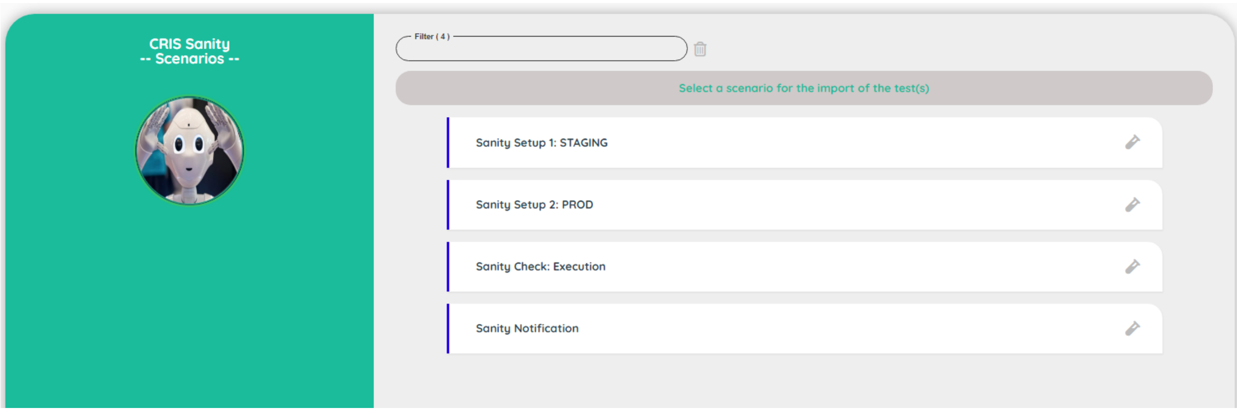
<u>Note:</u> The comment can be defined as a text for the Business or for the Designer (more technical). This is a simple way to indicate to the Business that want to review the scenario which steps are useful for him.
During the export, the field 'commentType' will be exported with the values:

- 1: Comment for the Business
- 0: Comment for the Designer (technical)

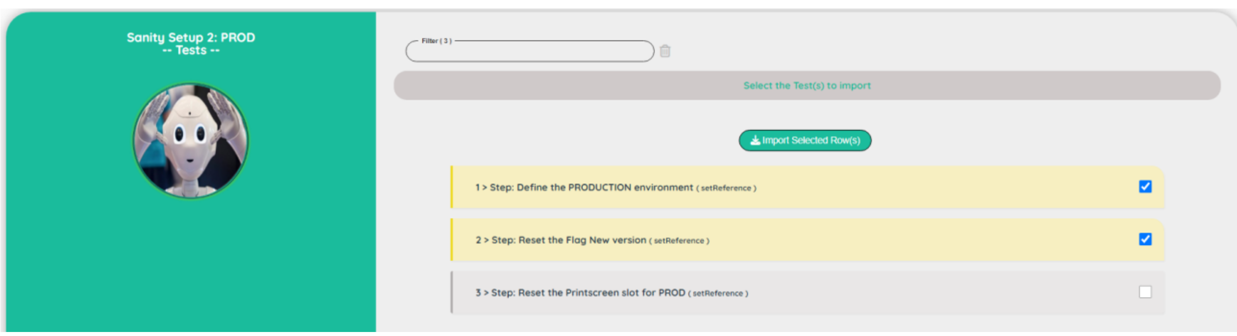
Import Test

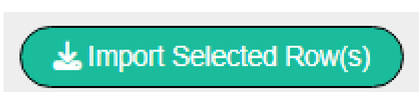
The Import Test will allow you to import a test from another scenario (but from the same project).

First select a scenario



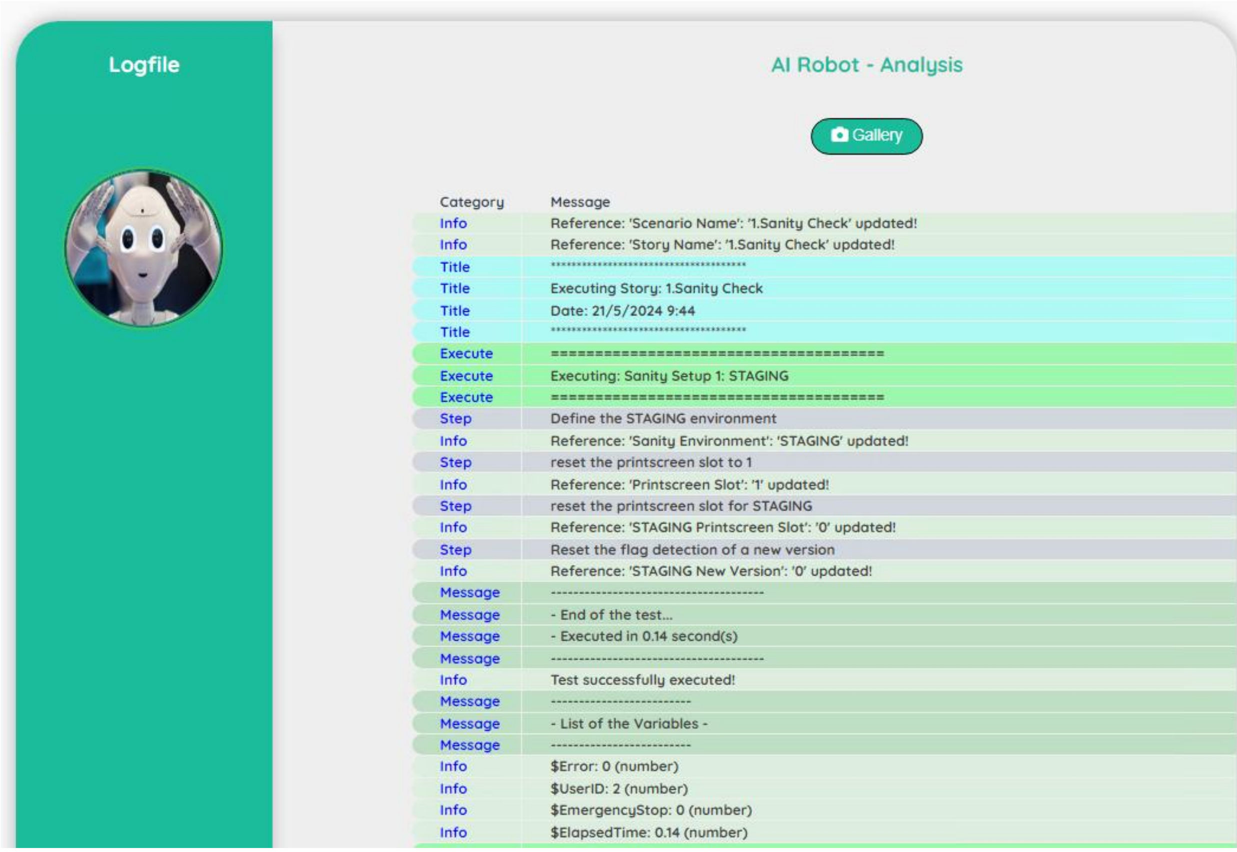
Select the test(s) that you want to import

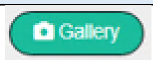


Click on the button 

Logfile

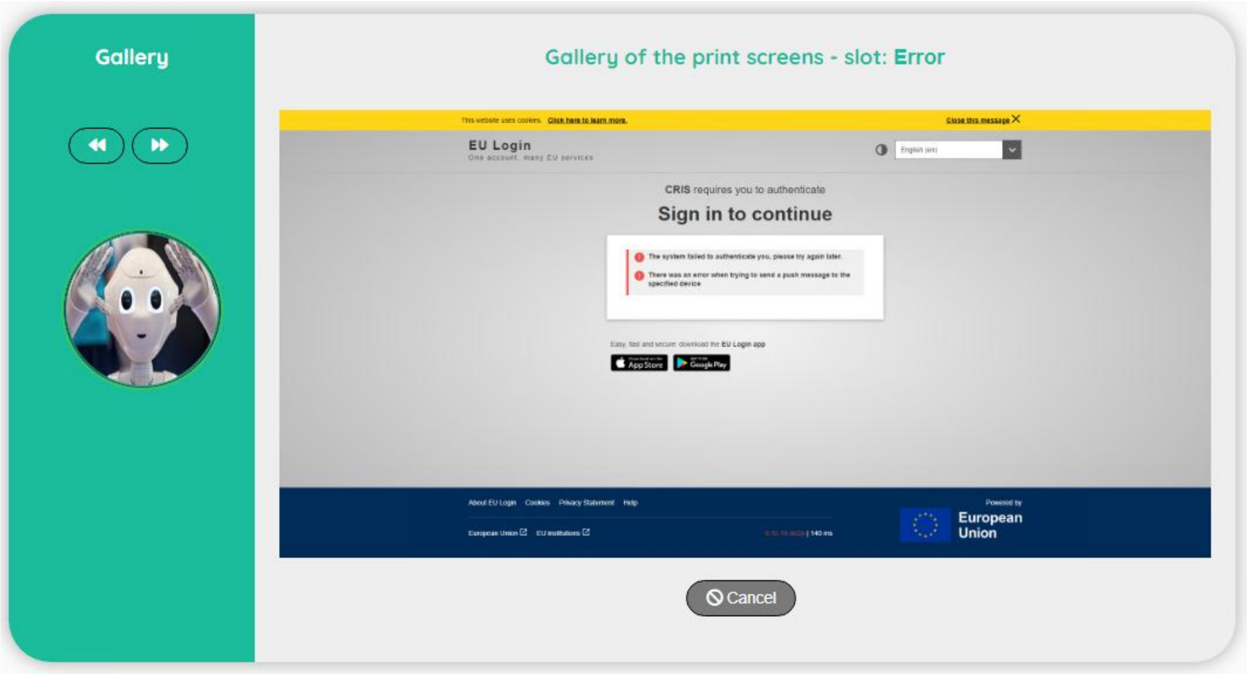
The logfile will show you all the steps executed with extra information.
It helps you to understand how the test has been executed.
Note: There is one logfile by user. You see always the logfile of the last execution

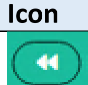
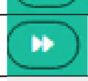


Topic	Icon	Comment
Gallery		Access to the print screen(s)

Gallery

The gallery will show you the print screen taken during the execution of the story.
The first print screen is done automatically by the Robot when an error occurs (it will show you the last error detected – in case of success of a story, you will see the last error detected)

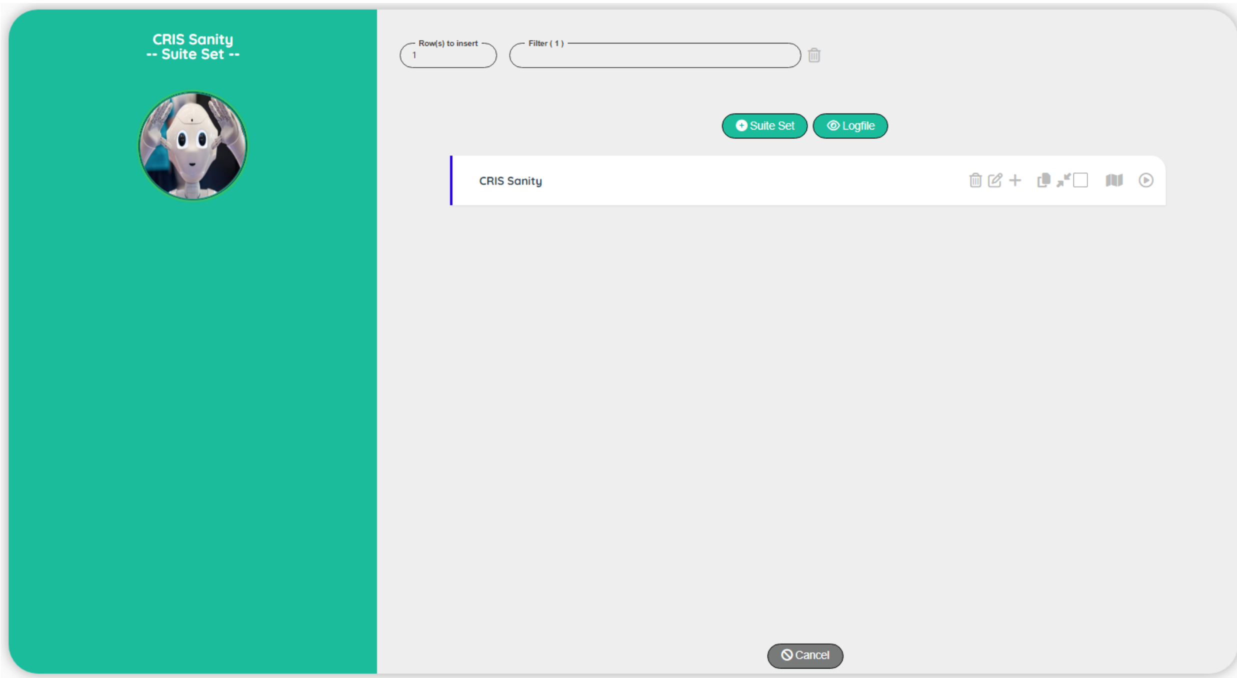


Topic	Icon	Comment
Previous		Go to the previous print screen
Next		Go to the next print screen
Cancel		Back to the Logfile

Suite Set

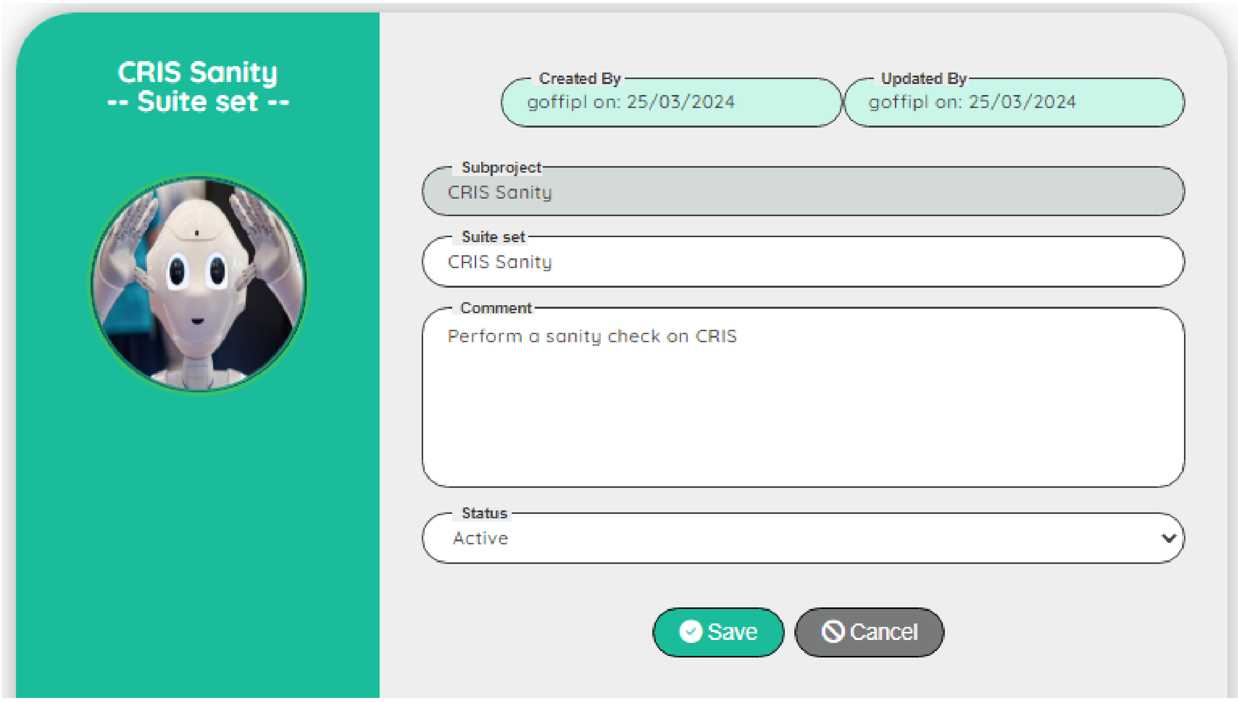
User Interface

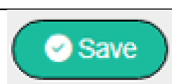
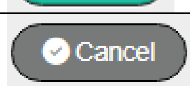
The Suite will allow you to group different scenarios (you can also consult the best practices guide for more information about the suite). A suite is composed from a suite set (parent) and suites (children)



Topic	Icon	Comment
Suite		Go to the screen suite
Execute		Execute a suite.
Add Suite set	Suite Set	Add a new Suite at the beginning of the list.
Cancel	Cancel	Back to the Dashboard screen.
Logfile	Logfile	Go to the screen to view the Log file.

Suite Set Edit

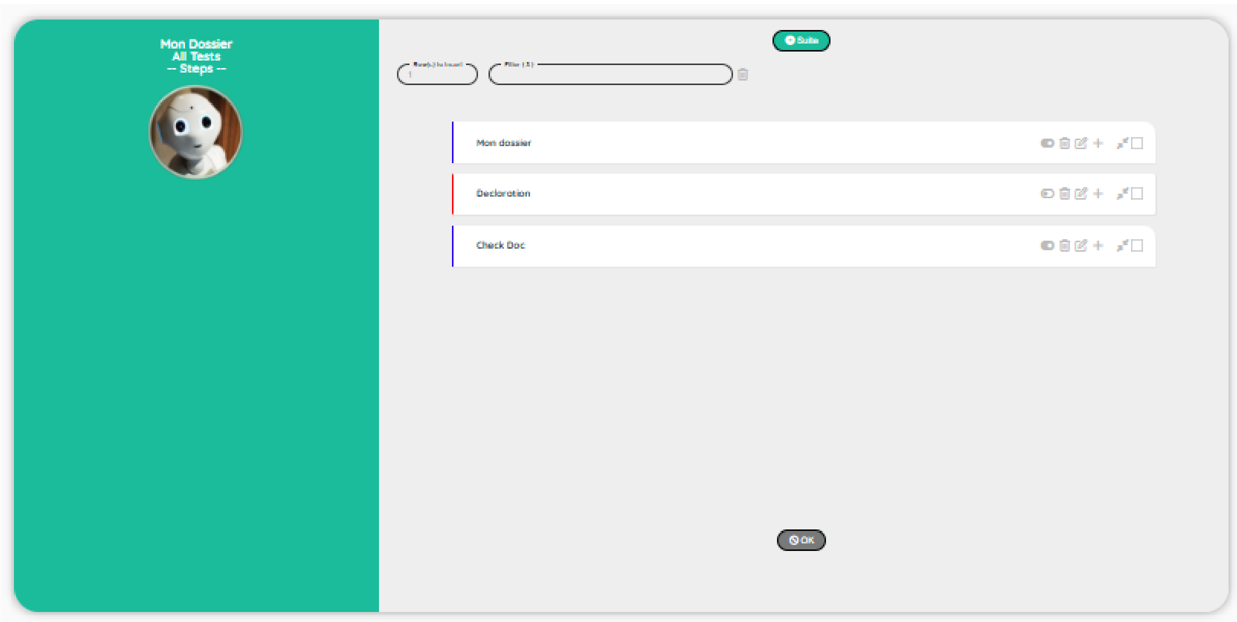


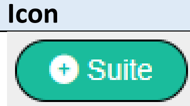
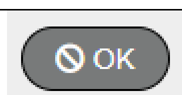
Topic	Icon	Comment
Subproject		For info: Name of the subproject
Suite set		Name of the suite set
Comment		Comment for the suite set
Active		Yes, No
Save		Save the edit
Cancel		Discard the edit

Suites

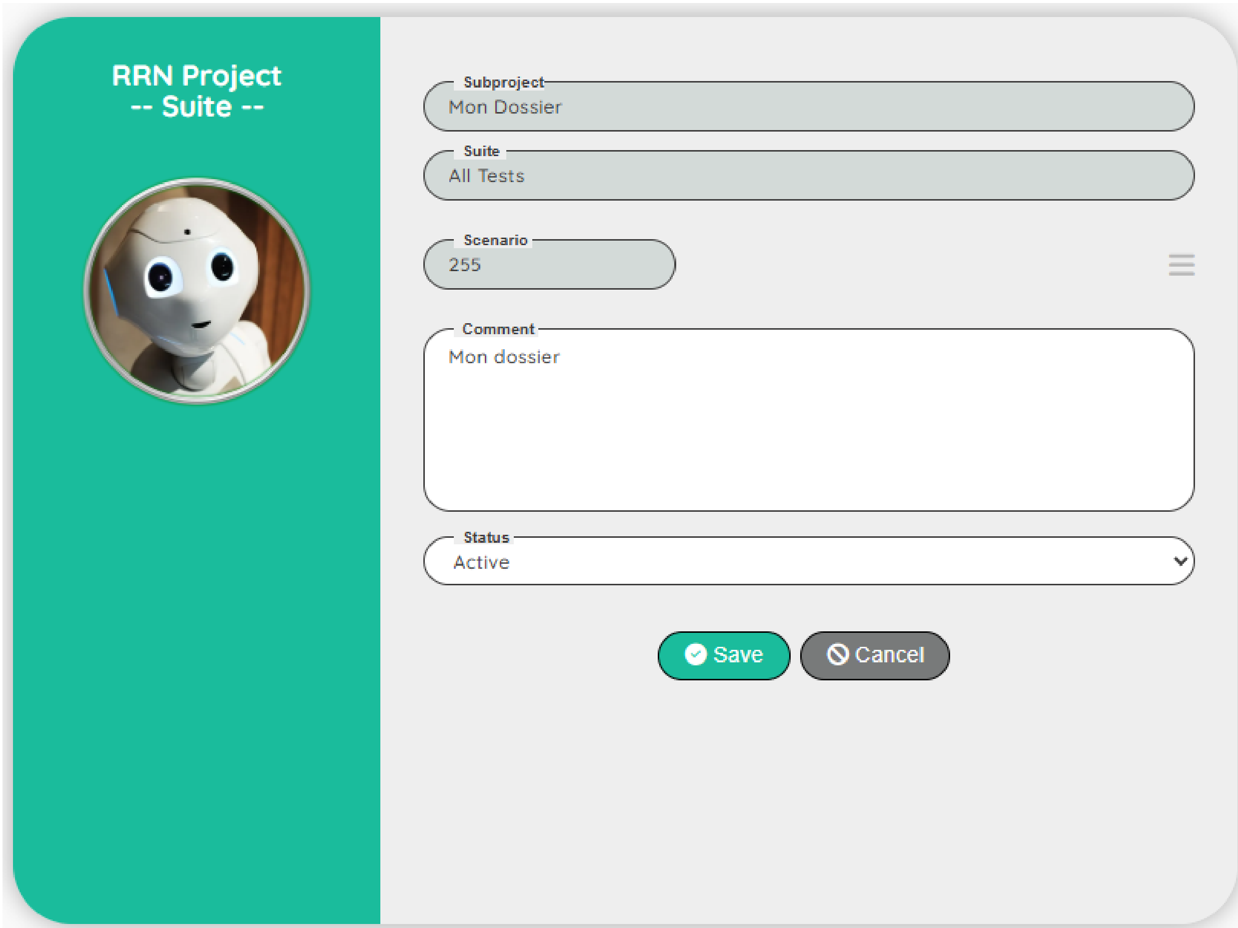
User Interface

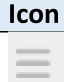
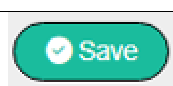
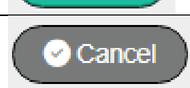
The Suites contains all the scenarios that you want to group, in order to execute multiple scenarios with one request



Topic	Icon	Comment
Add a Suite		Add a scenario at the beginning
Enable/Disable		Enable or Disable a suite.
Edit		Delete/Update/Insert/Move/Select.
OK		Back to the screen suite

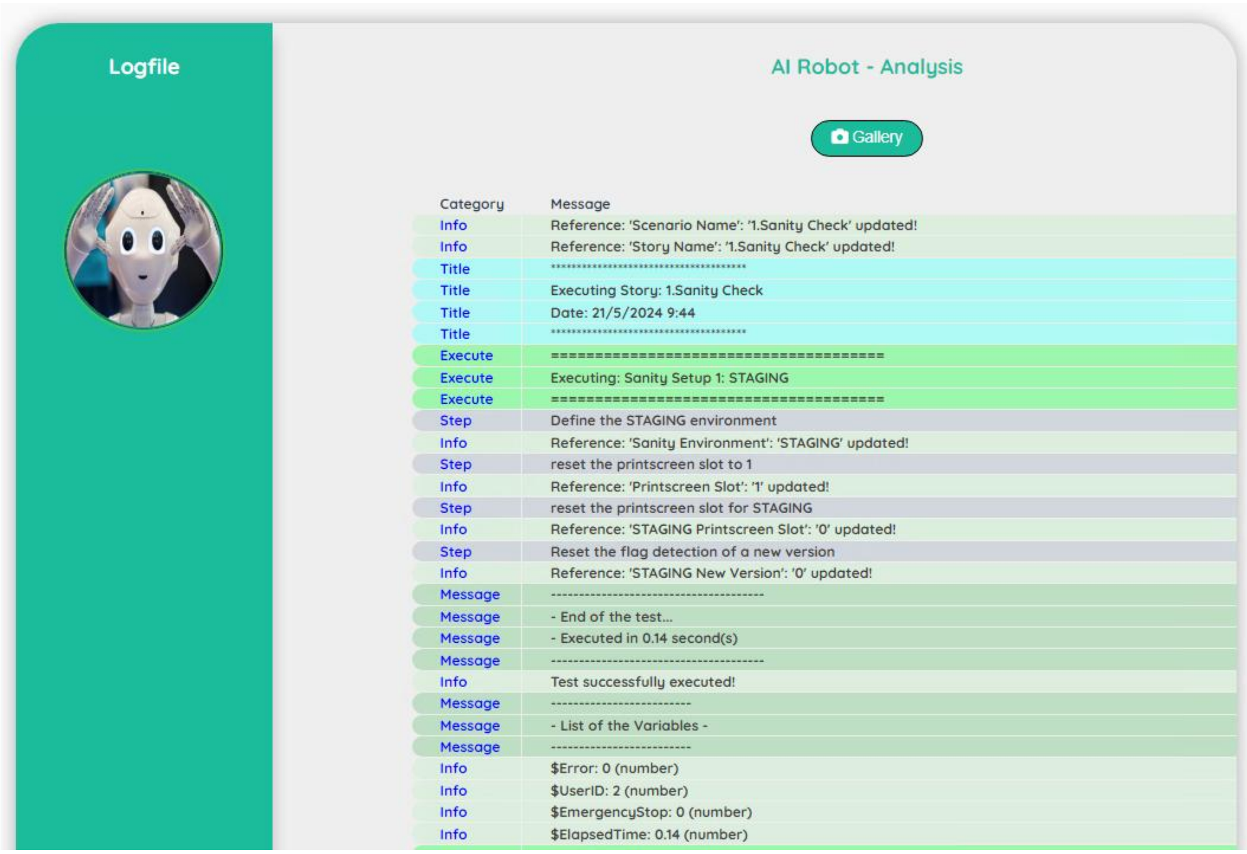
Suites Set Edit

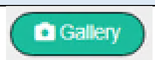


Topic	Icon	Comment
Select scenario		Select a scenario
Comment		Comment for the suites step
Active		Yes, No
Save		Save the edit
Cancel		Discard the edit

Logfile

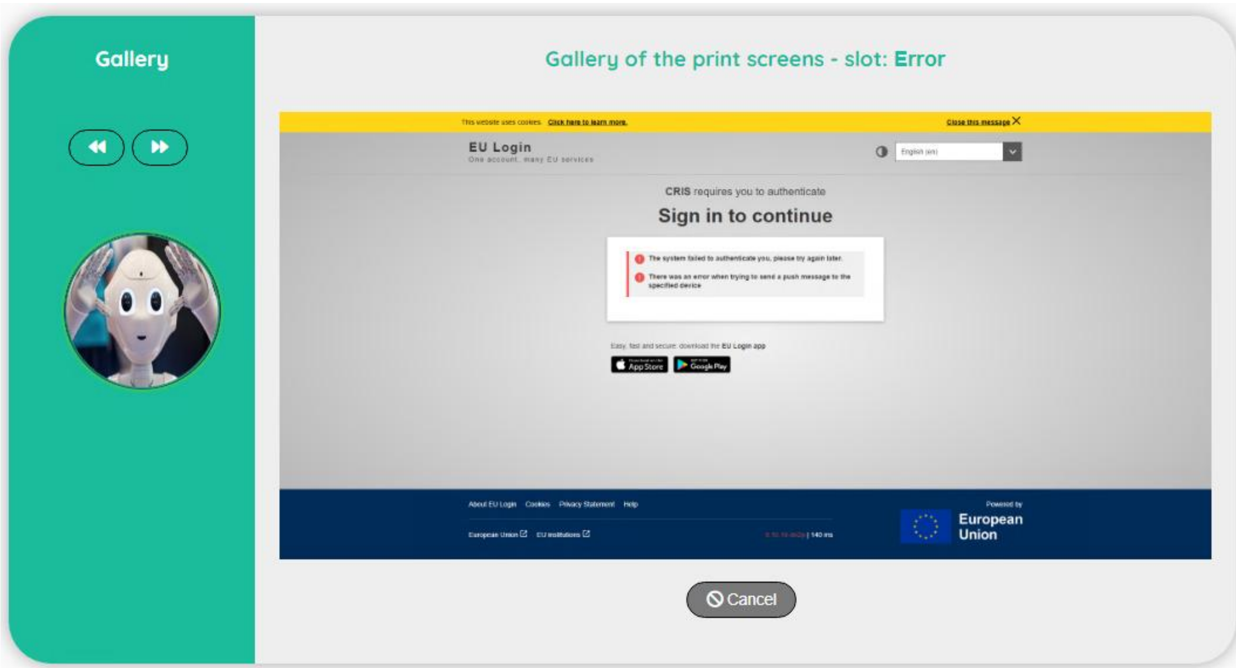
The logfile will show you all the steps executed with extra information.
It helps you to understand how the test has been executed.
Note: There is one logfile by user. You see always the logfile of the last execution

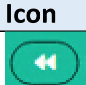
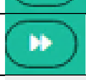


Topic	Icon	Comment
Gallery		Access to the print screen(s)

Gallery

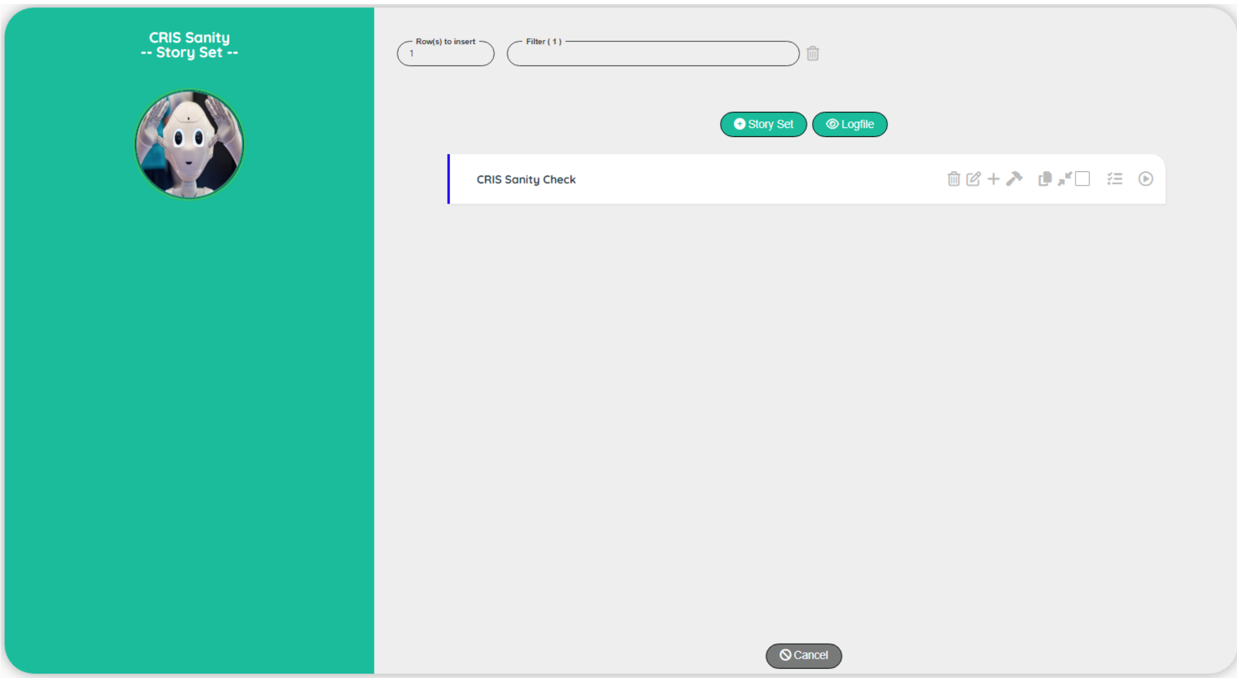
The gallery will show you the print screen taken during the execution of the story.
The first print screen is done automatically by the Robot when an error occurs (it will show you the last error detected – in case of success of a story, you will see the last error detected)

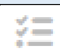


Topic	Icon	Comment
Previous		Go to the previous print screen
Next		Go to the next print screen
Cancel		Back to the Logfile

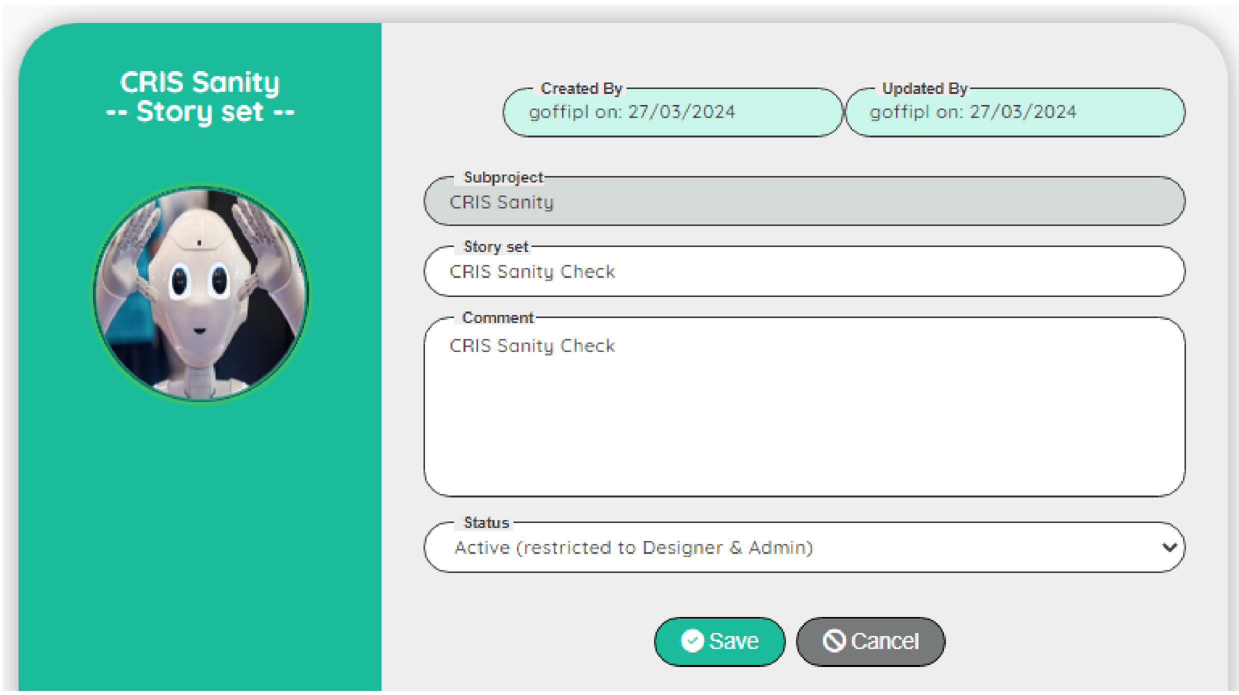
Story Set

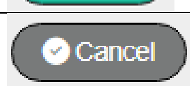
Story set



Topic	Icon	Comment
Stories		Go to the screen Stories.
Execute		Execute a Story.
Parameters		Add a story at the beginning of the list
Add Story		Add a story at the beginning of the list
Cancel		Back to the control panel
Logfile		Go to the screen to view the Log file.

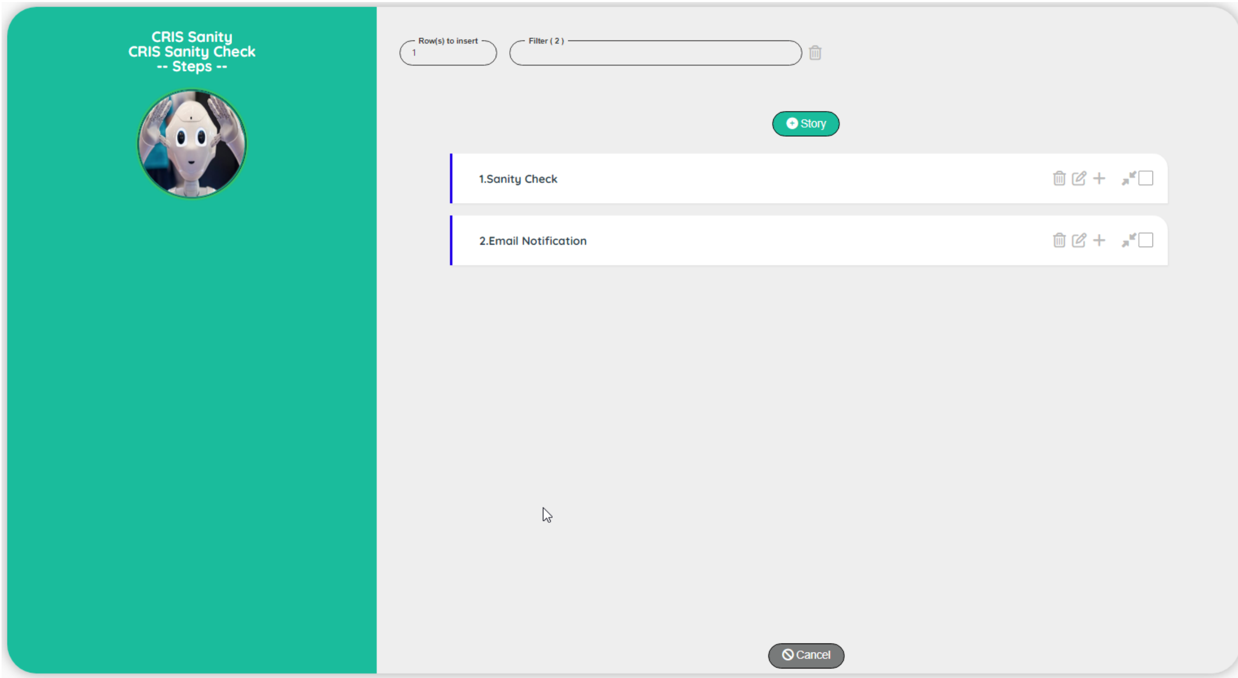
Story set Edit



Topic	Icon	Comment
Subproject		For info: Name of the subproject
Story set		Name of the story set
Comment		Comment for the story set
Active		Active, Not active, Publish
Save		Save the edit
Cancel		Discard the edit

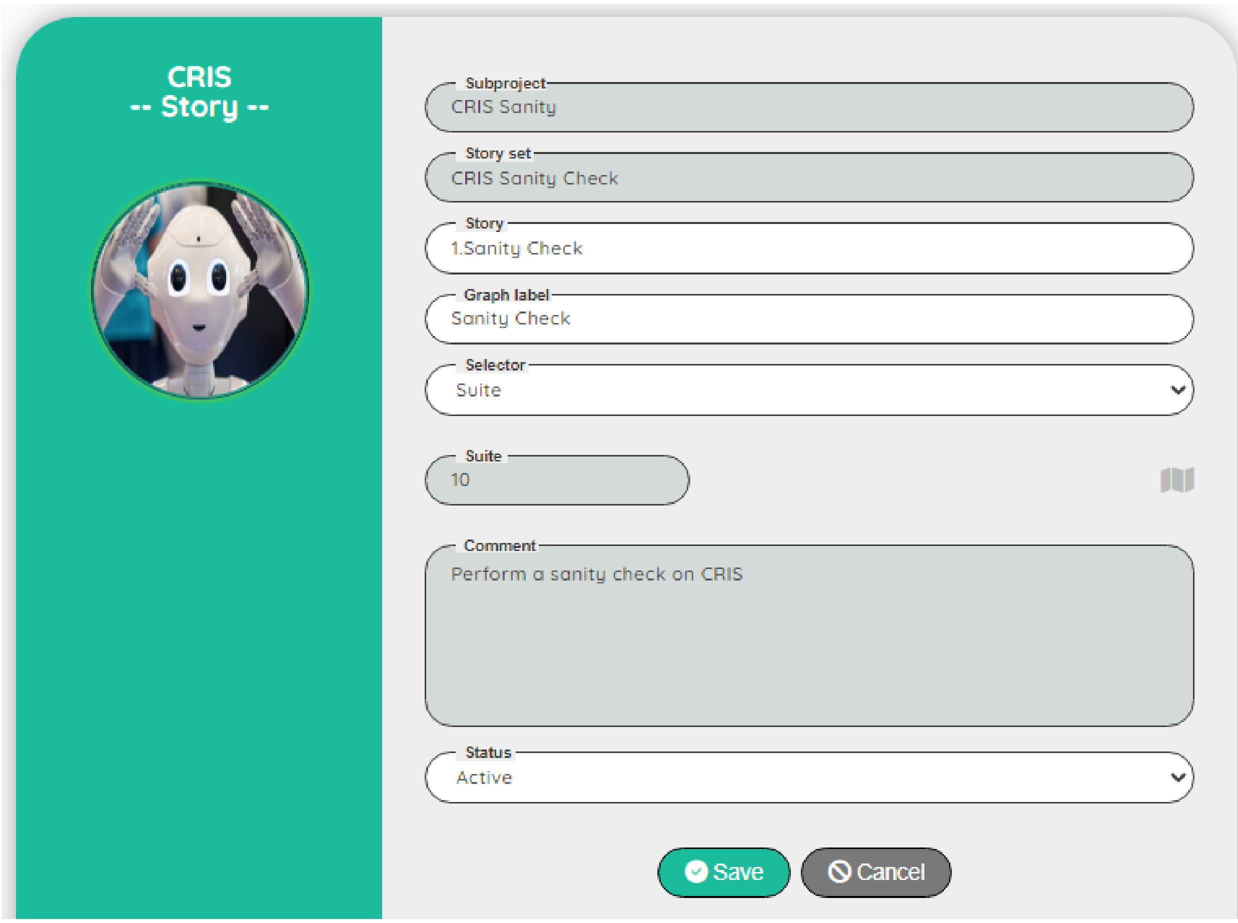
<u>Note</u>: The Tester can see only the Published story (Active and Published are visible by the Designer and the Administrator)

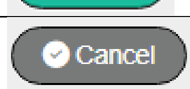
Stories



Topic	Icon	Comment
Add Story		Add a story at the beginning of the list
Cancel		Back to the Story set

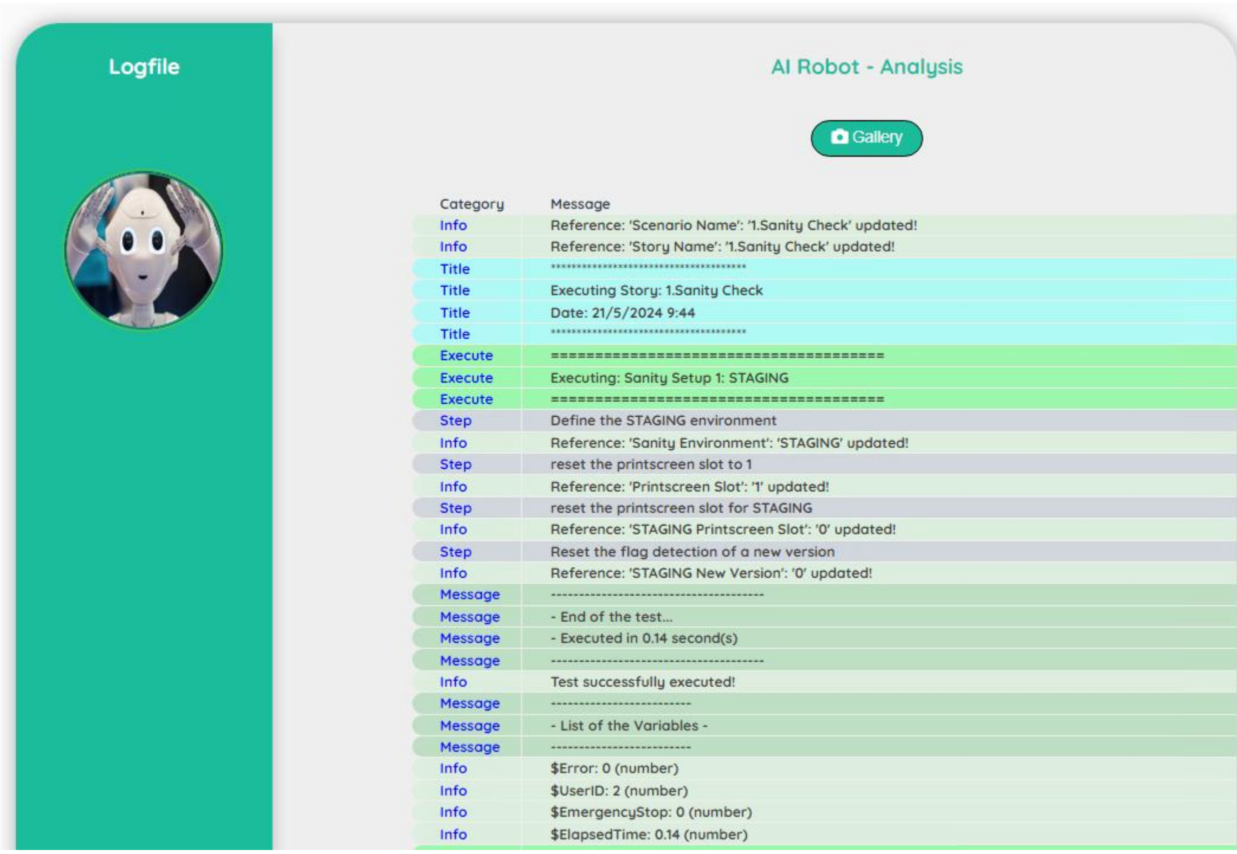
Stories Edit

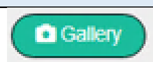


Topic	Icon	Comment
Subproject		For info: Name of the subproject
Story set		For info: Name of the story set
Story		Name of the story
Graph label		Scenario or Suite
Selector		Comment for the story set
Selector Info		ID and Comment on the selected selector
Active		Active, Not active, Publish
Save		Save the edit
Cancel		Discard the edit

Logfile

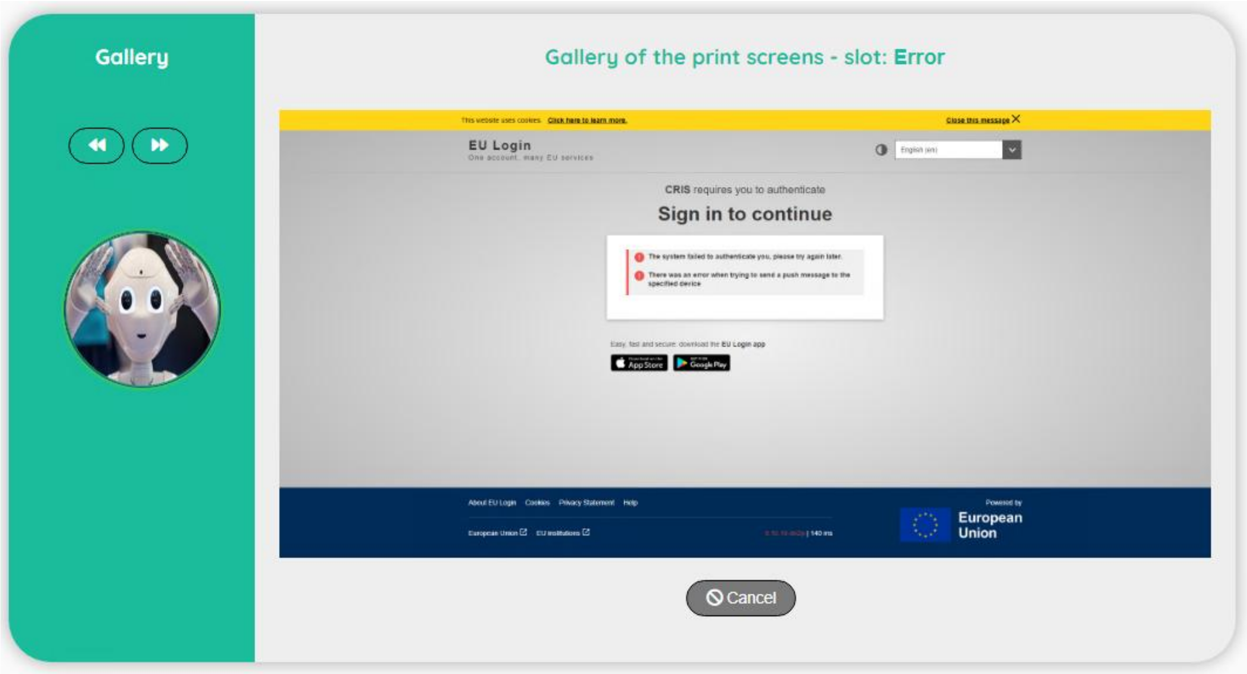
The logfile will show you all the steps executed with extra information.
It helps you to understand how the test has been executed.
Note: There is one logfile by user. You see always the logfile of the last execution



Topic	Icon	Comment
Gallery		Access to the print screen(s)

Gallery

The gallery will show you the print screen taken during the execution of the story.
The first print screen is done automatically by the Robot when an error occurs (it will show you the last error detected – in case of success of a story, you will see the last error detected)

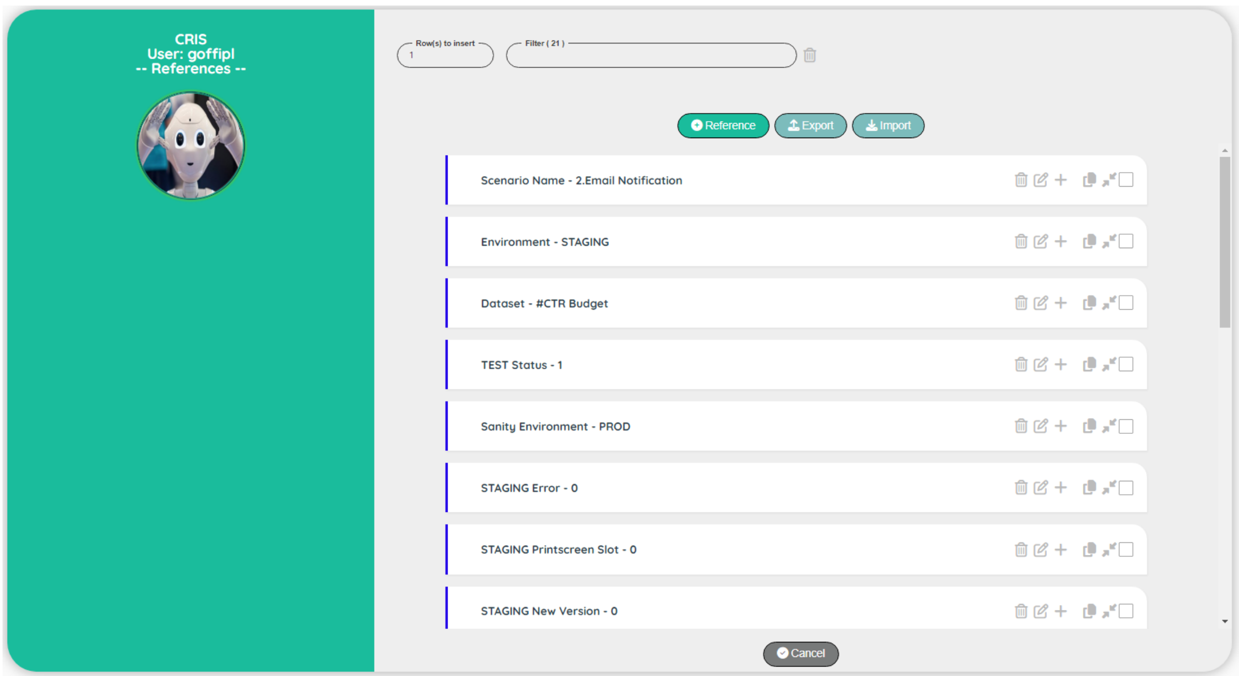


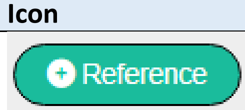
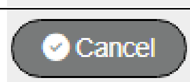
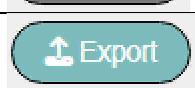
Topic	Icon	Comment
Previous		Go to the previous print screen
Next		Go to the next print screen
Cancel		Back to the Logfile

Reference

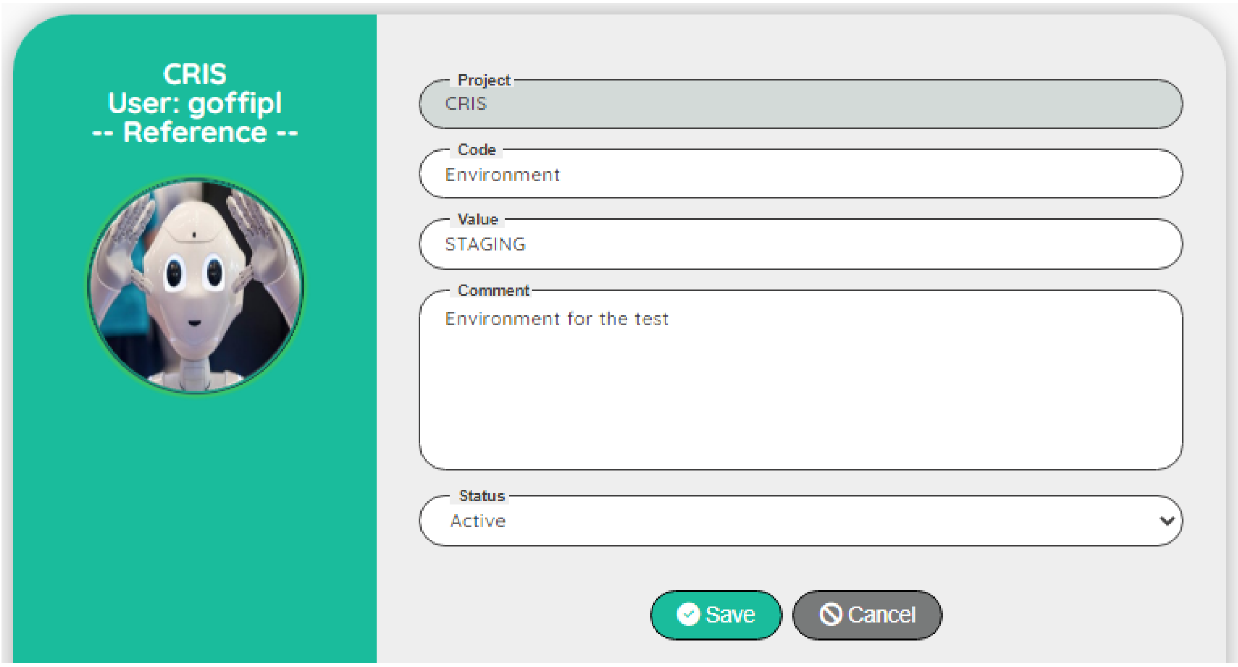
Reference

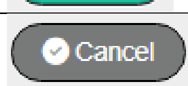
The reference will show all the values stored by the scenarios in order to exchange data.
A reference is own by a specific user for a project.



Topic	Icon	Comment
Add Reference		Add a reference at the beginning of the list.
Cancel		Back to the control panel.
Export		Go to the screen to export a reference.
Import		Go to the screen to import a reference.

Reference Edit

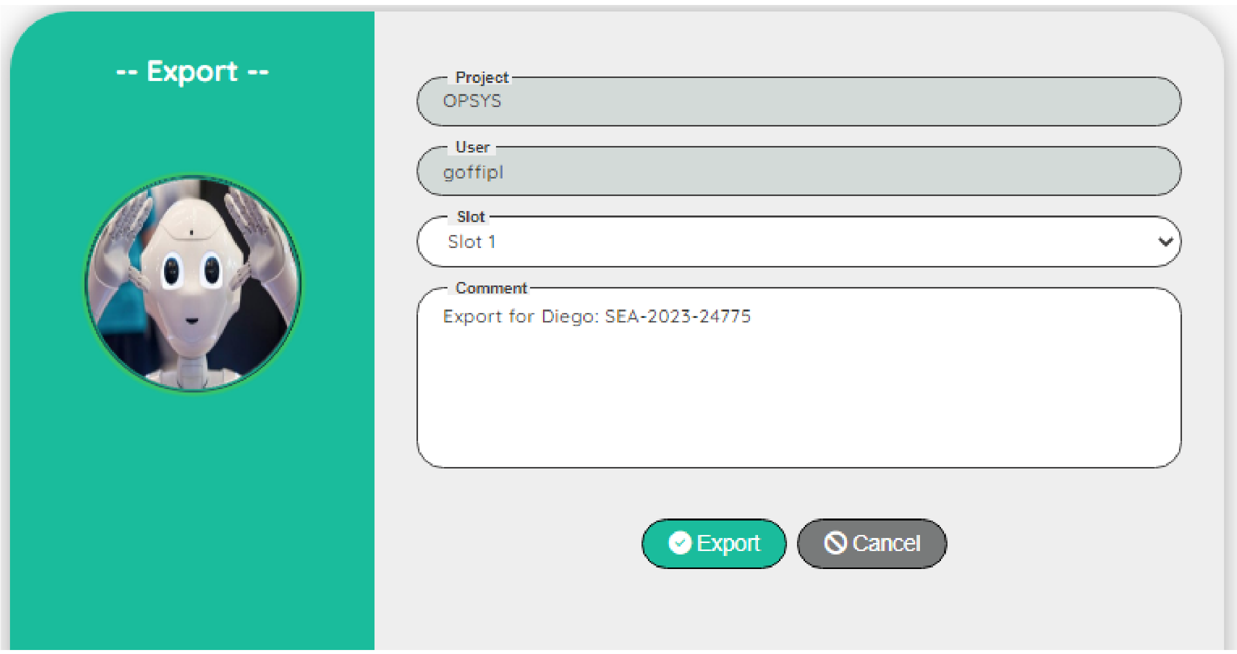


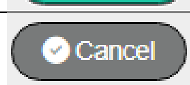
Topic	Icon	Comment
Project		For info: Name of the project
Code		Code of the reference
Value		Value of the reference
Comment		Comment for the reference
Active		Active, Not active
Save		Save the edit
Cancel		Discard the edit

Note: The reference 'Environment' is very important for the tool, be sure to create one and keep the value relevant to your tests ('Environment' is used in the Dashboard to display the graph with the performances).
Example of value for Environment could be for instance: PROD, ACC, TEST, STAGING….

Export Reference

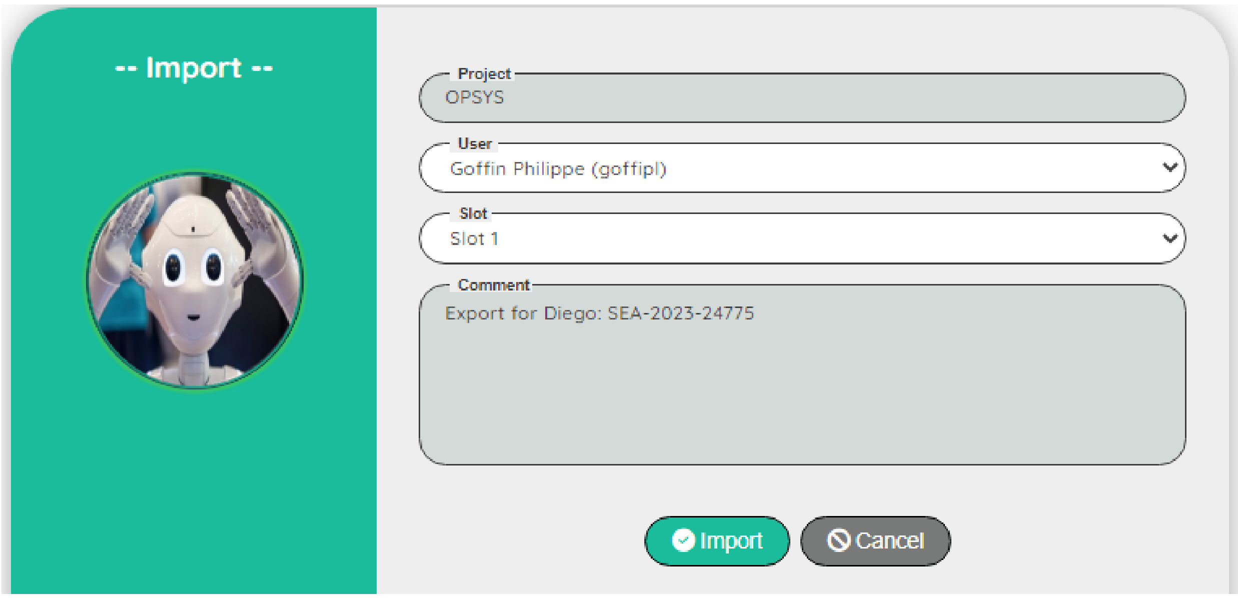
The export will allow you to take a backup or to share your reference with another colleague.

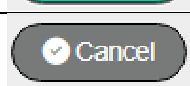


Topic	Icon	Comment
Project		For info: Name of the project
User		For info: Login of the user
Slot		10 slots are available to back up your references
Comment		Comment on the export
Export		Export the references
Cancel		Back to the control panel.

Import Reference

The import will allow you to restore a backup or get a reference from another colleague.
Be careful, you current references will be erased and replaced by the import



Topic	Icon	Comment
Project		For info: Name of the project
User		Select a user in the list
Slot		Select a slot
Comment		For info: Comment on the export
Import		Import the references
Cancel		Back to the control panel.

Specific Reference

When you create a new project, it can be a good idea to define some specific reference to impact your tests.

None of the specific references are mandatory (a default value will be used).
In the scenario, you can also write a step to update a specific reference (for instance the default timeout in seconds used by playwright to detect an element).

Be careful, the name of the reference is case sensitive!

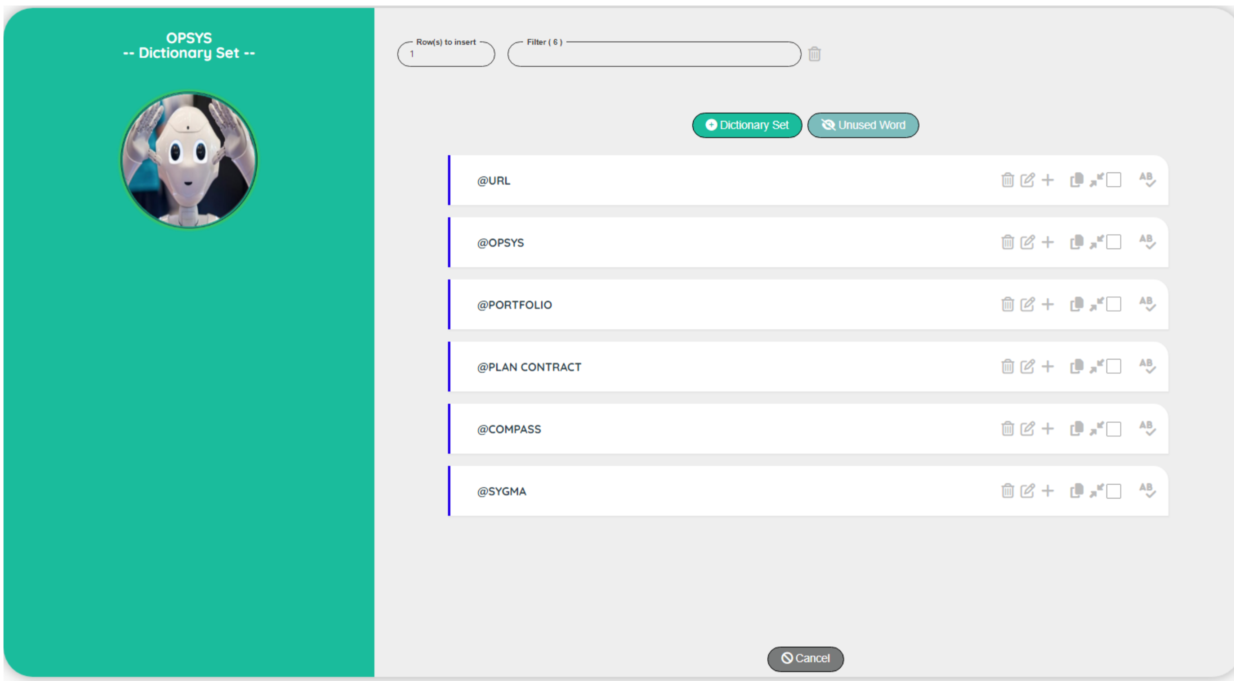
Reference	Default value	Comment
TimeOut	30	Playwright time out (in seconds) to detect an element
Headless	0	0: no headless browser (default), 1: headless browser
Browser	Chrome	Possible value: Chrome, Firefox or Safari
Device	<N/A>	A specific smartphone, a tablet or a desktop computer. Check the list of available devices on the playwright documentation. The device will impact the size of the browser!
Environment		Used by the performance graph (E.g.: ACC or PROD, INT….)
AllEnvironments		Used by the performance graph. A list of value comma separated (E.g.: **ACC,PROD**)

There are also technical references created by the tool during the execution of the tests.
These references are used by the tool to define context, to return the status of the execution, to restart at the right step after an error…

Dictionary

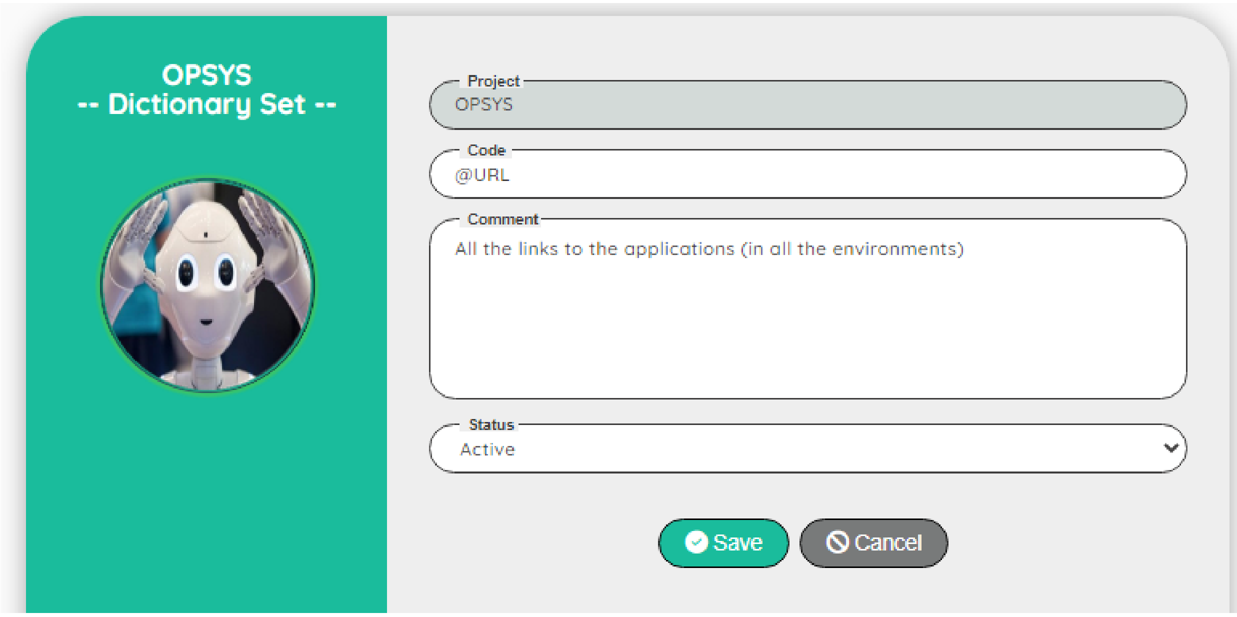
Dictionary set

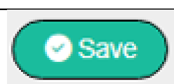
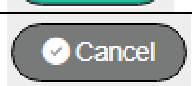
The dictionary is available for a project. It is composed of a dictionary set and a dictionary word.
You can store URL, xpath or even translation (there is a field language that you can use)
For the translation, the logic is the following: if you ask for an unknown language, the record with the language '*' will be used. So, you can for instance, use '*' for English and if a translation in another language is missing, the user will receive the English translation.



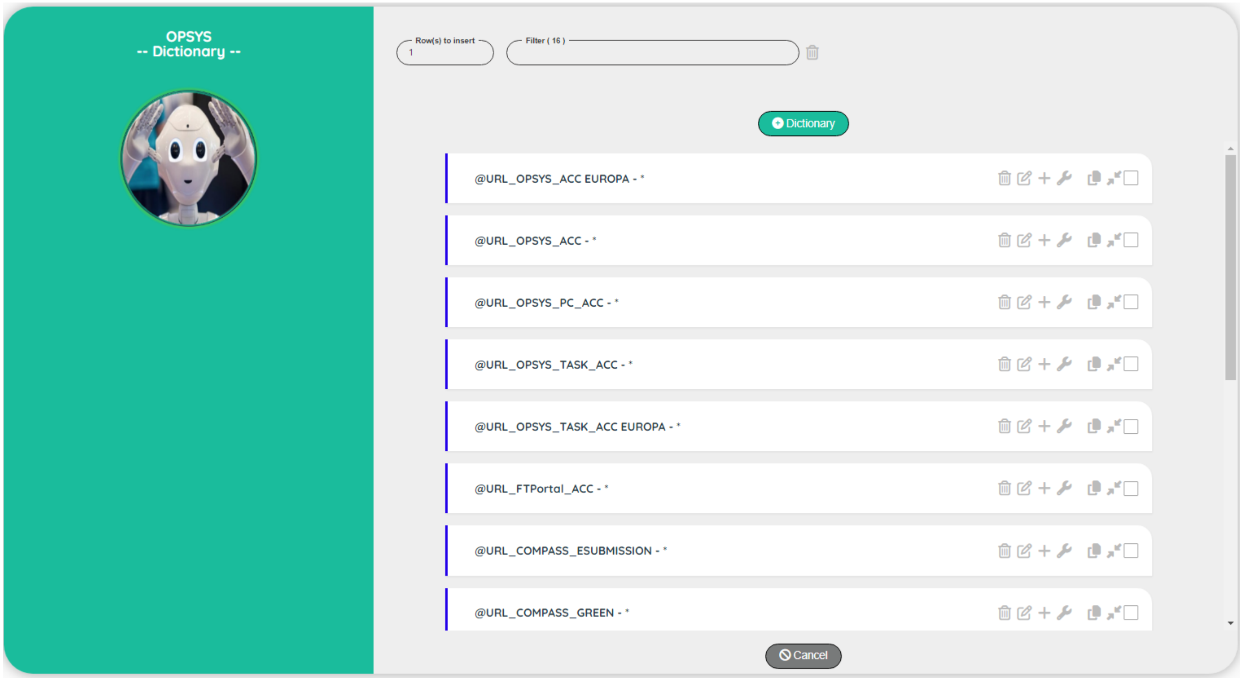
Topic	Icon	Comment
Dictionary	AB	Go to the screen dictionary word
Add Dictionary	Dictionary Set	Add a dictionary set at the beginning of the list.
Unused word	Unused Word	Go to the screen to export a reference.
Cancel	Cancel	Back to the control panel.

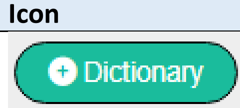
Dictionary set Edit



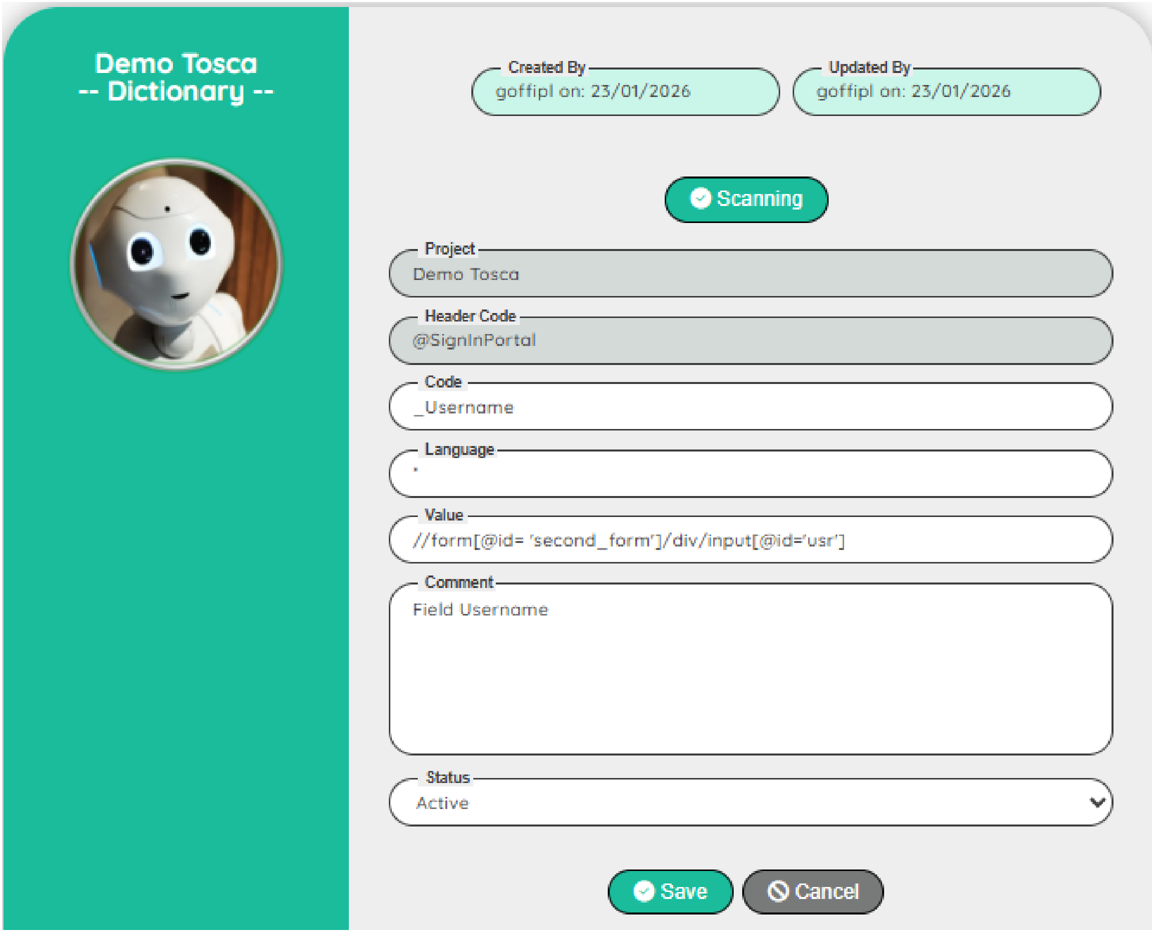
Topic	Icon	Comment
Project		For info: Name of the project
Code		Code of the dictionary must start by @
Comment		Comment for the dictionary
Active		Active, Not active
Save		Save the edit
Cancel		Discard the edit

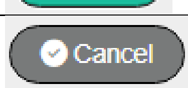
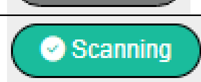
Dictionary word



Topic	Icon	Comment
Add Dictionary		Add a word at the beginning of the list.
Cancel		Back to the control panel.

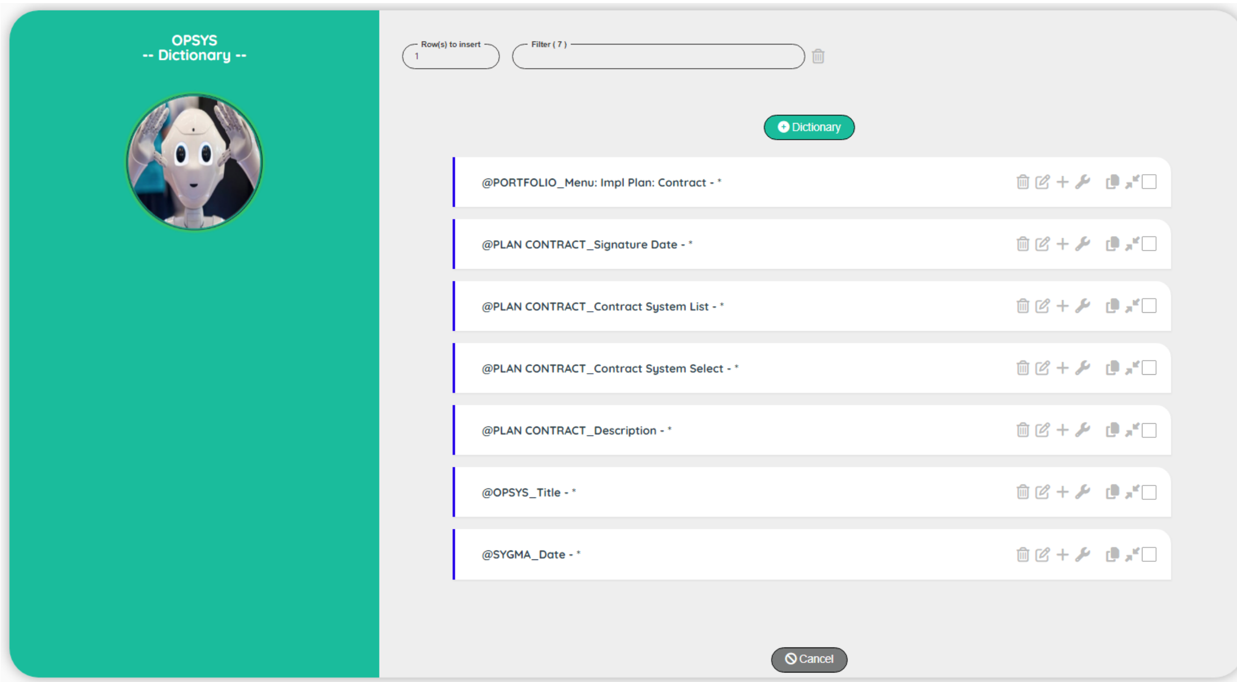
Dictionary word Edit

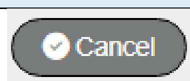


Topic	Icon	Comment
Project		For info: Name of the project
Dictionary set		For info: Name of the dictionary set
Code		Code of the dictionary must start by @
Language		* by default, but you can use an ISO code (EN, FR...)
Comment		Comment for the dictionary word
Active		Active, Not active
Save		Save the edit
Cancel		Discard the edit
Scanning		Scan a web page to detect elements

Dictionary unused word

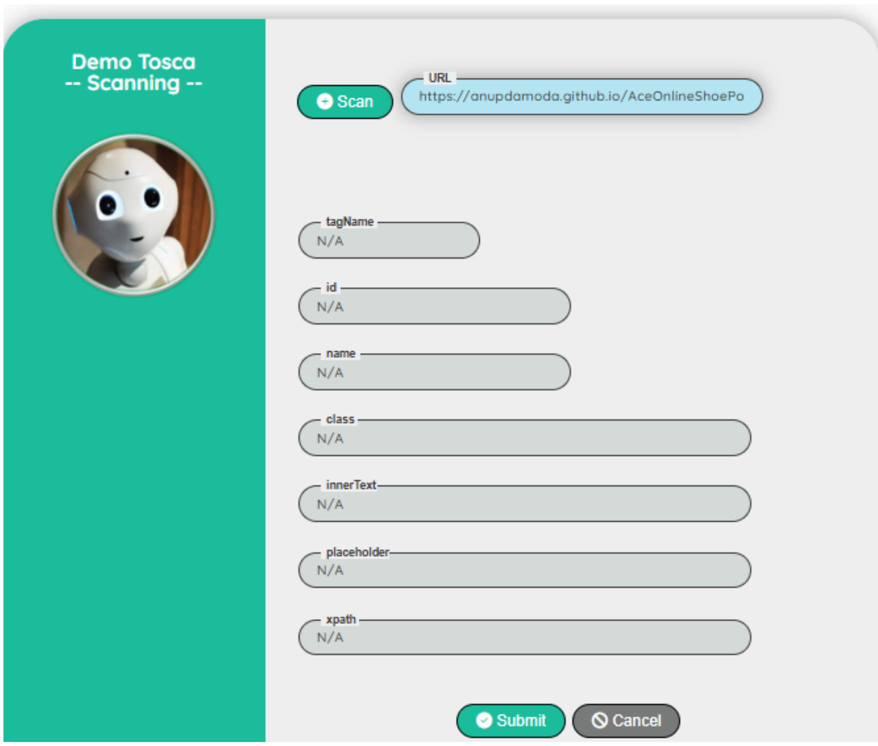
This screen will show you all the unused word(s).
Be careful, the detection can be fooled when you use a variable to replace a partial name of a word. For instance if you have @URL_Acceptance, it will be flagged as unused if you use the following step in your scenario: @URL_$Environment.



Topic	Icon	Comment
Cancel		Discard the edit

Dictionary scan web element

This screen will allow you to scan a web page to identify a specific web element.
From the Dictionary Edit screen, click on the button Scanning.

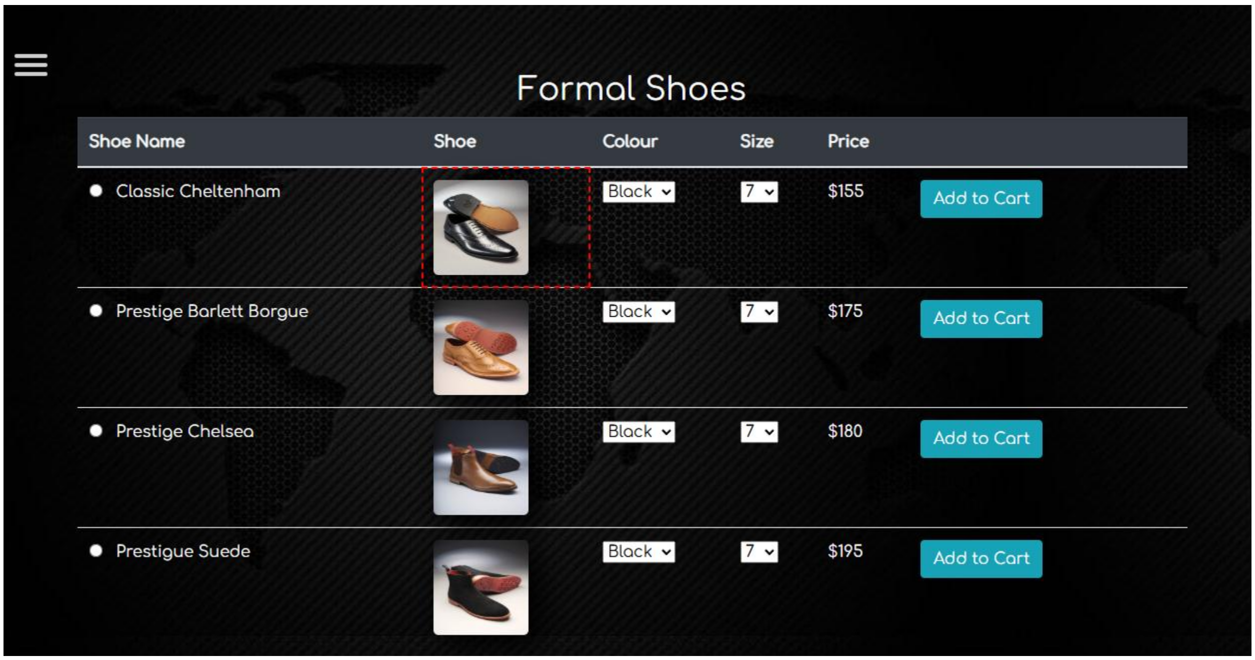


Topic	Icon	Comment
URL		The URL of the web page you want to scan
Scan		Scan a web page
Device		Web browser or Phone browser
Delay		Delay (in sec) before scanning elements

For the example, I used the link:
https://anupdamoda.github.io/AceOnlineShoePortal/FormalShoeslist.html

Use the mouse to select an element – hover the mouse on the element and a red dashed box will be displayed (for instance the picture of the shoes)

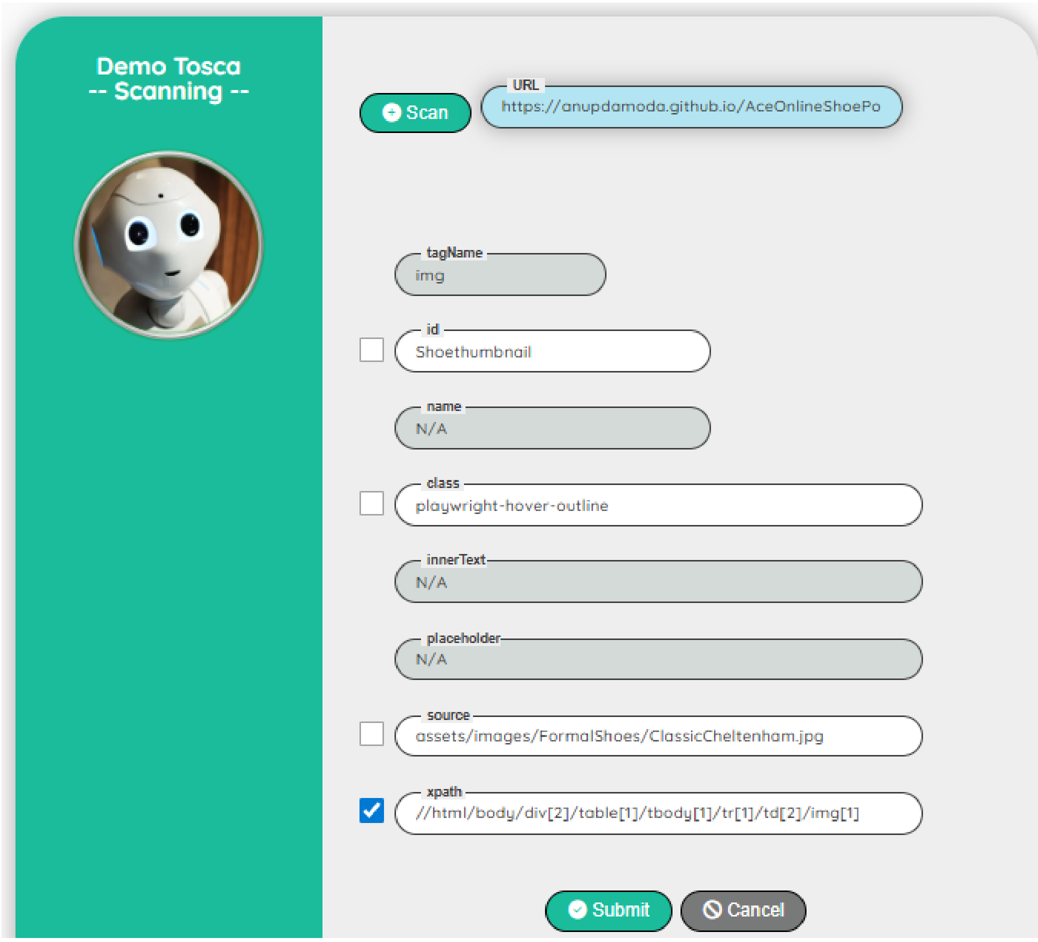
When you are satisfied, click with the mouse. The browser will be closed and you will back to the Robot.



Note 1: if the browser is not closed after the click, just close the external window.

Note 2: if you want to scan the Phone browser, you need first to launch Android Studio and start a phone with the Device Manager.

Depending of the element (image, button…) some attributes will be displayed or not.

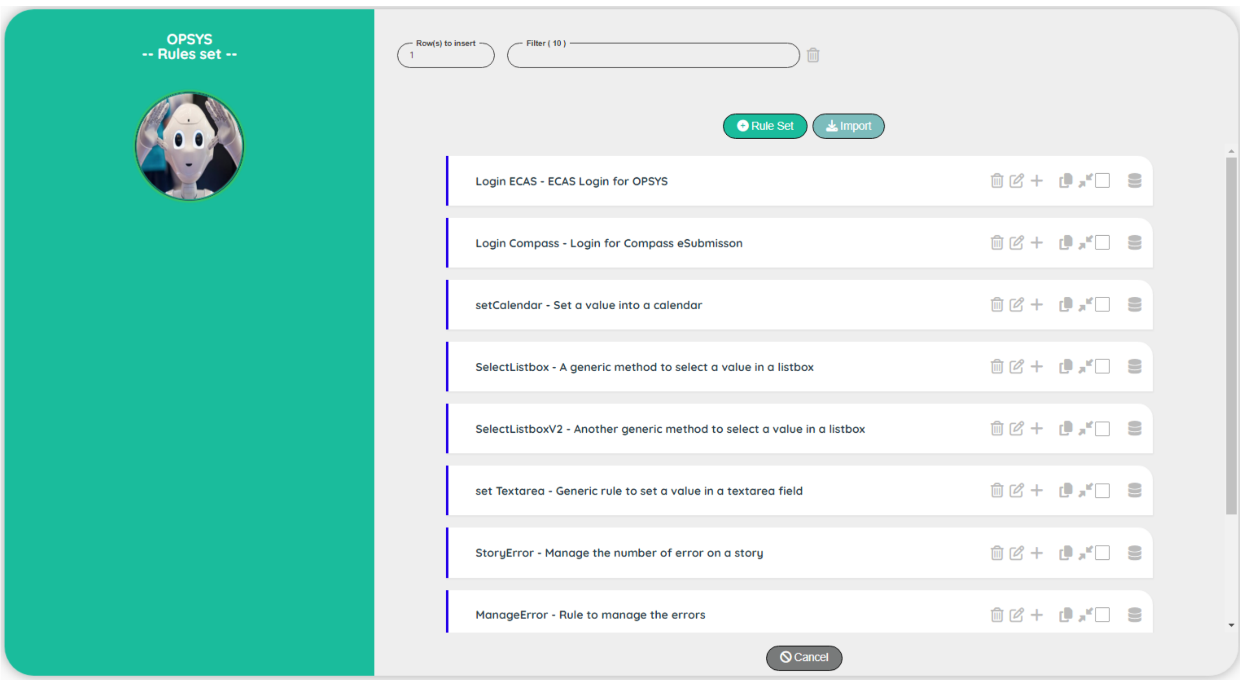


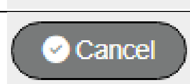
In this example, the robot was able to detect the Id, the class, the source and the xpath (selected by default). By clicking on the button Submit, the dictionary will be updated automatically!

Rules

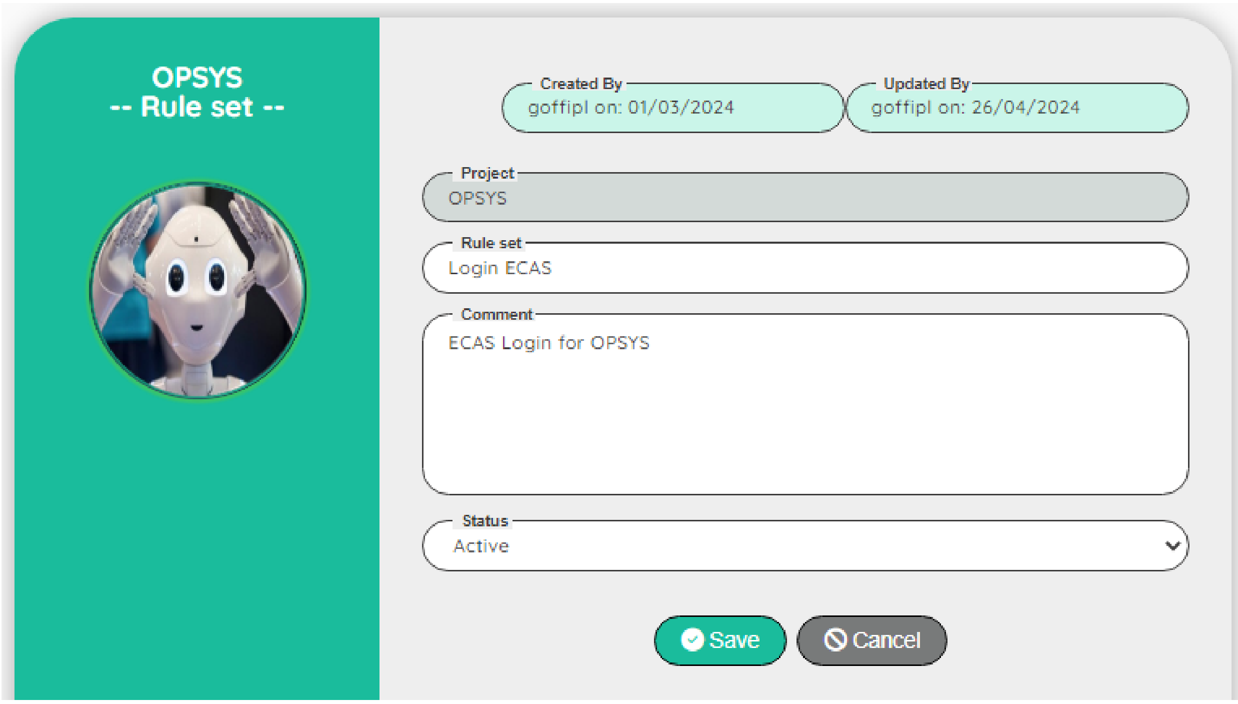
Rules set

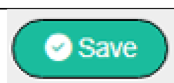
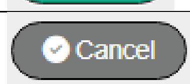
The rules are available for a project. It is composed of a rule set and rules.
The rules can be used by the robot to make a decision or by the Designer to define subroutine that can be reused. All the functions are available also in the rules.



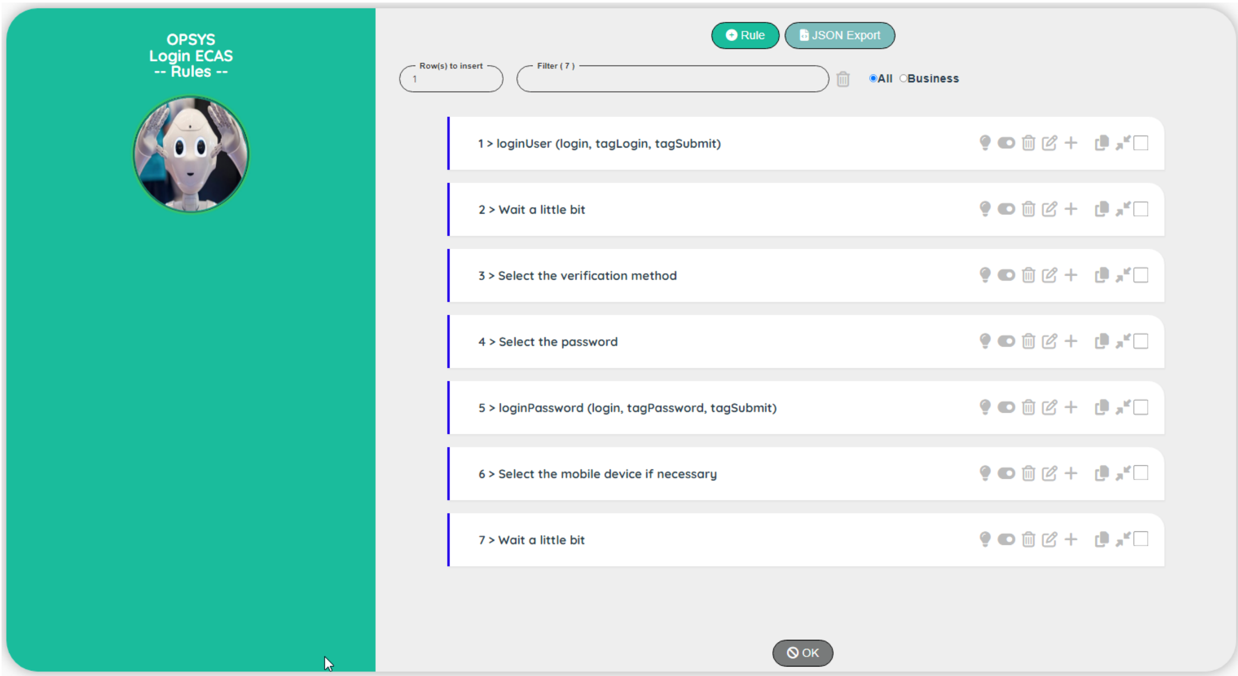
Topic	Icon	Comment
Rules		Go to the screen Rules
Add Rule set		Add a rule set at the beginning of the list.
Import rules		Import rules from another project
Cancel		Back to the control panel.

Rules set Edit



Topic	Icon	Comment
Project		For info: Name of the project
Rule set		Code of the rule set
Active		Active, Not active
Save		Save the edit
Cancel		Discard the edit

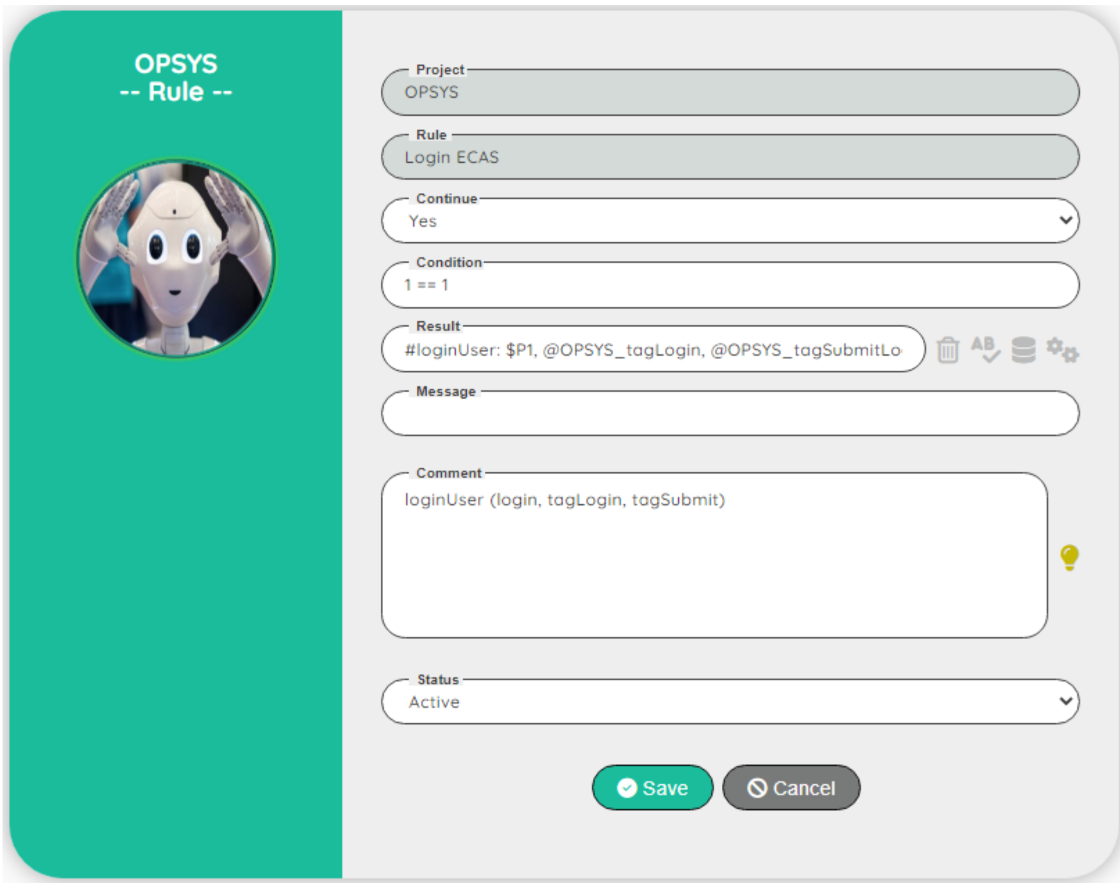
Rules

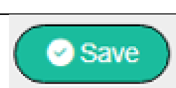
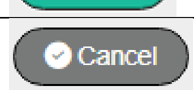


Topic	Icon	Comment
Add Rule	Rule	Add a rule at the beginning of the list.
JSON Export	JSON Export	Export rules into a .json file
Download	goffipl_rule.json	Download the .json file
Active/Inactive		Set the test Inactive/Active
Business/Designer		Set the comment type to Business / Designer (Technical)
Cancel	Cancel	Back to the rule set.

Note: To view how to upload a '.json' file into an Excel sheet, please refer to the section: How to process the .json file into Excel

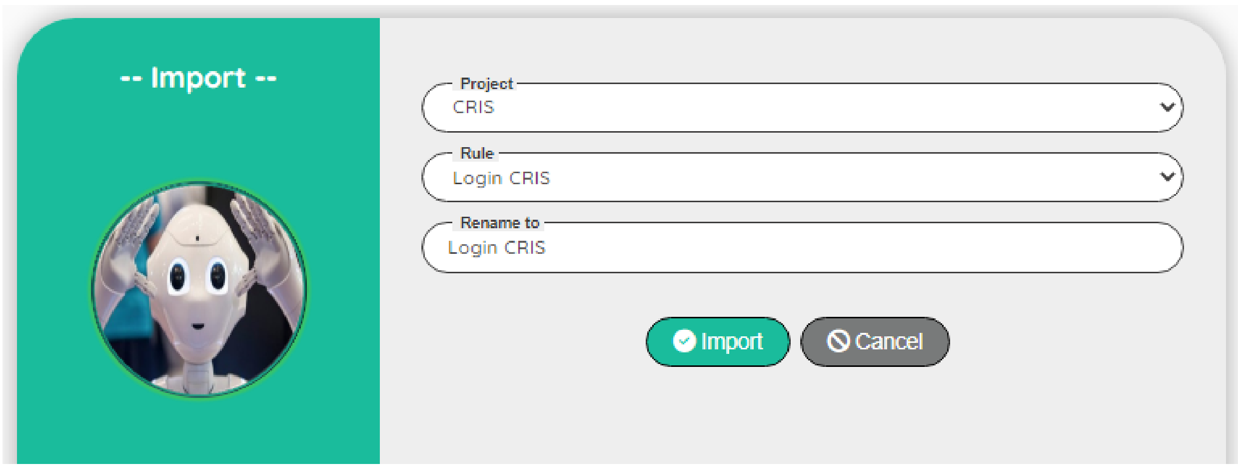
Rules Edit



Topic	Icon	Comment
Project		For info: Name of the project
Rule set		For info: Name of the rule set
Continue		Select Yes or Skip
Condition		Any valid JavaScript expression
Result		Any valid JavaScript expression or an existing function with its parameters (function starts with #)
Comment	 Comment for the Business  Comment for the Designer	Comment on the test
Active		Active, Not active
Save		Save the edit
Cancel		Discard the edit

Import rule set (including rules)

This screen will allow you to import the rules from another project



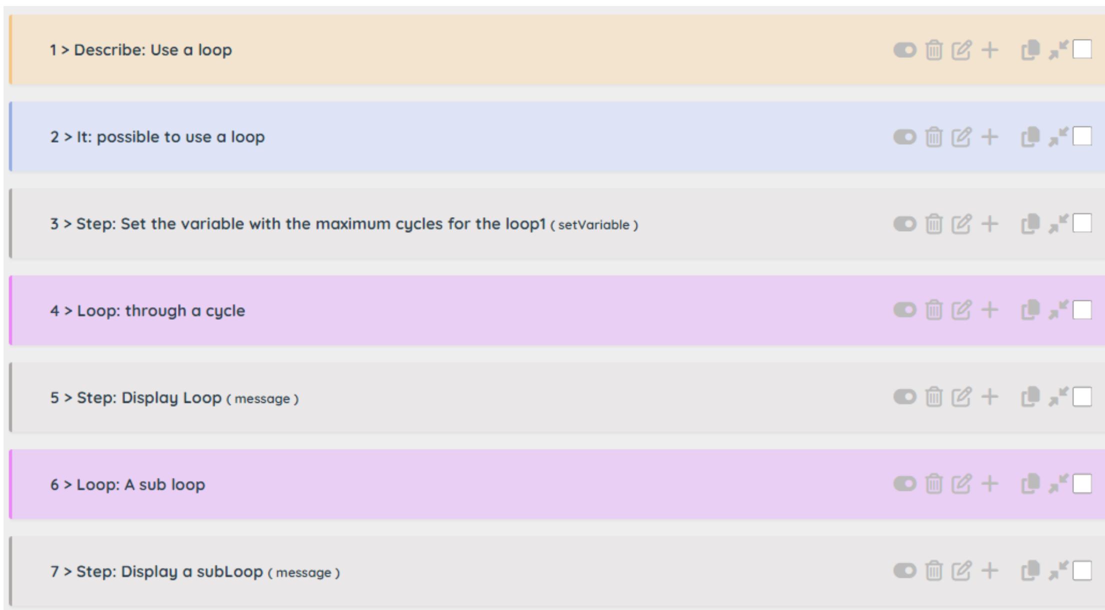
Topic	Icon	Comment
Project		Select a project
Rule set		Select a rule
Rename to		If necessary, rename the rule
Import		Import the rules
Cancel		Discard the import

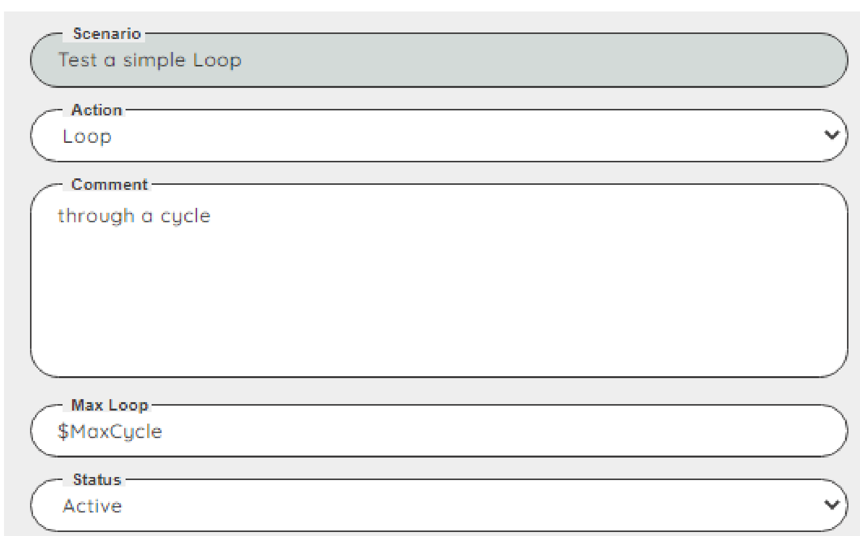
Loop

Principle

The Loops are very useful when you need to repeat a process.
Example: Upload a set of documents, key a set of experts data…

To define a loop, you need to use the instruction Loop





Loop has a parameter: Max Loop to define the number of loops.
Note: If a 'It block' is defined, the loop is inside the block (and so, depends on the condition of a skipIt or a skipDescribe)

You can control the loop thanks to the variable $Loop or $Loop1 (it's equivalent).
For instance, you can enter a value with specific data from the dataset like this:
setValue @Portal_LumpSum, **#$Dataset_Lump Sum Comment$Loop** to key data from the dataset:

- #SEA-2023_Lump Sum Comment**1**: This is the first comment
- #SEA-2023_Lump Sum Comment**2**: This is the second comment
- #SEA-2023_Lump Sum Comment**3**: This is the third comment

You can define a nested loop if you define a second Loop inside the first one

In the image above, the line 6 define a nested loop.
If the maximum loop for the first one is 2 and 3 for the second, we will have the following sequence:

(1) → 1.1, 1.2, 1.3 **(2)** → 2.1, 2.2, 2.3

Break a loop

Note: Be careful, the update of the variable used in the parameter of the Loop (Max Loop) during the process has no effect as the number of loops is evaluated during the definition of the loop.
If you need to break a rule in progress, you can use a skipIt (but the loop will continue its cycle without any process)

End of a loop

There are three ways to define the end of a loop:

1) There is no more step
   As in the image above, the loop 1 and the loop 2 will be defined until the line 7

2) There is a Describe
   When a Describe is detected, the Robot understand that the current loop is ended.
   This is the most common way to indicate the end of a loop

3) There is an End Loop
   When you have a skipIt before the loop, the existence of the loop will depend of the skipIt condition. In the case you have nested loop, you cannot use a Describe to end a loop.
   In this case, you need to use End Loop. This instruction will be ignore if a skipIt is triggered.

Examples of loop

A simple case: One loop inside a scenario

Action	Comment	Loop
Describe	Manage the deliveries	
It	Possible to manage deliveries	
Step	Get the number of deliveries from the dataset into the variable $NbDelivery	
Loop	Through all the deliveries (max loop = $NbDelivery)	1
Step	Add a new delivery	1
Step	Enter the name from the dataset using #$Dataset_Delivery$Loop	1
It	Possible to continue inside a loop	1
Step	…	1
Describe	Manage the document (End of the loop on the deliveries)	
It	….	

Two independent loops inside a scenario

Action	Comment	Loop
Describe	Manage the deliveries	
It	Possible to manage deliveries	
Step	Get the number of deliveries from the dataset into the variable $NbDelivery	
Loop	Through all the deliveries (max loop = $NbDelivery)	1
Step	Add a new delivery	1
Step	Enter the name from the dataset using #$Dataset_Delivery$Loop	1
It	Possible to continue inside a loop	1
Step	…	1
Describe	Manage the document (End of the loop on the deliveries)	
It	Possible to manage document	
Step	Get the number of documents from the dataset into the variable $NbDocument	
Loop	Through all the documents (max loop = $NbDocument)	1
Step	Add a new document	1
Step	Enter the name of the document from the dataset using #$Dataset_Document$Loop	1
It	Possible to continue inside a loop	1
Step	…	1
Describe	Manage the save data (End of the loop on the document)	
It	….	

Two nested loops inside a scenario

Action	Comment	Loop
Describe	Manage the contractors	
It	Possible to manage contractors	
Step	Get the number of contractors from the dataset into the variable $NbContractor	
Loop	Through all the contractors (max loop = $NbContractor)	1
Step	Get the name of the contractor in the variable $Name	1
It	Possible to continue inside a loop	1
Step	…	1
It	Possible to manage Experts of the contractor	1
Step	Get the number of experts from the dataset into the variable $NbExpert	1
Loop	Through all the expert (max loop = $NbExpert)	2
Step	Add a new Expert	2
Step	Enter the name of the expert from the dataset using #$Dataset_Expert$Loop	2
It	Possible to continue inside a loop	2
Step	…	2
Describe	Save the contractor (end of the loop 2, back to loop 1)	1
It	Possible to save the contractor	1
Step	…	1
Describe	Finalise the test (end of the loop 1)	
It	Possible to finalise the test	

Two nested loops inside a scenario with a conditional skipIt (we need to use End Loop)

Action	Comment	Loop
Describe	Manage the contractors	
It	Possible to manage contractors	
Step	Get the number of contractors from the dataset into the variable $NbContractor	
Loop	Through all the contractors (max loop = $NbContractor)	1
Step	Get the name of the contractor in the variable $Name	1
It	Possible to continue inside a loop	1
Step	…	1
It	Possible to manage Experts of the contractor	1
Step	Skip the test if the contractor is 'ArtComputer' (using $Name)	1
Step	Get the number of experts from the dataset into the variable $NbExpert	1
Loop	Through all the expert (max loop = $NbExpert)	2
Step	Add a new Expert	2
Step	Enter the name of the expert from the dataset using #$Dataset_Expert$Loop	2
End Loop	End of the loop 2, back to loop 1	1
It	Possible to save the contractor	1
Step	…	1
Describe	Finalise the test (end of the loop 1)	
It	Possible to finalise the test	

Performance

Principle

The Robot is not designed to measure the performance of an application (like load runner for instance) however, you can measure the elapsed time between two points in the application.

You have also the possibility to compare the last measure with the average of the last 10 tests. It can be useful after a new deployment for instance.

You can set as many timers you want in a scenario by respecting the following rules:

- Each timer as a unique identifier (topic).
- The identifier used during the closure must be equal to the identifier of the start.

In the dashboard, you have the possibility to display the performance of a story or focus on the performance of a specific step in a story.

To measure elapsed time, you can use the following functions:

startTimer	topic		Start a timer to measure the
stopTimer	environment	topic	Store the elapsed time in the database

Define the measure

If your scenario is well structured, it should be very easy to identify the bloc of actions thanks to the "Describe" and the "It".

If you want to measure the global performance of a test, I suggest to start the measure, just after the login.

Note: If you want to measure the performance of the login, take into account that the measure is done by scenario. You cannot display a global measure of the login of your application!

To indicate the starting point of the measure, use the function startTimer() with the name of the topic (a unique identifier). Keep the name short!
Note: The topic is case-insensitive.

To indicate the end of the measure, use the function stopTimer() with the parameters: Environment (Example: ACC, TEST or PROD) and the name of the topic used in the startTime function.

Warning: be careful when using the function pause() because you will increase artificially the performance.
Also, increasing a pause of an existing scenario will impact the global performance!

Mechanism

The stopTimer will measure the elapsed time and store the value in the database.
The Robot will perform the following tasks:

Shift the 10 existing measures up and store the elapsed time at the measure 11

ID	T1	T2	T3	T4	…	T11	T12	T13	…
1	M1	M1	M1	M1		M1	M2	M3	
2			M2	M2		M2	M3	M4	
3				M3		M3	M4	M5	
4						M4	M5	M6	
5						M5	M6	M7	
6						M6	M7	M8	
7						M7	M8	M9	
8						M8	M9	M10	
9						M9	M10	M11	
10						M10	M11	M12	
11	M1	M2	M3	M4		M11	M12	M13	

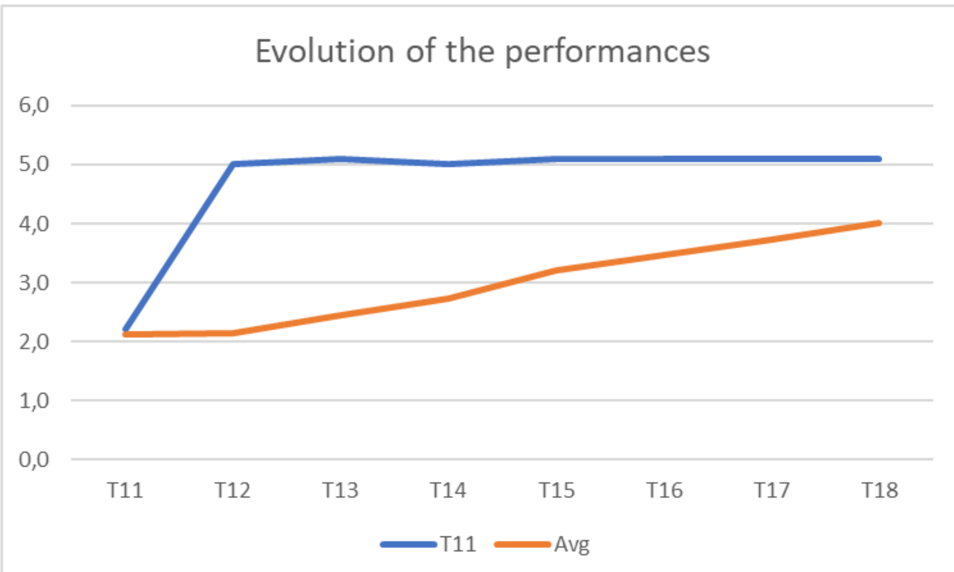
With Txx: Timing and Mxx: Measure

Note: we use a cheat with the first and the second measure, to keep the coherence of the data

Using this mechanism will reduce excessive measurement variations.
However, be careful when you need to perform a measure after a new deployment, if the elapsed time is suddenly high, the impact will not be very visible at first!

Example:

ID	T11	T12	T13	T14	T15	T16	T17	T18
1	2,0	2,1	2,2	2,1	2,2	2,3	2,0	2,1
2	2,1	2,2	2,1	2,2	2,3	2,0	2,1	2,2
3	2,2	2,1	2,2	2,3	2,0	2,1	2,2	2,1
4	2,1	2,2	2,3	2,0	2,1	2,2	2,1	2,2
5	2,2	2,3	2,0	2,1	2,2	2,1	2,2	5,0
6	2,3	2,0	2,1	2,2	2,1	2,2	5,0	5,1
7	2,0	2,1	2,2	2,1	2,2	5,0	5,1	5,0
8	2,1	2,2	2,1	2,2	5,0	5,1	5,0	5,1
9	2,2	2,1	2,2	5,0	5,1	5,0	5,1	5,1
10	2,1	2,2	5,0	5,1	5,0	5,1	5,1	5,1
11	2,2	5,0	5,1	5,0	5,1	5,1	5,1	5,1
Avg	2,1	2,2	2,4	2,7	3,2	3,5	3,7	4,0



To quickly highlight the variation, we need to show on the same graph the average and the last measure.
As you can see on the graph, at T12 we suddenly increase the elapsed time!

Performance with the Robot

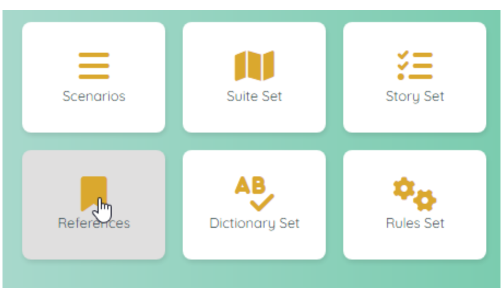
You can visualize the performances of a scenario in the Dashboard.

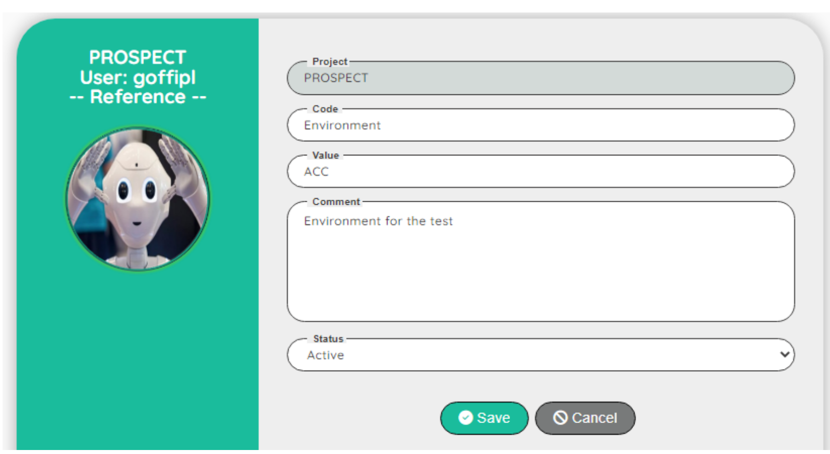
At the upper left, you can see an icon with a graph, each click will display a different graph.

	Graph of the execution of the story (status)
	Graph of the performance of all the scenarios of the story
	Graph of the performance of all the scenarios of a specific step of a story

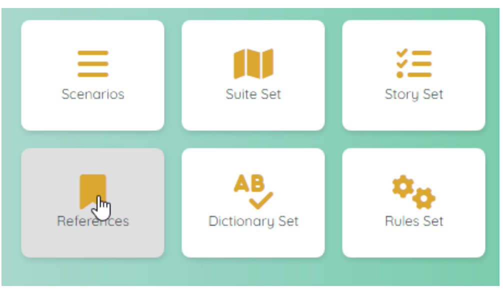
Note 1: The step is a kind of zoom on the graph to highlight specific scenarios.

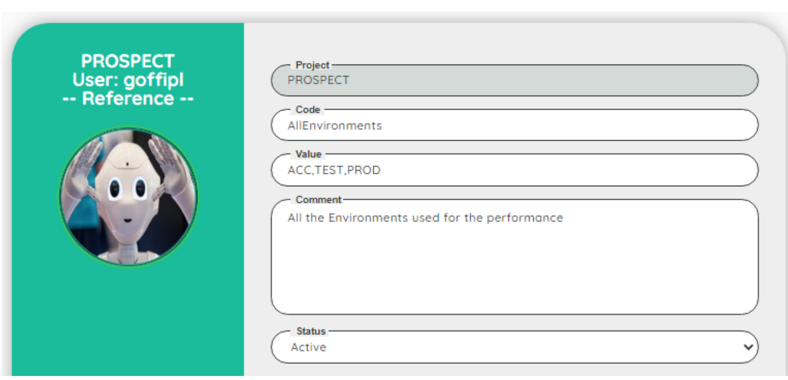
Note 2: In the Dashboard, the Robot can use the value ‘Environment’ of the reference data.
Be sure that you have defined the correct environment before displaying the performance!
If no reference ‘Environment’ can be found, you will not be able to select the performance graph!



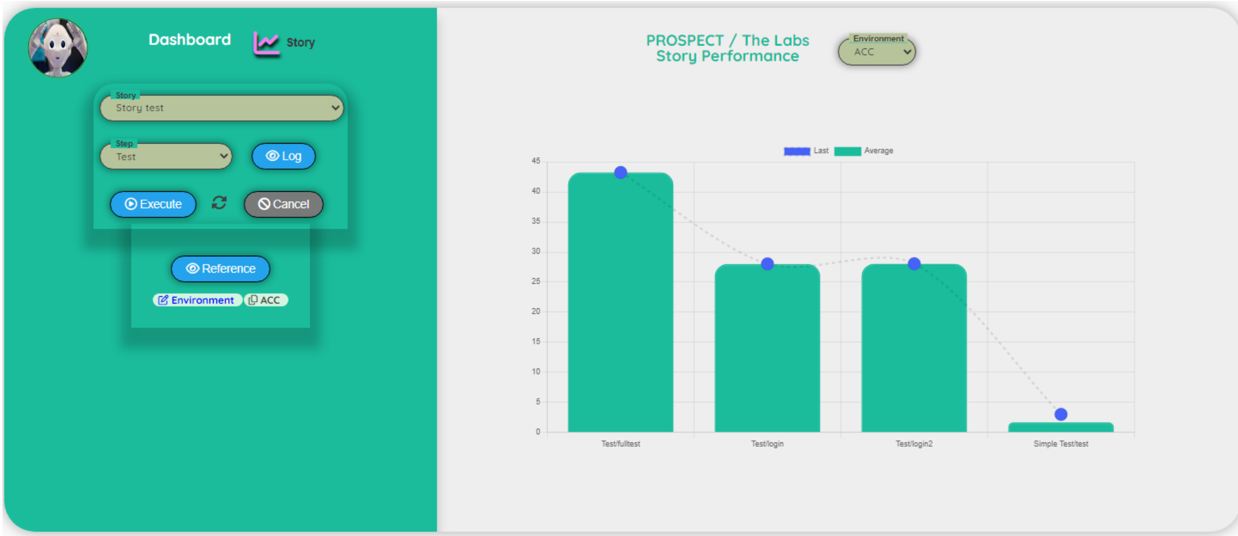


Note 3: In the Dashboard, the Robot can use the value 'AllEnvironments' of the reference data. If the reference doesn't exist, there is no error, but the list will be limited to the current environment.





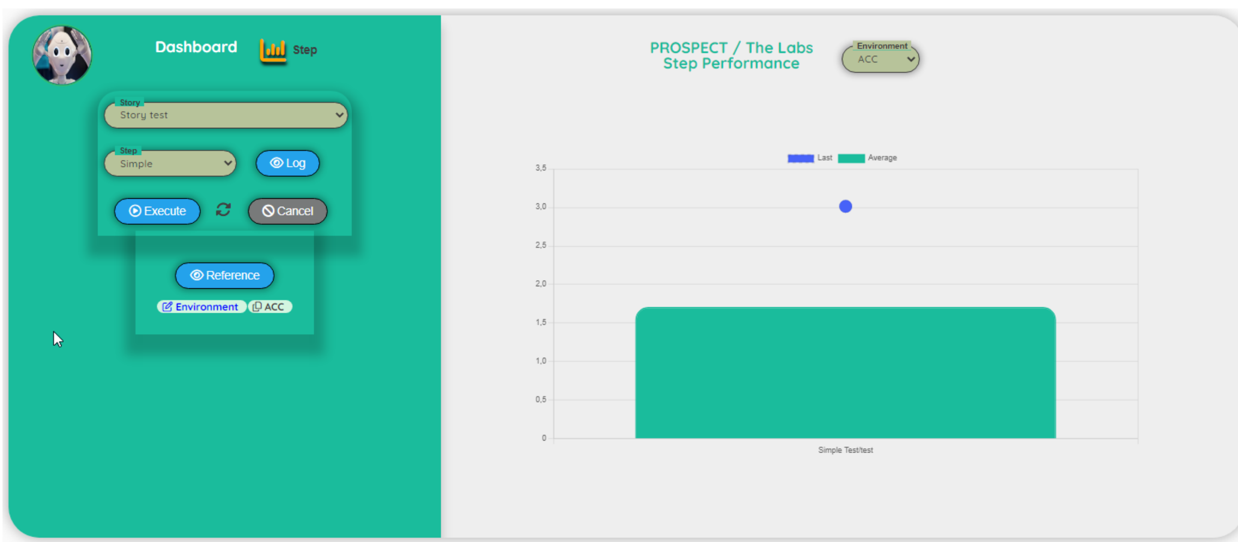
Example: Performance of the story



At this point the selection of a specific step has no impact on the graph.
The first three bars represent the step: ‘Test’ and the last one, the step: ‘Simple’.

Example: Performance of the step.

In this case, we decided to highlight the step: Simple to have a better view of the gap between the average and the last measure.
The step graph will act as a zoom.



Example: Performance detail (story or step).

This graph is useful to view the variance of the measures.
As we use up to 10 measures to compute the average, it's also important to check the coherence of the measures. Although, the quality of the average will be better if you have already run the test a few time.

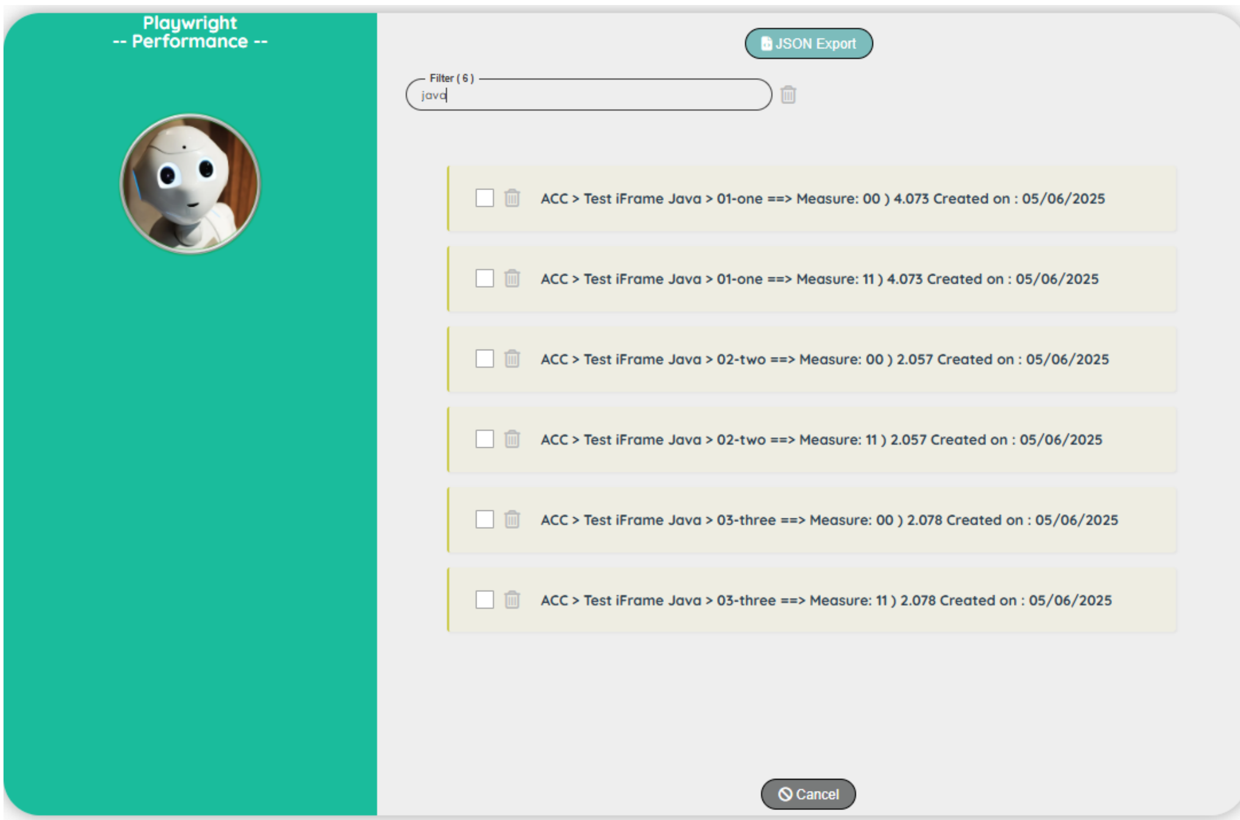
From 1 to 10 (in green), the data that will be used for the computation of the average.
In blue the last measure recorded.

You can change the environment to get a quick comparison of the performances.

In this case the performances in PROD

Performance with Admin right

If you are connected to the application as Admin, you can export the performance data into a Json file.

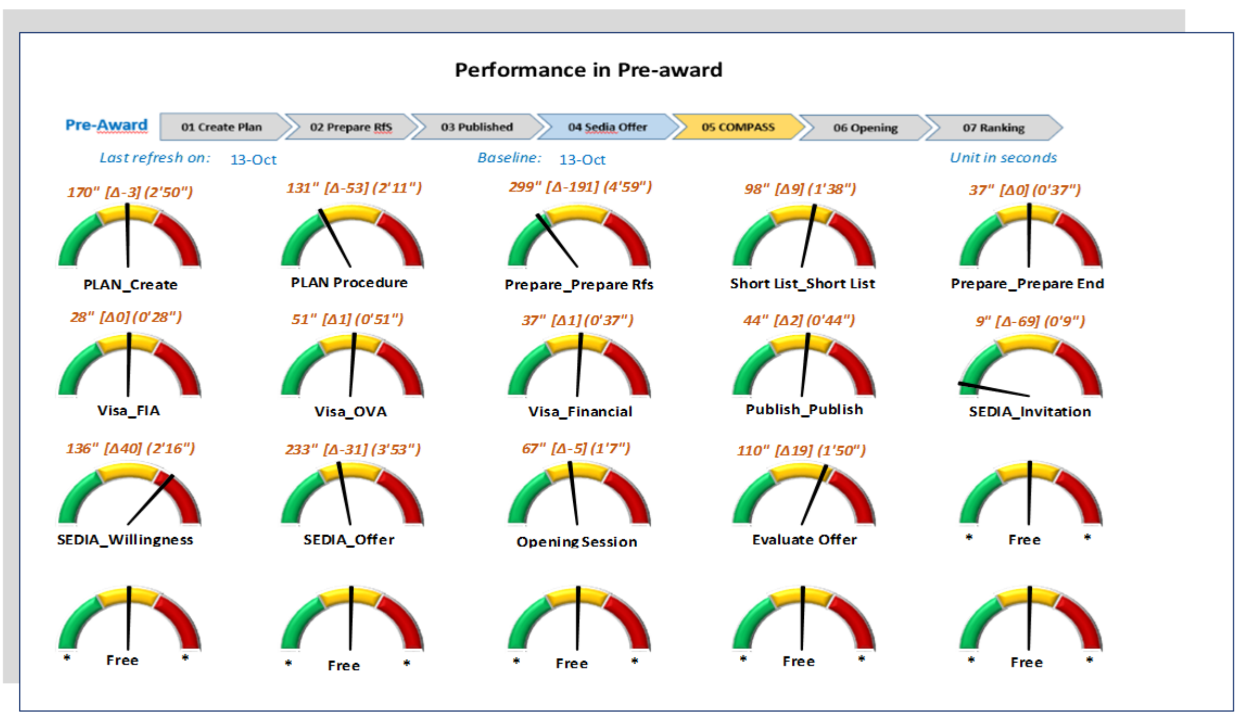


The .Json file has the following structure.
You can see how to import a .Json file into Excel in the Chapter: Test / How to process the .json file into Excel

```json
{
    "Performance": [
        {
            "performanceID": 111,
            "projectID": 85,
            "project": "PROSPECT",
            "scenarioID": 105,
            "scenario": "Test",
            "space": "ACC",
            "topic": "fulltest",
            "created": "11/12/2024",
            "sequenceID": 1,
            "sequenceID2": "01",
            "measure": 43.236
        },
```

Example of graphs of performance designed in Excel

This was designed for a previous project to show the evolution of the performances in the different business categories.



Testing mobile devices

Testing mobile devices is a bit tricky.
You need to use two different methods depending on the type of test.

Android Apps

To test a pure **Android Apps**, you need to use a Virtual phone created with the system: Google Play Store (see below).
The Robot will launch a Chrome browser with a picture (print screen) of the real Android Phone (refreshed every 5 seconds). You must interact with the Chrome browser (not with the virtual Android phone). You can move the mouse over an element to analyse.
In the Robot user interface, you can switch between a normal click behaviour or a submit behaviour (the element is analysed, the browser is closed and the attributes of the element are stored in the dictionary)

Android Web page

In the case, you need to test an **Android Web page**, you need to use a Virtual phone created with the system: Google APIs (see below).
There is no hover event available, so you cannot preview a frame around the element to scan (the frame is only visible after you clicked!)

Testing mobile devices (web browser)

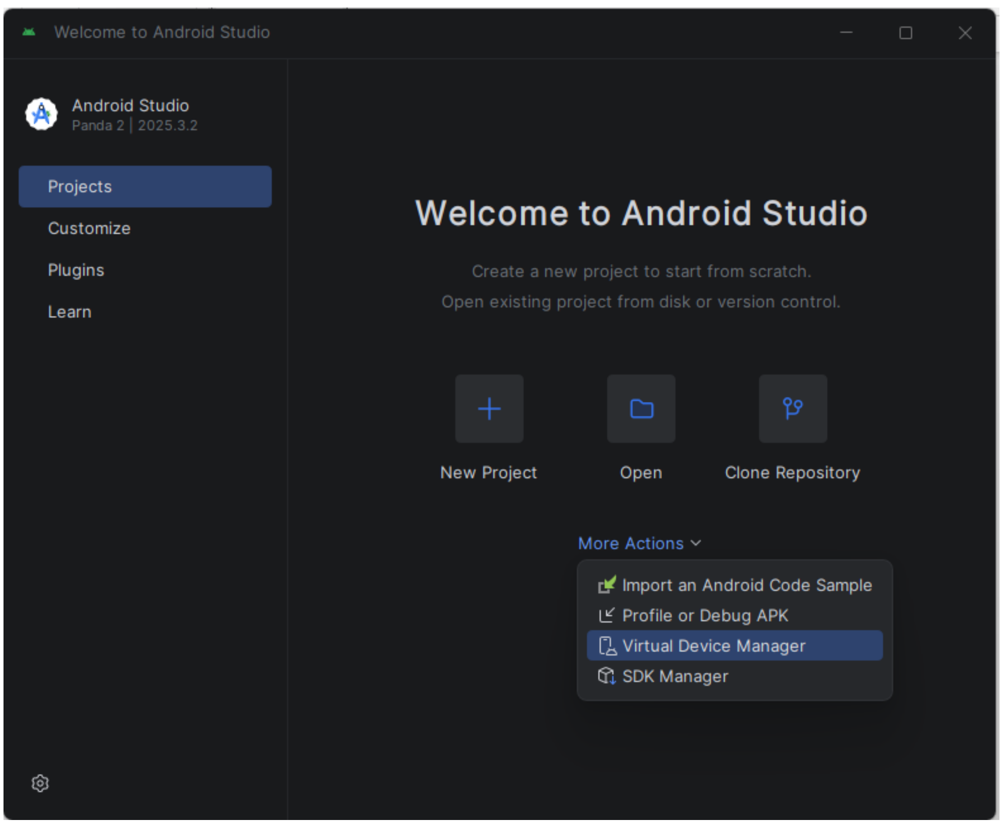
Android Studio - Installation

To test a mobile Android device, you need to carefully setup your environment !

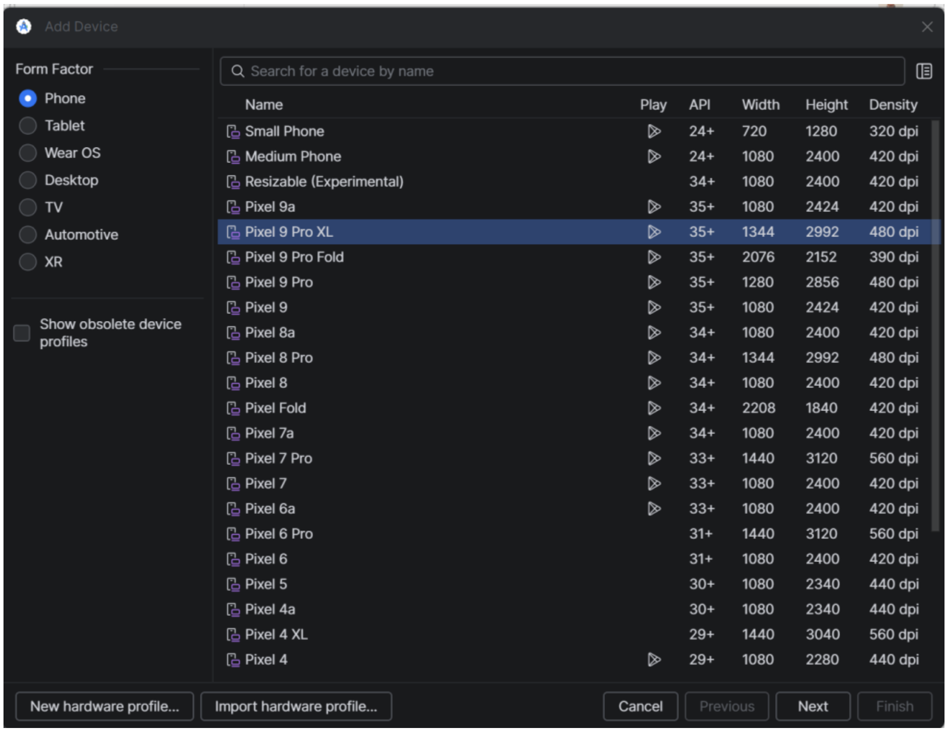
Download: Android Studio: https://developer.android.com/studio

Android Studio is a platform for the developers. It offers an emulator for different Android devices like: Phone, Tablet, Wear OS (e.g. watches), Desktop, TV, Automotive, XR.

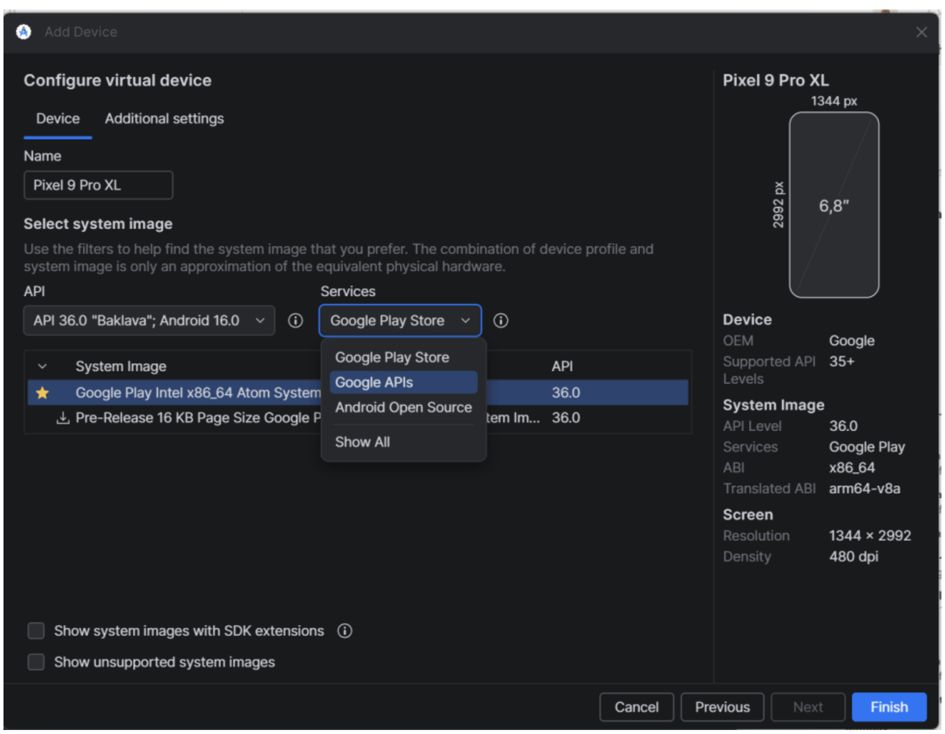
When you have installed Android Studio, click on the menu: More Actions and select "Virtual Device Manager"



Click on the “+” to add a new device and select a phone (e.g. Pixel 9 Pro XL)

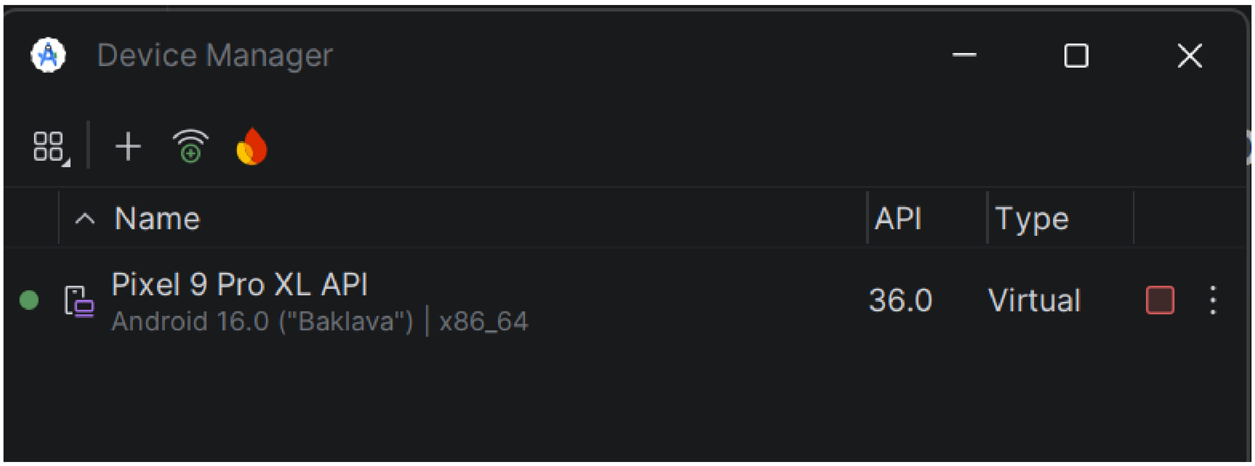


Click on Next, and now **a very important step**: you need to select **Google APIs** rather than Google Play Store



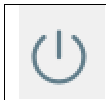
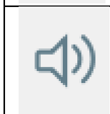
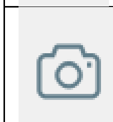
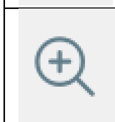
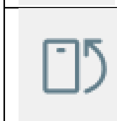
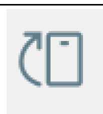
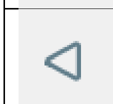
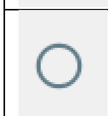
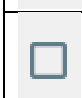
Select the system image: Google APIs…. Atom System Image to enable the button Finish.

This step is very important because the Google Chrome from the Play Store contains a lot of security that will block Playwright from exchanging information (during the handshake protocol).



When you see a green dot next to the selected device, you are ready to go…

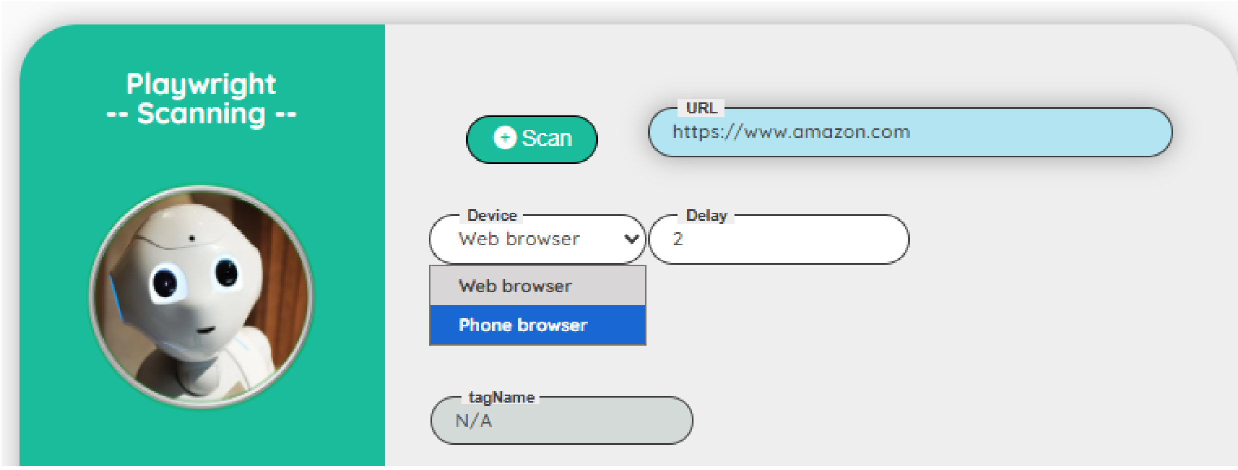
The menu on the right can be used to control the phone:

	Switch On/Off the device
 	Increase / decrease the volume of the device
	Take a snapshot
	Zoom on the screen
 	Rotate the device portrait to landscape
	Back
	Home
	Overview (?)

The Home button is very important because in a lot of scenarios, you will expect that the device is in the Home position.

<u>Note</u>: this configuration will not work to test Android Apps (in this case, you need to follow instruction on the next section)

In the Robot tool, you can use the **Scan** with the option **Phone browser** to help you to identify an element from your Phone browser.



You can increase the delay (before scanning) to help you to setup the webpage as you want. For instance: acknowledge a message, go to another part of the be site….

Click on the Scan button to start the scanner.

You can go directly on the Virtual Android device to click on the element to scan.

<u>Note:</u> Unlike desktop browsers, the Android environment does not handle the hover event. Therefore, the scanner's behavior will be slightly different: the green frame to highlight the element will only appear after clicking on it.

Testing mobile devices (Apps)

If you need to test Android Apps from the Play Store, you need to install a Phone emulator based on Play Store.

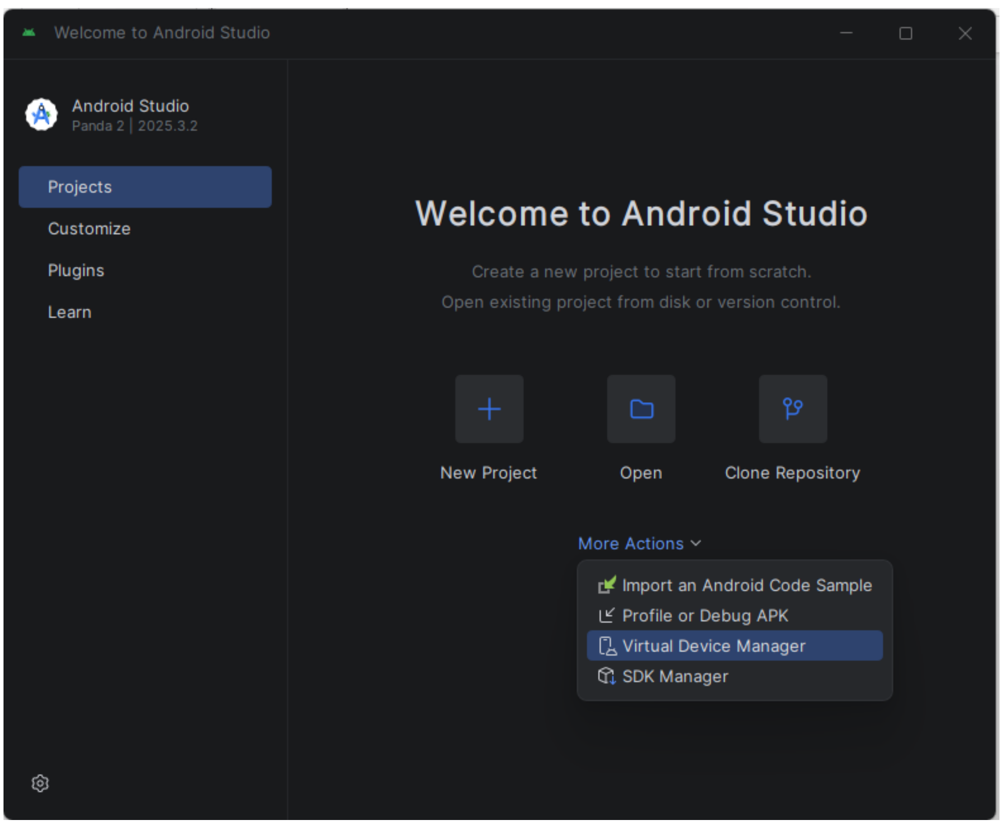
Android Studio - Installation

To test a mobile Android device, you need to carefully setup your environment !

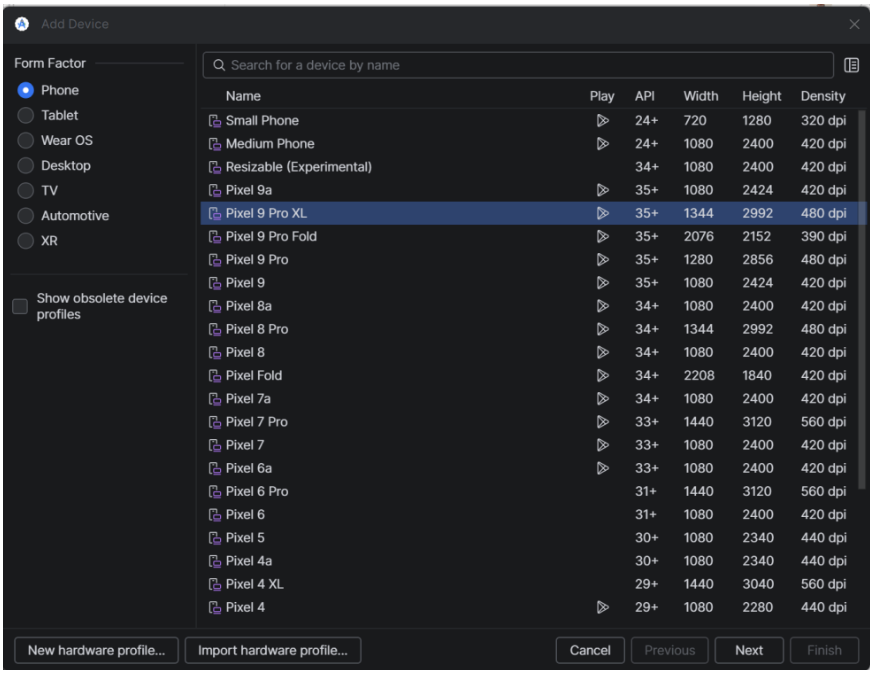
Download: Android Studio: https://developer.android.com/studio

Android Studio is a platform for the developers. It offers an emulator for different Android devices like: Phone, Tablet, Wear OS (e.g. watches), Desktop, TV, Automotive, XR.

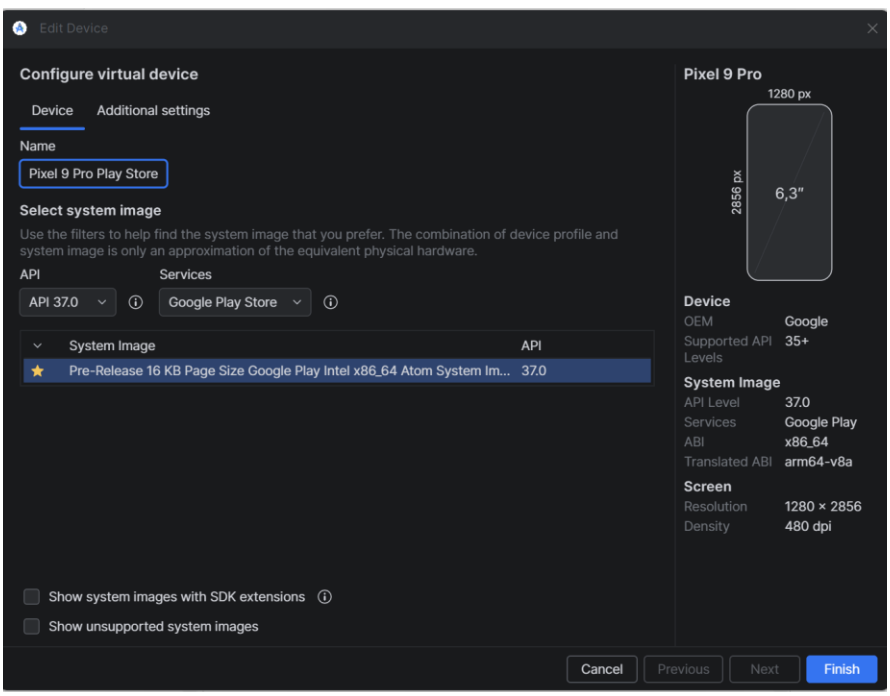
When you have installed Android Studio, click on the menu: More Actions and select "Virtual Device Manager"



Click on the “+” to add a new device and select a phone (e.g. Pixel 9 Pro XL)



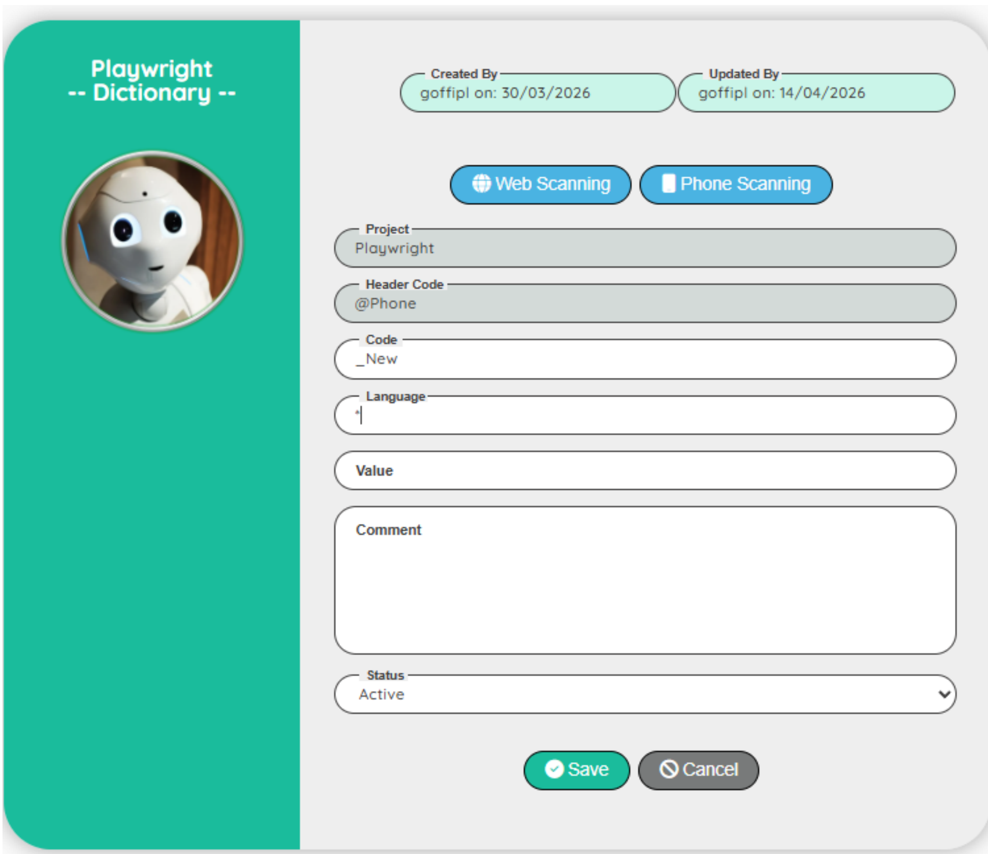
Click on Next, and now **a very important step**: you need to select **Google Play Store** rather than Google APIs.



Android Studio - Scanner

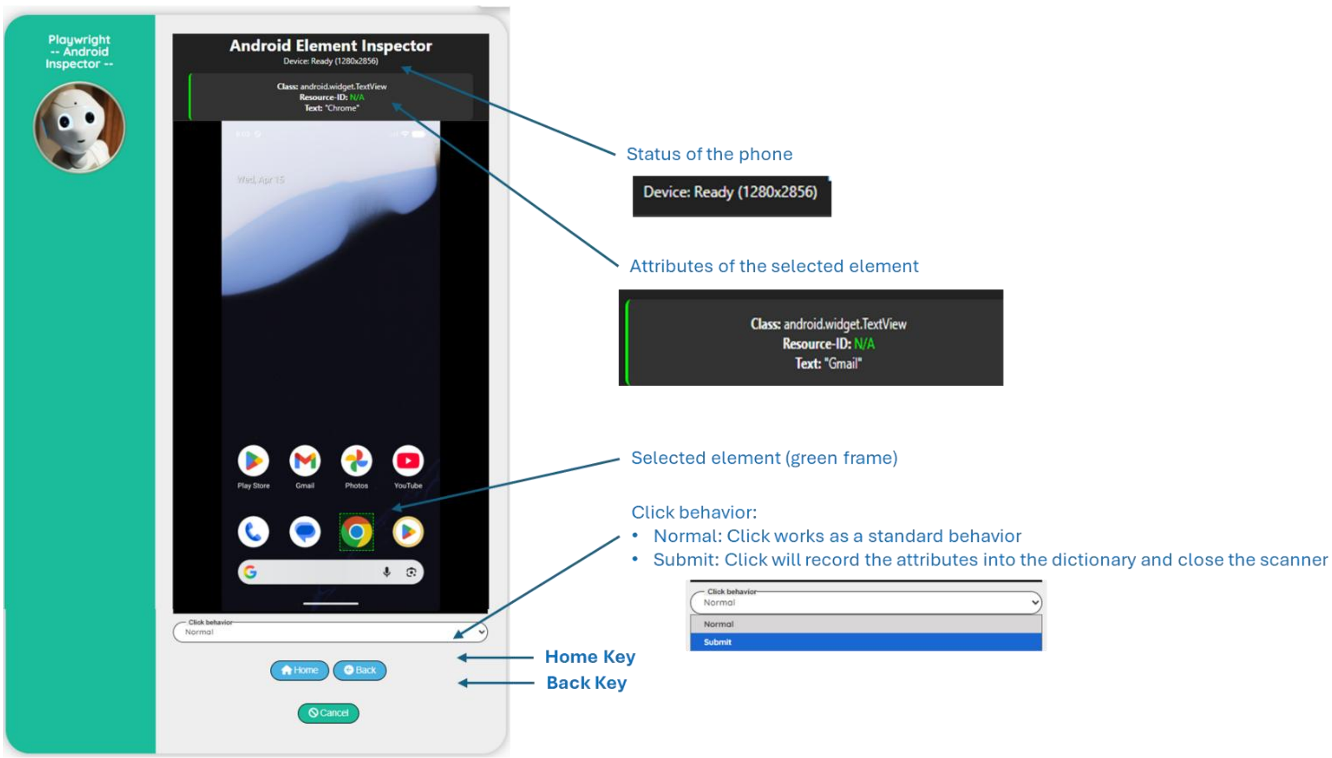
Go to the Dictionary screen (Designer role) and create a new entry.
Be sure you launch Android Studio and start a virtual phone based on Google Play Store.

Click on the 'Phone Scanning' button

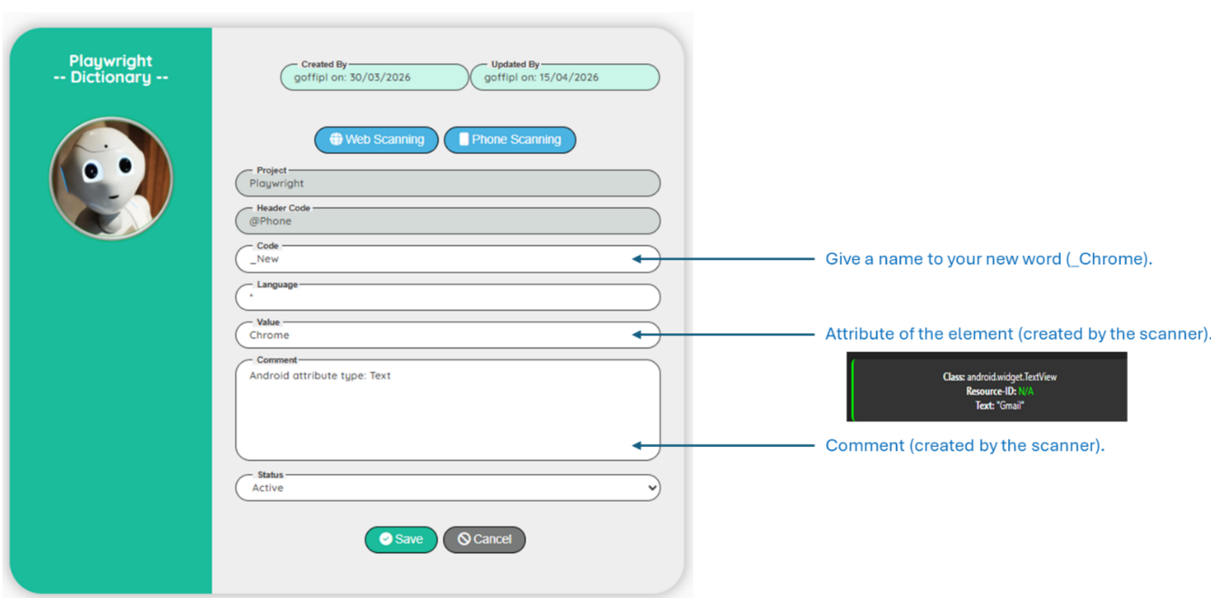


Now you can see the view of your virtual phone in the Robot user interface!
You can play directly from the Robot interface or from the virtual phone (Android Studio).

Change the behavior of the click to 'Submit', select the element to analyze and click on it to store the attributes directly in the dictionary.

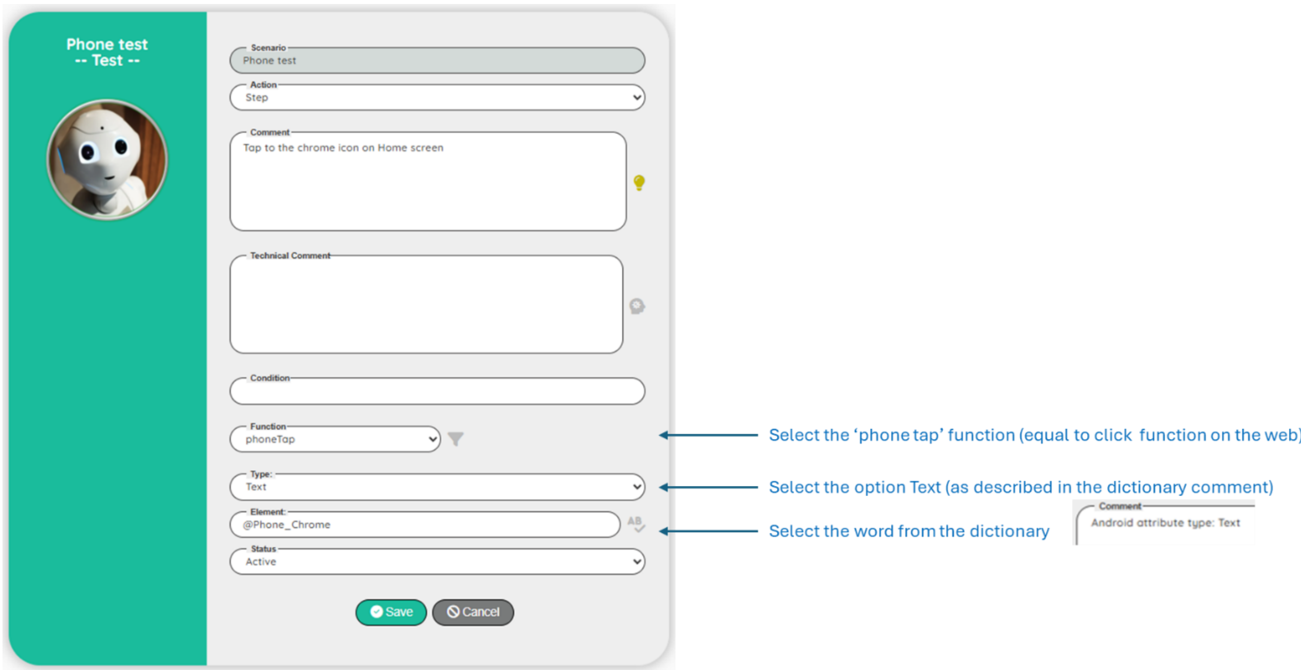


You can see the result of the scanner stored in the fields Value and Comment



Phone functions

Use a phone function. For instance, phoneTap that is used to simulate the tap (click) on the screen of the phone.
Select the word from the dictionary (you can click on it to see the detail with the comment)
Select the type (Text or Element) you can see this information in the dictionary comment.



Mobile testing test case

Steps	Expected result
Click on the home key	Screen returns to the home page
Click on the google icon (bottom on the screen)	Screen shows the Google page
Click on the Google bar and enter a link + Enter	Screen shows the new web page
Take a print screen of the web page	A print screen is visible in the robot

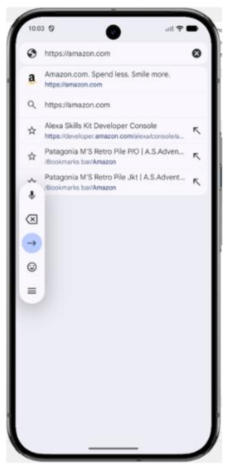
Execute the phone test case

Click on the Home key.

phonePress(Home)

Click on the Google icon.

phoneTap(text, @Phone_Chrome)

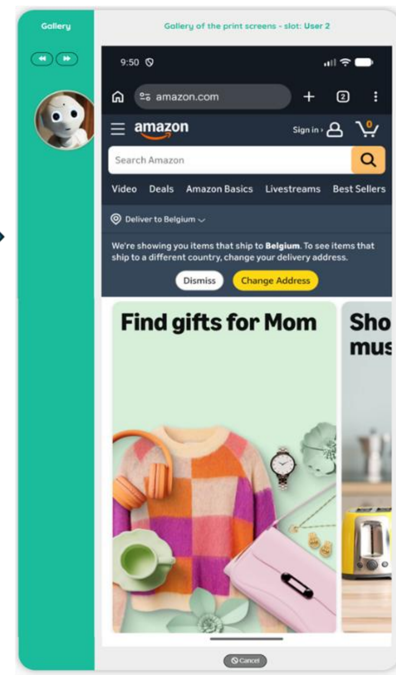


Click on the Google bar and key: https://amazon.com and type ENTER.

phoneTap(element, @Phone_Searchbar)
phoneFill(phone, @Phone_Searchbar, https://amazon.com)
phonePress(Enter)

Take a print capture in the slot 2

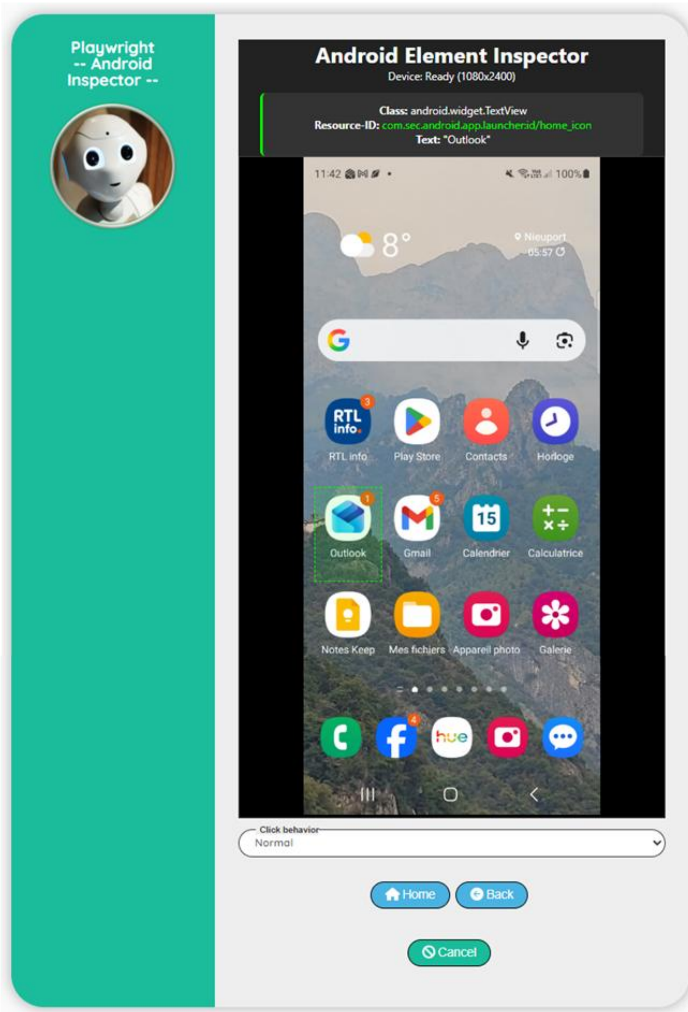
phoneCapture(phone, 2, Normal)



Mobile tests: Using Vizor to connect your own smartphone

Visor a software that can be used to image your smartphone (Android or iPhone)

https://www.vysor.io/download/



I haven't been able to get the scanner working with my iPhone. The smartphone is visible in **Vyzor**, but communication with the device doesn't seem to work!.

After further research, it appears that it is impossible to use an iPhone because iOS does not expose anything similar to ADB (the connector used to communicate with an Android device).

Mobile testing: Conclusions

A very nice implementation of mobile testing within the Robot tool ecosystem.

However, mobile testing is more complex than testing for a traditional web application.
A mobile application can be:

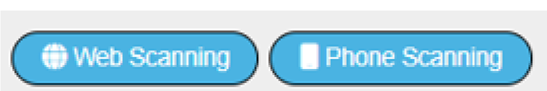
- A mobile application
- A mobile browser application

For each category, you must use a specific virtual phone to establish the correct connection.

If you want to use a 'Mobile application', you need to create a phone based on the "Google Play Store" service.
If you want to use a 'Mobile browser application', you need to create a phone based on the "Google APIs" service.

This is the reason why you need to specify the type of phone to use
(like in this example: **phoneFill**(phone, @Phone_Searchbar, https://amazon.com)

Also, when you want to scan a page to detect elements, you need to use the right scanner



Web Scanning is used to scan Desktop browser or Phone browser,
Phone Scanning is used to scan only Mobile application

Testing ODBC databases

When preparing your tests, you may need to access a database to clear records from a previous test (for example, cleaning your shopping cart on the eCommerce site you are testing).

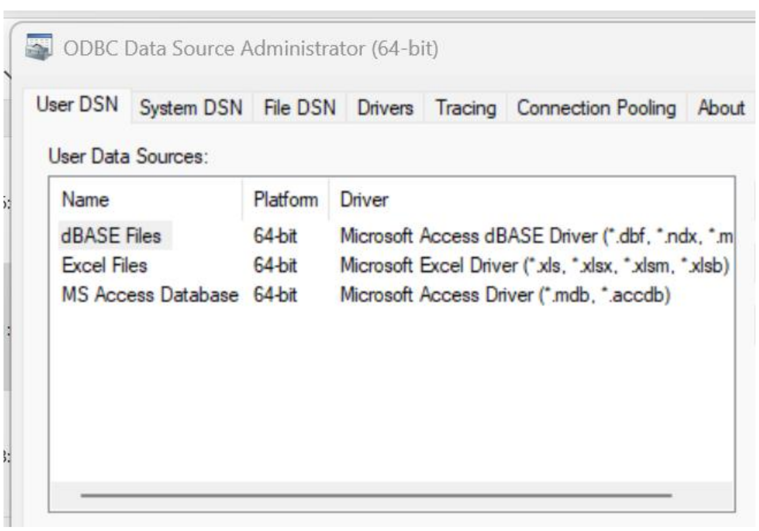
Database supported

- SQL Server
- Oracle
- DB2
- MySQL / MariaDB
- PostgreSQL
- Snowflake
- Teradata
- SAP HANA
- Any ODBC-compliant DB

Prerequisite

Before you can use the database, you must have an ODBC driver installed for the type of database you want to use.

To check the ODBC driver is installed, On windows, use the command Win + R → odbcad32



In this example, we would like to use MySQL ODBC driver, but as you can see in the above picture, there is no MySQL ODBC driver available.

Installing the ODBC driver:

On a terminal, type the following command:

node -p "process.arch"

Output	Meaning
X64	Install **64-bit ODBC**
Ia32	Install **32-bit ODBC**

Most modern systems are **x64**.

Download MySQL Connector/ODBC

Go to the official page:

🔗 **https://dev.mysql.com/downloads/connector/odbc/**

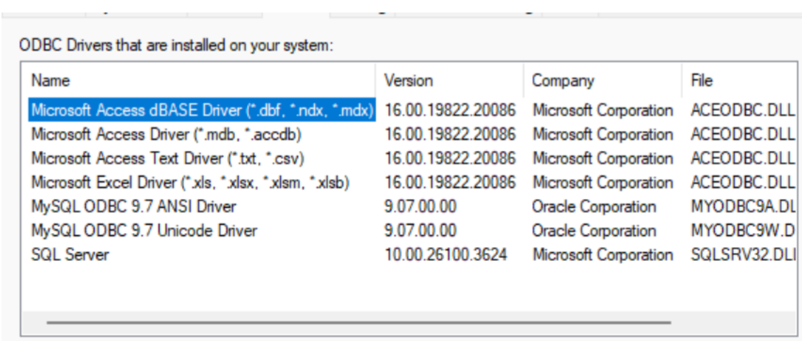
Verify Installation

Press: Win + R → odbcad32

Go to Drivers tab and you must see one or more of:

- MySQL ODBC 8.0 ANSI Driver
- MySQL ODBC 8.0 Unicode Driver

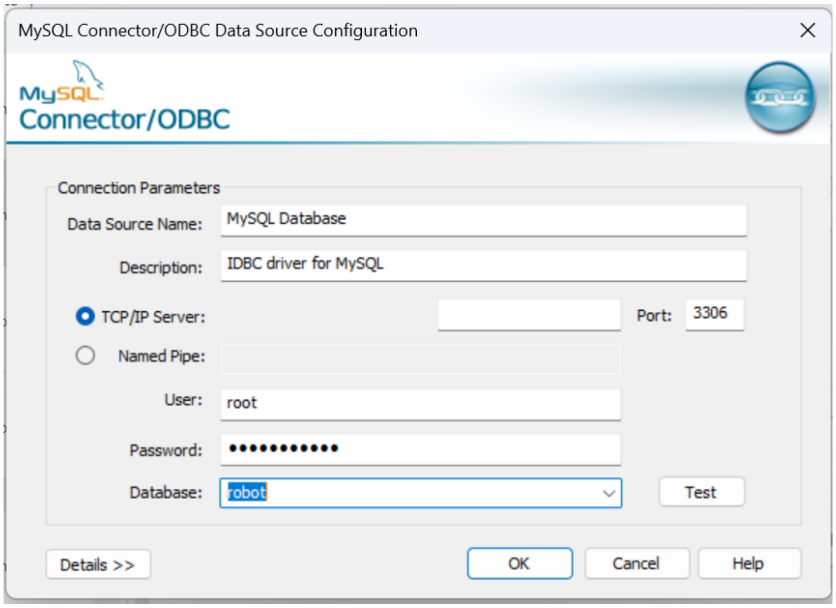
If you don't → wrong architecture installed.


ODBC Drivers that are installed on your system:

Name	Version	Company	File
Microsoft Access dBASE Driver (*.dbf, *.ndx, *.mdx)	16.00.19822.20086	Microsoft Corporation	ACEODBC.DLL
Microsoft Access Driver (*.mdb, *.accdb)	16.00.19822.20086	Microsoft Corporation	ACEODBC.DLL
Microsoft Access Text Driver (*.txt, *.csv)	16.00.19822.20086	Microsoft Corporation	ACEODBC.DLL
Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlsb)	16.00.19822.20086	Microsoft Corporation	ACEODBC.DLL
MySQL ODBC 9.7 ANSI Driver	9.07.00.00	Oracle Corporation	MYODBC9A.DL
MySQL ODBC 9.7 Unicode Driver	9.07.00.00	Oracle Corporation	MYODBC9W.D
SQL Server	10.00.26100.3624	Microsoft Corporation	SQLSRV32.DLI

Testing the connection

Go to the User DSN tab and click on the button Add...

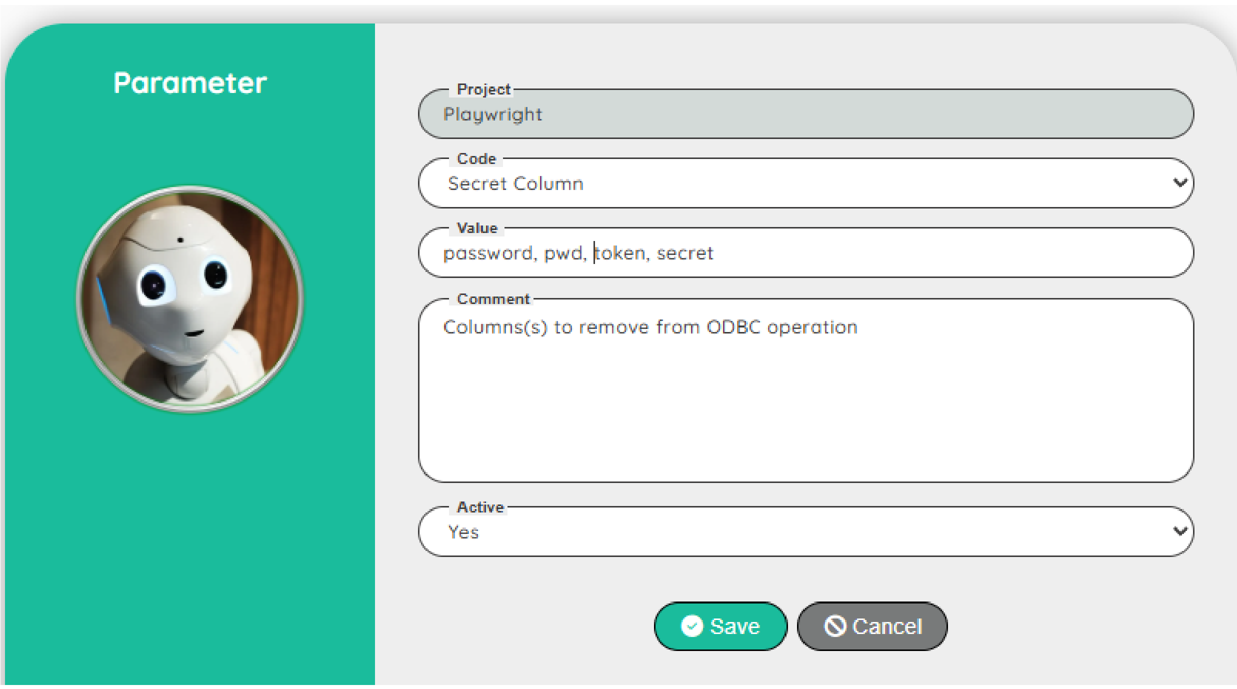


Enter the definition of your database and click on the Test button.
You must receive the message: Connection successful

Parameters

When using the ODBC_ExecuteSQL function, the result is stored in the database.
However, if you don't want to store sensitive data, the **Administrator** can define a global project parameter to exclude some columns from the SQL result.
By default, if there is no parameter defined for the project, the following columns will be excluded: **password, token, secret**

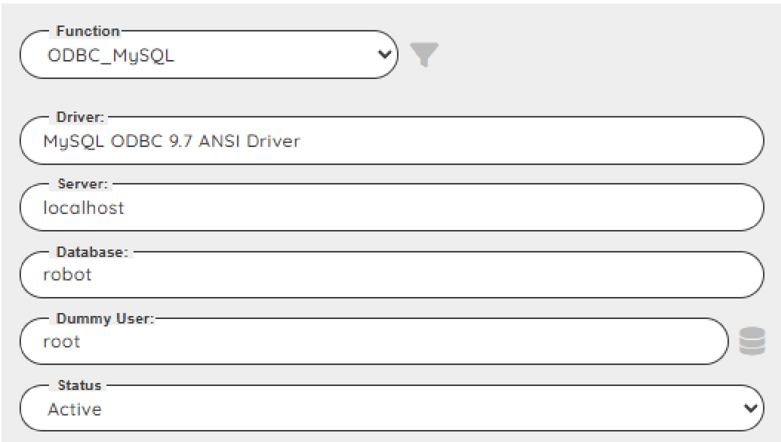
If you need to exclude specific column(s), create a project parameter: Secret Column



Be sure to include the default column(s) if relevant!

Example of a select on a MySQL database

Define the ODBC connection string

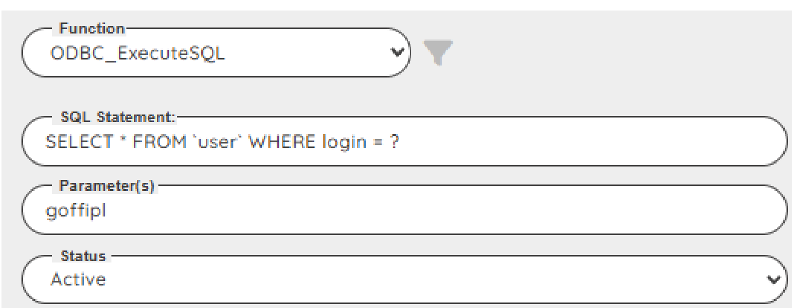


Be careful with the name of the driver, be sure that it matches your setup:

Press: Win + R → odbcad32
Go to Drivers tab and check the name of the driver.

We use the dummy user to define the user Id and the password of the database.
(the password can be crypted)

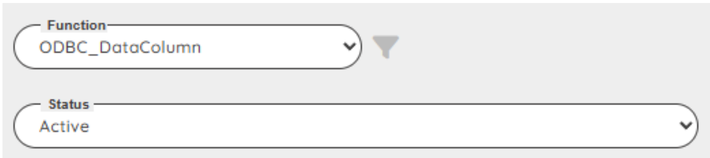
Select a record from the database

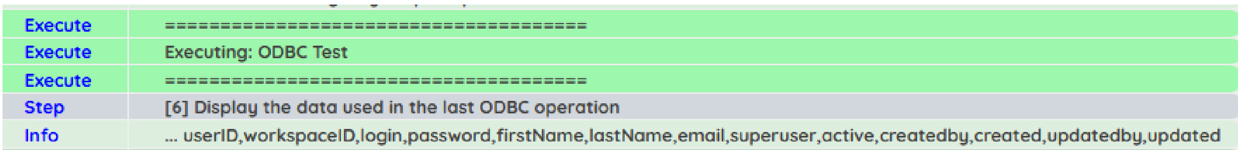


The result is now store into the Robot database

Get the name of the columns from the previous select

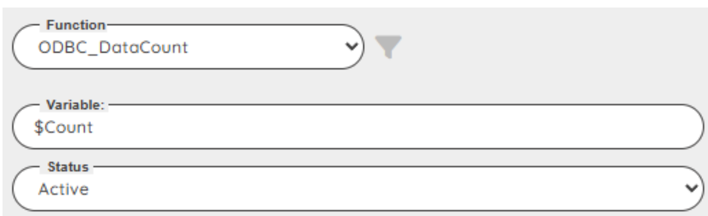
This function returns only information for the designer in order to define the next steps

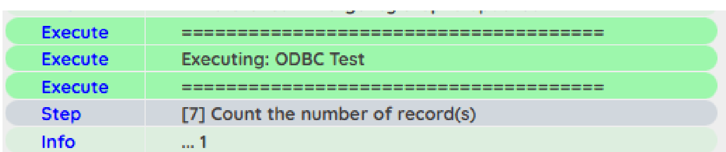




Get the number of records

Before we can extract the value of a record, we need to check the number of record(s)

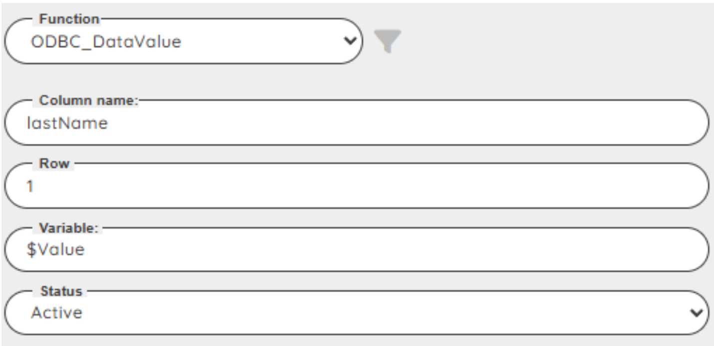


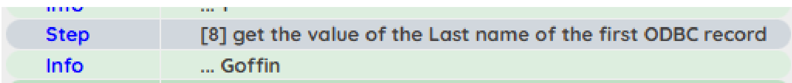


Note: as we store the number of record(s) in a variable, we can very easily create a loop to process all the record(s) of the ODBC Select

Get the last name of the user

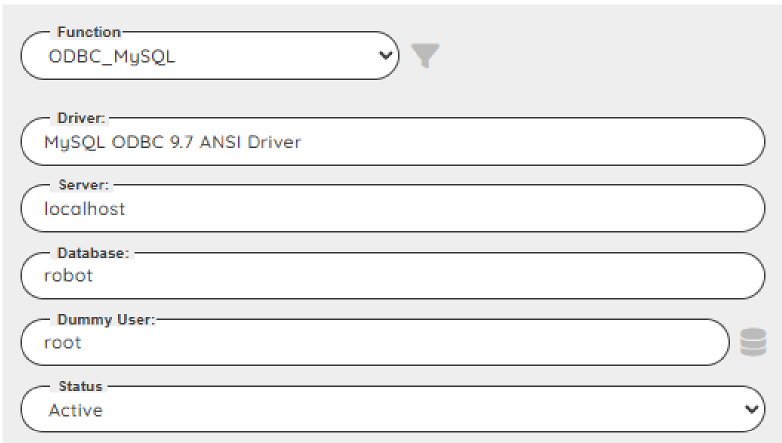
Now that we know the name of the columns, we can extract the value(s)



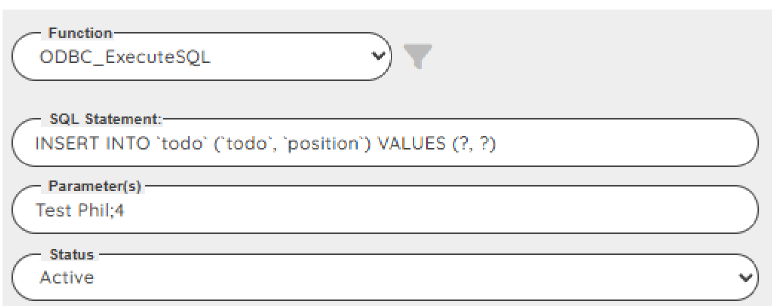


Example of a DML statement on a MySQL database

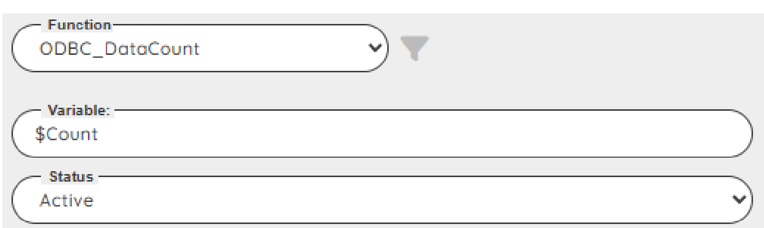
Define the ODBC connection string



Insert a new record into a table todo



Check if the record has been added



Delete the new record